

**Elektrilised meditsiiniseadmed. Osa 1: Üldised nõuded
esmasele ohutusele ja olulistele toimimisnäitajatele**

**Medical electrical equipment -- Part 1: General
requirements for basic safety and essential performance**

EESTI STANDARDI EESSÕNA

NATIONAL FOREWORD

See Eesti standard EVS-EN 60601-1:2006/A1:2013 sisaldab Euroopa standardi EN 60601-1:2006/A1:2013 ingliskeelset teksti.	This Estonian standard EVS-EN 60601-1:2006/A1:2013 consists of the English text of the European standard EN 60601-1:2006/A1:2013.
Standard on jõustunud sellekohase teate avaldamisega EVS Teatajas.	This standard has been endorsed with a notification published in the official bulletin of the Estonian Centre for Standardisation.
Euroopa standardimisorganisatsioonid on teinud Euroopa standardi rahvuslikele liikmetele kättesaadavaks 04.10.2013.	Date of Availability of the European standard is 04.10.2013.
Standard on kättesaadav Eesti Standardikeskusest.	The standard is available from the Estonian Centre for Standardisation.

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ICS 11.040

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**Medical electrical equipment -
Part 1: General requirements for basic safety and essential performance
(IEC 60601-1:2005/A1:2012)**

Appareils électromédicaux -
Partie 1: Exigences générales pour la
sécurité de base et les performances
essentielles
(CEI 60601-1:2005/A1:2012)

Medizinische elektrische Geräte -
Teil 1: Allgemeine Festlegungen für die
Sicherheit einschließlich der wesentlichen
Leistungsmerkmale
(IEC 60601-1:2005/A1:2012)

This amendment A1 modifies the European Standard EN 60601-1:2006; it was approved by CENELEC on 2013-09-24. CENELEC members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this amendment the status of a national standard without any alteration.

Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the CEN-CENELEC Management Centre or to any CENELEC member.

This amendment exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CENELEC member into its own language and notified to the CEN-CENELEC Management Centre has the same status as the official versions.

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CENELEC

European Committee for Electrotechnical Standardization
Comité Européen de Normalisation Electrotechnique
Europäisches Komitee für Elektrotechnische Normung

CEN-CENELEC Management Centre: Avenue Marnix 17, B - 1000 Brussels

Foreword

The text of document 62A/805/FDIS, future IEC 60601-1:2005/A1, prepared by SC 62A, "Common aspects of electrical equipment used in medical practice", of IEC TC 62, "Electrical equipment in medical practice" was submitted to the IEC-CENELEC parallel vote and approved by CENELEC as EN 60601-1:2006/A1:2013.

The following dates are fixed:

- latest date by which the document has to be implemented at national level by publication of an identical national standard or by endorsement (dop) 2014-06-24
- latest date by which the national standards conflicting with the document have to be withdrawn (dow) 2018-12-24

*In the foreword of EN 60601-1:2006, **replace** the first sentence of the third paragraph by:*

This European Standard supersedes EN 60601-1:1990 and its amendments, EN 60601-1-1:2001 and EN 60601-1-4:1996 + A1:1999.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CENELEC [and/or CEN] shall not be held responsible for identifying any or all such patent rights.

This document has been prepared under a mandate given to CENELEC by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For the relationship with EU Directive(s) see informative Annex ZZ, which is an integral part of this document.

Endorsement notice

The text of the International Standard IEC 60601-1:2005/A1:2012 was approved by CENELEC as a European Standard without any modification.

Replace the Bibliography of EN 60601-1:2006 by:

IEC 60073	NOTE	Harmonized as EN 60073.
IEC 60086-1	NOTE	Harmonized as EN 60086-1.
IEC 60127-6	NOTE	Harmonized as EN 60127-6.
IEC 60309-1	NOTE	Harmonized as EN 60309-1.
IEC 60332-1-2	NOTE	Harmonized as EN 60332-1-2.
IEC 60332-2-2	NOTE	Harmonized as EN 60332-2-2.
IEC 60317-43	NOTE	Harmonized as EN 60317-43.
IEC 60601-1-1:2000	NOTE	Harmonized as EN 60601-1-1:2001 (not modified).
IEC 60601-1-4:1996	NOTE	Harmonized as EN 60601-1-4:1996 + A1:1999 (not modified).
IEC 60601-1-11	NOTE	Harmonized as EN 60601-1-11.
IEC 60601-2-22	NOTE	Harmonized as EN 60601-2-22.
IEC 60601-2-49:2001	NOTE	Harmonized as EN 60601-2-49:2001 (not modified).

IEC 60695-1-10	NOTE	Harmonized as EN 60695-1-10.
IEC 60721 series	NOTE	Harmonized in EN 60721 series.
IEC 60990	NOTE	Harmonized as EN 60990.
IEC 61000-4-11	NOTE	Harmonized as EN 61000-4-11.
IEC 61010 series	NOTE	Harmonized in EN 61010 series.
IEC 61010-1:2010	NOTE	Harmonized as EN 61010-1:2010 (not modified).
IEC 61140:2001	NOTE	Harmonized as EN 61140:2002 (not modified).
IEC 61558-1	NOTE	Harmonized as EN 61558-1.
IEC 61558-2-4	NOTE	Harmonized as EN 61558-2-4.
IEC 61558-2-23	NOTE	Harmonized as EN 61558-2-23.
IEC 62079:2001	NOTE	Harmonized as EN 62079:2001 (not modified).
IEC 62353	NOTE	Harmonized as EN 62353.
IEC 62471:2006	NOTE	Harmonized as EN 62471:2008 (modified).
IEC 80001-1:2010	NOTE	Harmonized as EN 80001-1:2011 (not modified).
ISO 407	NOTE	Harmonized as EN ISO 407.
ISO 7396-1	NOTE	Harmonized as EN ISO 7396-1.
ISO 8041	NOTE	Harmonized as EN ISO 8041.
ISO 13485	NOTE	Harmonized as EN ISO 13485.
ISO 15001	NOTE	Harmonized as EN ISO 15001.

Annex ZA (normative)

Normative references to international publications with their corresponding European publications

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

NOTE When an international publication has been modified by common modifications, indicated by (mod), the relevant EN/HD applies.

Replace Annex ZA of EN 60601-1:2006 by :

<u>Publication</u>	<u>Year</u>	<u>Title</u>	<u>EN/HD and IEC/ISO Year</u>	
IEC 60065 (mod)	2001	Audio, video and similar electronic apparatus - Safety requirements	EN 60065	2002
A1 (mod)	2005		+ corr. March A1	2006
			+ corr. August A11	2007
			A2	2008
A2 (mod)	2010		A12	2010
				2012
IEC 60068-2-2	2007	Environmental testing Part 2: Tests - Test B: Dry heat	EN 60068-2-2	2007
IEC 60079-0 (mod)	- ¹⁾	Explosive atmospheres - Part 0: Equipment - General requirements	EN 60079-0	2012
IEC 60079-2	- ¹⁾	Explosive atmospheres - Part 2: Equipment protection by pressurized enclosure "p"	EN 60079-2	2007
IEC 60079-5	- ¹⁾	Explosive atmospheres - Part 5: Equipment protection by powder filling "q"	EN 60079-5	2007
IEC 60079-6	- ¹⁾	Explosive atmospheres - Part 6: Equipment protection by oil immersion "o"	EN 60079-6	2007
IEC 60083	- ¹⁾	Plugs and socket-outlets for domestic and similar general use standardized in member countries of IEC	IEC 60083	2009
IEC 60085	- ¹⁾	Electrical insulation - Thermal evaluation and designation	EN 60085	2008
IEC 60086-4	- ¹⁾	Primary batteries Part 4: Safety of lithium batteries	EN 60086-4	2007
IEC 60112	- ¹⁾	Method for the determination of the proof and the comparative tracking indices of solid insulating materials	EN 60112	2003
IEC 60127-1	2006	Miniature fuses Part 1: Definitions for miniature fuses and general requirements for miniature fuse-links	EN 60127-1	2006

¹⁾ Undated reference, converted to dated reference in this European Standard.

<u>Publication</u>	<u>Year</u>	<u>Title</u>	<u>EN/HD and IEC/ISO Year</u>	
IEC 60227-1 ²⁾	1993	Polyvinyl chloride insulated cables of rated voltages up to and including 450/750 V		
A1	1995			
A2	1998	Part 1: General requirements	HD 21.1 S4	2002
IEC 60245-1 ³⁾	2003	Rubber insulated cables - Rated voltages up to and including 450/750 V		2003
		Part 1: General requirements		
IEC 60252-1	- ¹⁾	AC motor capacitors	EN 60252-1	2011
		Part 1: General - Performance, testing and rating - Safety requirements - Guidance for installation and operation		
IEC 60320-1	- ¹⁾	Appliance couplers for household and similar general purposes	EN 60320-1	2001
		Part 1: General requirements		
IEC 60335-1 (mod)	2010	Household and similar electrical appliances - Safety	EN 60335-1	2012
		Part 1: General requirements		
IEC 60364-4-41 (mod)	2005	Low-voltage electrical installations	HD 60364-4-41	2006
		Part 4-41: Protection for safety - Protection against electric shock		
IEC 60384-14	2005	Fixed capacitors for use in electronic equipment	EN 60384-14	2005
		Part 14: Sectional specification - Fixed capacitors for electromagnetic interference suppression and connection to the supply mains		
IEC 60417	Data base	Graphical symbols for use on equipment available from http://www.graphical-symbols.info/equipment	IEC 60417	2004
IEC 60445	- ¹⁾	Basic and safety principles for man-machine interface, marking and identification - Identification of equipment terminals, conductor terminations and conductors	EN 60445	2010
IEC 60447	- ¹⁾	Basic and safety principles for man-machine interface, marking and identification - Actuating principles	EN 60447	2004
IEC 60529	1989	Degrees of protection provided by enclosures (IP Code)	EN 60529	1991
			+ corr. May	1993
A1	1999		A1	2000
IEC 60601-1-2	- ¹⁾	Medical electrical equipment	EN 60601-1-2	2007
		Part 1-2: General requirements for basic safety and essential performance - Collateral standard: Electromagnetic compatibility - Requirements and tests		
IEC 60601-1-3	- ¹⁾	Medical electrical equipment -	EN 60601-1-3	2008
		Part 1-3: General requirements for basic safety and essential performance - Collateral standard: Radiation protection in diagnostic X-ray equipment	+ corr. March	2010
A1	- ¹⁾		A1	2013

²⁾ HD 21.1 S4:2002, *Cables of rated voltages up to and including 450/750 V and having thermoplastic insulation - Part 1: General requirements*, which is related to, but not directly equivalent with, IEC 60227-1, applies instead.

³⁾ HD 22.1 S4:2002, *Cables of rated voltages up to and including 450/750 V and having cross-linked insulation - Part 1: General requirements*, which is related to, but not directly equivalent with, IEC 60245-1, applies instead.

<u>Publication</u>	<u>Year</u>	<u>Title</u>	<u>EN/HD and IEC/ISO</u>	<u>Year</u>
IEC 60601-1-6	- ¹⁾	Medical electrical equipment Part 1-6: General requirements for basic safety and essential performance - Collateral standard: Usability	EN 60601-1-6	2010
IEC 60601-1-8	- ¹⁾	Medical electrical equipment Part 1-8: General requirements for basic safety and essential performance - Collateral Standard: General requirements, tests and guidance for alarm systems in medical electrical equipment and medical electrical systems	EN 60601-1-8 + corr. March	2007 2010
A1	- ¹⁾		A1	2013
IEC 60664-1	2007	Insulation coordination for equipment within low-voltage systems Part 1: Principles, requirements and tests	EN 60664-1	2007
IEC 60695-11-10	- ¹⁾	Fire hazard testing Part 11-10: Test flames - 50 W horizontal and vertical flame test methods	EN 60695-11-10	1999
A1				2003
IEC 60730-1 (mod)	2010	Automatic electrical controls for household and similar use Part 1: General requirements	EN 60730-1	2011
IEC 60825-1	2007	Safety of laser products Part 1: Equipment classification and requirements	EN 60825-1	2007
IEC 60851-3	2009	Winding wires - Test methods Part 3: Mechanical properties	EN 60851-3	2009
IEC 60851-5	2008	Winding wires - Test methods Part 5: Electrical properties	EN 60851-5	2008
IEC 60851-6	1996	Winding wires - Test methods	EN 60851-6	1996
A1	1997	Part 6: Thermal properties	A1	1997
IEC/TR 60878	2003	Graphical symbols for electrical equipment in medical practice	IEC/TR 60878	2003
IEC 60884-1	- ¹⁾	Plugs and socket-outlets for household and similar purposes Part 1: General requirements	IEC 60884-1	2013
IEC 60950-1 (mod)	2001	Information technology equipment - Safety Part 1: General requirements	EN 60950-1 + corr. April A11	2001 2004 2004
IEC 61058-1 (mod)	2000	Switches for appliances		
A1	2001	Part 1: General requirements	EN 61058-1	2002
A2	2007		A2	2008
IEC 61558-2-1	- ¹⁾	Safety of power transformers, power supplies, reactors and similar products - Part 2-1: Particular requirements and tests for separating transformers and power supplies incorporating separating transformers for general applications	EN 61558-2-1	2007
IEC 61672-1	- ¹⁾	Electroacoustics - Sound level meters Part 1: Specifications	EN 61672-1	2003
IEC 61672-2	- ¹⁾	Electroacoustics - Sound level meters Part 2: Pattern evaluation tests	EN 61672-2	2003

<u>Publication</u>	<u>Year</u>	<u>Title</u>	<u>EN/HD and IEC/ISO</u>	<u>Year</u>
IEC 61965	- ¹⁾	Mechanical safety of cathode ray tubes	EN 61965	2003
IEC 62133	- ¹⁾	Secondary cells and batteries containing alkaline or other non-acid electrolytes – Safety requirements for portable sealed secondary cells, and for batteries made from them, for use in portable applications	EN 62133	2013
IEC 62304	2006	Medical device software – Software lifecycle processes	EN 62304 + corr. November	2006 2008
ISO 780	- ¹⁾	Packaging - Pictorial marking for handling of goods	EN ISO 780	1999
ISO 1853	- ¹⁾	Conducting and dissipative rubbers, vulcanized or thermoplastic - Measurement of resistivity	ISO 1853	2011
ISO 2878	- ¹⁾	Rubber, vulcanized - Antistatic and conductive products - Determination of electrical resistance	ISO 2878	2011
ISO 2882	- ¹⁾	Rubber, vulcanized - Antistatic and conductive products for hospital use - Electrical resistance limits	ISO 2882	1997
ISO 3746	- ¹⁾	Acoustics - Determination of sound power levels of noise sources using sound pressure - Survey method using an enveloping measurement surface over a reflecting plane	EN ISO 3746	2010

<u>Publication</u>	<u>Year</u>	<u>Title</u>	<u>EN/HD and IEC/ISO Year</u>	
ISO 3864-1	2002	Graphical symbols - Safety colours and safety signs Part 1: Design principles for safety signs in workplaces and public areas	ISO 3864-1	2011
ISO 7000	2004	Graphical symbols for use on equipment – Collection of symbols	ISO 7000	2004
ISO 7010	2011	Graphical symbols – Safety colours and safety signs – Registered safety signs	EN ISO 7010	2012
ISO 9614-1	- ¹⁾	Acoustics – Determination of sound power levels of noise sources using sound intensity – Measurement at discrete points	EN ISO 9614-1	2009
ISO 10993 all parts	- ¹⁾	Biological evaluation of medical devices	See list below	
ISO 10993-1	2009	Biological evaluation of medical devices - Part 1: Evaluation and testing within a risk management process	EN ISO 10993-1	2009
+ corr June	2010			
ISO 10993-2	2006	Biological evaluation of medical devices - Part 2: Animal welfare requirements	EN ISO 10993-2	2006
ISO 10993-3	2003	Biological evaluation of medical devices - Part 3: Tests for genotoxicity, carcinogenicity and reproductive toxicity	EN ISO 10993-3	2003
ISO 10994-4	2002	Biological evaluation of medical devices - Part 4: Selection of tests for interactions with blood		
A1	2006		EN ISO 10993-4	2009
ISO 10993-5	2009	Biological evaluation of medical devices - Part 5: Tests for in vitro cytotoxicity	EN ISO 10993-5	2009
ISO 10993-6	2007	Biological evaluation of medical devices - Part 6: Tests for local effects after implantation	EN ISO 10993-6	2009
ISO 10993-7	2008	Biological evaluation of medical devices - Part 7: Ethylene oxide sterilization residuals	EN ISO 10993-7	2008
+ corr November	2009		+ AC	2009
ISO 10993-9	2009	Biological evaluation of medical devices - Part 9: Framework for identification and quantification of potential degradation products	EN ISO 10993-9	2009
ISO 10993-10	2010	Biological evaluation of medical devices - Part 10: Tests for irritation and skin sensitization	EN ISO 10993-10	2010
ISO 10993-11	2006	Biological evaluation of medical devices - Part 11: Tests for systemic toxicity	EN ISO 10993-11	2009
ISO 10993-12	2012	Biological evaluation of medical devices - Part 12: Sample preparation and reference materials	EN ISO 10993-12	2012
ISO 10993-13	2010	Biological evaluation of medical devices - Part 13: Identification and quantification of degradation products from polymeric medical devices	EN ISO 10993-13	2010
ISO 10993-14	2001	Biological evaluation of medical devices - Part 14: Identification and quantification of degradation products from ceramics	EN ISO 10993-14	2009
ISO 10993-15	2000	Biological evaluation of medical devices - Part 15: Identification and quantification of degradation products from metals and alloys	EN ISO 10993-15	2009
ISO 10993-16	2010	Biological evaluation of medical devices - Part 16: Toxicokinetic study design for degradation products and leachables	EN ISO 10993-16	2010

<u>Publication</u>	<u>Year</u>	<u>Title</u>	<u>EN/HD and IEC/ISO</u>	<u>Year</u>
ISO 10993-17	2002	Biological evaluation of medical devices - Part 17: Establishment of allowable limits for leachable substances	EN ISO 10993-17	2009
ISO 10993-18	2005	Biological evaluation of medical devices - Part 18: Chemical characterization of materials	EN ISO 10993-18	2009
ISO/TS 10993-19	2006	Biological evaluation of medical devices - Part 19: Physico-chemical, morphological and topographical characterization of materials	ISO/TS 10993-19	2006
ISO/TS 10993-20	2006	Biological evaluation of medical devices - Part 20: Principles and methods for immunotoxicology testing of medical devices	ISO/TS 10993-20	2006
ISO 11135-1	2007	Sterilization of health care products – Ethylene oxide – Part 1: Requirements for development, validation and routine control of a sterilization process for medical devices	EN ISO 11135-1	2007
ISO 11137-1	2006	Sterilization of health care products – Radiation – Part 1: Requirements for development, validation and routine control of a sterilization process for medical devices	EN ISO 11137-1	2006
ISO 13857	2008	Safety of machinery – Safety distances to prevent hazard zones being reached by the upper and lower limbs	EN ISO 13857	2008
ISO 14971	2007	Medical devices – Application of risk management to medical devices	EN ISO 14971	2012
ISO 15223-1	2012	ISO 15223-1:2012, Medical devices – Symbols to be used with medical device labels, labelling and information to be supplied – Part 1: General requirements	EN ISO 15223-1	2012
ISO 17665-1	2006	Sterilization of health care products – Moist heat – Part 1: Requirements for the development, validation and routine control of a sterilization process for medical devices	EN ISO 17665-1	2006
ISO 23529	- ¹⁾	Rubber – General procedures for preparing and conditioning test pieces for physical test methods	ISO 23529	2010
ISO 80000-1	2009	Quantities and units – Part 1: General	EN ISO 80000-1	2013

Annex ZZ (informative)

Relationship between this European Standard and the Essential Requirements of EU Directive 93/42/EEC on Medical Devices

This European Standard has been prepared under a mandate given to CENELEC by the European Commission and the European Free Trade Association to provide a means of conforming to the Essential Requirements given in Annex I of the EC Directives 93/42/EEC as amended by 2007/47/EC.

General Guidance:

Once this standard will be cited in the Official Journal of the European Union under that Directive, compliance with the clauses of this standard given in Table ZZ.1 confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding Essential Requirements (ERs) of that Directive and associated EFTA regulations.

NOTE 1 This standard is intended to be applied in its entirety only. Selected clauses or subclauses may be not applicable due to the specific type of equipment under consideration. It is necessary to understand and apply Clauses 1 to 5. It is also recommended to understand and apply those clauses which contain general requirements related to a specific subclause. Elements of the standard that are not cited in Table ZZ.1 may be relevant for the appropriate fulfilment of certain essential requirements through indirect reference, and for safety and performance aspects of the device, that are not addressed through essential requirements.

NOTE 2 Where a reference from a clause of this standard to the risk management process is made, the risk management process needs to be in compliance to the MDD (Directive 93/42/EEC amended by 2007/47/EC). This means that risks have to be reduced "as far as possible", "to a minimum", "to the lowest possible level", "minimized" or "removed", according to the wording of the corresponding essential requirement.

NOTE 3 With respect to note 4 of clause 4.2.2 General requirement for risk management, the manufacturer's policy for determining **acceptable risk** must be in compliance with essential requirements 1, 2, 5, 6, 7, 8, 9, 11 and 12 of the directive.

NOTE 4 References in the clauses 3 to 17 or in the Annexes of this standard specify whether the normative references listed in Clause 2 as cited in Annex ZA are to be applied in whole or in part.

NOTE 5 This Annex ZZ is based on Normative References according to Annex ZA, replacing the references in the core text.

WARNING: Other requirements and other EU Directives and Regulations may be applicable to the product(s) falling within the scope of this standard.

Table ZZ.1: Relationship between Essential Requirements of Directive 93/42/EEC amended by 2007/47/EC, and Clauses and Subclauses of this standard

No.	Essential Requirement	Coverage
I.		
1.	General Guidance note 2 and 3 shall be observed	
1	The devices must be designed and manufactured in such a way that, when used under the conditions and for the purposes intended, they will not compromise the clinical condition or the safety of patients, or the safety and health of users or, where applicable, other persons, provided that any risks which may be associated with their intended use constitute acceptable risks when weighed against the benefits to the patient and are compatible with a high level of protection of health and safety. This shall include:	Not completely covered But If the manufacturer follows this standard in his design and manufacturing process, this European Standard gives a valuable set of technical requirements to assist in fulfilling this ER for equipment in the scope of this standard.
	- reducing, as far as possible, the risk of use error due to the ergonomic features of the device and the environment in which the device is intended to be used (design for patient safety), and	Not covered See EN/IEC 60601-1-6, EN/IEC 62366, EN/IEC 60601-1-11 and EN/IEC 60601-1-12
	- consideration of the technical knowledge, experience, education and training and where applicable the medical and physical conditions of intended users (design for lay, professional, disabled or other users).	Covered only for accompanying documents by: 7.9.1 Paragraphs 4 and 5, intended operator
2.	General Guidance note 2 and 3 shall be observed	
2	The solutions adopted by the manufacturer for the design and construction of the devices must conform to safety principles, taking account of the generally acknowledged state of the art. In selecting the most appropriate solutions, the manufacturer must apply the following principles in the following order:	1st paragraph: Covered only in respect of the following and under the condition that 2nd paragraph (including the following 3 bullets) is taken into account: 8 Protection against electrical hazards from ME equipment 9 Protection against mechanical hazards of ME equipment and ME systems 15 Construction of me equipment 2nd paragraph (including the following 3 bullets) Not covered in the normative text.
	- eliminate or reduce risks as far as possible (inherently safe design and construction),	
	- where appropriate take adequate protection measures including alarms if necessary, in relation to risks that cannot be eliminated,	
	- inform users of the residual risks due to any shortcomings of the	

No.	Essential Requirement	Coverage
	protection measures adopted.	
3	The devices must achieve the performances intended by the manufacturer and be designed, manufactured and packaged in such a way that they are suitable for one or more of the functions referred to in Article 1 (2) (a), as specified by the manufacturer.	Not covered
4	The characteristics and performances referred to in Sections 1, 2 and 3 must not be adversely affected to such a degree that the clinical conditions and safety of the patients and, where applicable, of other persons are compromised during the lifetime of the device as indicated by the manufacturer, when the device is subjected to the stresses which can occur during normal conditions of use.	Not covered However, the standard provides a procedure for the generation of information that is necessary to document that the device is in compliance with this ER.
5.	General Guidance note 2 and 3 shall be observed	
5	The devices must be designed, manufactured and packed in such a way that their characteristics and performances during their intended use will not be adversely affected during transport and storage taking account of the instructions and information provided by the manufacturer.	Covered only in respect of the following: Instructions and information provided by the manufacturer 7.2.17 Marking on protective packaging 7.9.3.1 Technical description 15.3.7 Environmental influences
6.	General Guidance note 2 and 3 shall be observed	
6	Any undesirable side-effect must constitute an acceptable risk when weighed against the performances intended.	Not covered.
6a	Demonstration of conformity with the essential requirements must include a clinical evaluation in accordance with Annex X.	Not covered
II.		
7	Chemical, physical and biological properties	General Guidance note 2 and 3 shall be observed
7.1	The devices must be designed and manufactured in such a way as to guarantee the characteristics and performances referred to in Section I (3) on the 'General requirements'. Particular attention must be paid to:	Not covered
	- the choice of materials used, particularly as regards toxicity and, where appropriate, flammability,	Partially covered in respect of the following: Toxicity: 11.7 Biocompatibility, the manufacturer should apply the appropriate part of the EN ISO 10993 series 13.1.2 Emissions, deformation of Enclosure or

No.	Essential Requirement	Coverage
		exceeding maximum temperature Flammability: 11.2 Fire prevention 11.3 Constructional requirements for fire enclosures 11.4 ME equipment and ME systems intended for use with flammable anaesthetics Annex G Protection against hazards of ignition of flammable anaesthetic mixtures
	- the compatibility between the materials used and biological tissues, cells and body fluids, taking account of the intended purpose of the device,	Not covered The manufacturer should apply the appropriate part of the EN ISO 10993 series
	- where appropriate, the results of biophysical or modeling research whose validity has been demonstrated beforehand.	Not covered
7.2	The devices must be designed, manufactured and packed in such a way as to minimize the risks posed by contaminants and residues to the persons involved in the transport, storage and use of the devices and to the patients, taking account of the intended purpose of the product. Particular attention must be paid to the tissues exposed and to the duration and frequency of exposure.	Not covered
7.3	The devices must be designed and manufactured in such a way that they can be used safely with the materials, substances and gases with which they enter into contact during their normal use or during routine procedures;	Covered only for the physical properties dealt with in Subclauses: 11.2.2 ME equipment and ME systems used in conjunction with oxygen rich environments 11.2.3 Single fault conditions related to oxygen rich environments and 11.6.1, 11.6.2, 11.6.3, 11.6.4, 11.6.6, 11.6.7, 11.6.8 (Overflow, spillage, leakage, cleaning, disinfection, sterilization and compatibility with substances used)
	if the devices are intended to administer medicinal products they must be designed and manufactured in such a way as to be compatible with the medicinal products concerned according to the provisions and restrictions governing these products and that their performance is maintained in accordance with the intended use.	Not covered
7.4	Where a device incorporates, as an integral part, a substance which, if used separately, may be considered to be a	Not covered

No.	Essential Requirement	Coverage
	medicinal product as defined in Article 1 of Directive 2001/83/EC and which is liable to act upon the body with action ancillary to that of the device, the quality, safety and usefulness of the substance must be verified by analogy with the methods specified in Annex I to Directive 2001/83/EC.	
	For the substances referred to in the first paragraph, the notified body shall, having verified the usefulness of the substance as part of the medical device and taking account of the intended purpose of the device, seek a scientific opinion from one of the competent authorities designated by the Member States or the European Medicines Agency (EMA) acting particularly through its committee in accordance with Regulation (EC) No 726/2004 on the quality and safety of the substance including the clinical benefit/risk profile of the incorporation of the substance into the device. When issuing its opinion, the competent authority or the EMA shall take into account the manufacturing process and the data related to the usefulness of incorporation of the substance into the device as determined by the notified body. Where a device incorporates, as an integral part, a human blood derivative, the notified body shall, having verified the usefulness of the substance as part of the medical device and taking into account the intended purpose of the device, seek a scientific opinion from the EMA, acting particularly through its committee, on the quality and safety of the substance including the clinical benefit/risk profile of the incorporation of the human blood derivative into the device. When issuing its opinion, the EMA shall take into account the manufacturing process and the data related to the usefulness of incorporation of the substance into the device as determined by the notified body. Where changes are made to an ancillary substance incorporated in a device, in particular related to its manufacturing process, the notified body shall be informed of the changes and shall consult the relevant medicines competent authority (i.e. the one involved in the initial consultation), in	Not covered

No.	Essential Requirement	Coverage
	order to confirm that the quality and safety of the ancillary substance are maintained. The competent authority shall take into account the data related to the usefulness of incorporation of the substance into the device as determined by the notified body, in order to ensure that the changes have no negative impact on the established benefit/risk profile of the addition of the substance in the medical device. When the relevant medicines competent authority (i.e. the one involved in the initial consultation) has obtained information on the ancillary substance, which could have an impact on the established benefit/risk profile of the addition of the substance in the medical device, it shall provide the notified body with advice, whether this information has an impact on the established benefit/risk profile of the addition of the substance in the medical device or not. The notified body shall take the updated scientific opinion into account in reconsidering its assessment of the conformity assessment procedure.	
7.5	The devices must be designed and manufactured in such a way as to reduce to a minimum the risks posed by substances leaking from the device.	Covered in respect of the following: 9.7 Pressure vessels and parts subject to pneumatic and hydraulic pressure, 11.6.1 Protection against overflow, spillage, leakage, ingress of water or particulate matter, cleaning, disinfection and sterilization, compatibility with substances 11.6.2 Overflow 15.4.9 Oil containers
	Special attention shall be given to substances which are carcinogenic, mutagenic or toxic to reproduction, in accordance with Annex I to Council Directive 67/548/EEC of 27 June 1967 on the approximation of laws, regulations and administrative provisions relating to the classification, packaging and labeling of dangerous substances.	Not covered
	If parts of a device (or a device itself) intended to administer and/or remove medicines, body liquids or other substances to or from the body, or devices intended for transport and storage of such body fluids or substances, contain phthalates which	Not covered

No.	Essential Requirement	Coverage
	are classified as carcinogenic, mutagenic or toxic to reproduction, of category 1 or 2, in accordance with Annex I to Directive 67/548/EEC, these devices must be labeled on the device itself and/or on the packaging for each unit or, where appropriate, on the sales packaging as a device containing phthalates.	
	If the intended use of such devices includes treatment of children or treatment of pregnant or nursing women, the manufacturer must provide a specific justification for the use of these substances with regard to compliance with the essential requirements, in particular of this paragraph, within the technical documentation and, within the instructions for use, information on residual risks for these patient groups and, if applicable, on appropriate precautionary measures.	Not covered
7.6	Devices must be designed and manufactured in such a way as to reduce, as much as possible, risks posed by the unintentional ingress of substances into the device taking into account the device and the nature of the environment in which it is intended to be used.	Not covered
8	Infection and microbial contamination	General Guidance note 2 and 3 shall be observed
8.1	The devices and manufacturing processes must be designed in such a way as to eliminate or reduce as far as possible the risk of infection to the patient, user and third parties. The design must allow easy handling and, where necessary, minimize contamination of the device by the patient or vice versa during use.	Not covered
8.2	Tissues of animal origin must originate from animals that have been subject to veterinary controls and surveillance adapted to the intended use of the tissues.	Not covered
	Notified Bodies shall retain information on the geographical origin of the animals.	Not covered
	Processing, preservation, testing and handling of tissues, cells and substances of animal origin must be carried out so as to provide optimal security. In particular safety with regard to viruses and other <i>transmissible</i> agents must be addressed by	Not covered

No.	Essential Requirement	Coverage
	implementation of validated methods of elimination or viral inactivation in the course of the manufacturing process.	
8.3	Devices delivered in a sterile state must be designed, manufactured and packed in a non-reusable pack and/or according to appropriate procedures to ensure that they are sterile when placed on the market and remain sterile, under the storage and transport conditions laid down, until the protective packaging is damaged or opened.	Not covered
8.4	Devices delivered in a sterile state must have been manufactured and sterilized by an appropriate, validated method.	Not covered
8.5	Devices intended to be sterilized must be manufactured in appropriately controlled (e.g. environmental) conditions.	Not covered
8.6	Packaging system for non-sterile devices must keep the product without deterioration at the level of cleanliness stipulated and, if the devices are to be sterilized prior to use, minimize the risk of microbial contamination;	Covered in respect of 7.2.17 Marking aspects of protective packaging
	the packaging system must be suitable taking account of the method of sterilization indicated by the manufacturer.	Not covered
8.7	The packaging and/or label of the device must distinguish between identical or similar products sold in both sterile and non-sterile condition.	Not covered
9	Construction and environmental properties	General Guidance note 2 and 3 shall be observed
9.1	If the device is intended for use in combination with other devices or equipment, the whole combination, including the connection system must be safe and must not impair the specified performances of the devices.	Covered in respect of the following: 9.1 Mechanical hazards 16.3 Power supply 16.5 Separation devices 16.6 Leakage currents 16.8 Interruption of power supply
	Any restrictions on use must be indicated on the label or in the instructions for use.	Covered by 16.2 Accompanying documents of an ME system
9.2	Devices must be designed and manufactured in such a way as to remove or minimize as far as is possible:	
	- the risk of injury, in connection with their physical features, including the volume/pressure ratio, dimensional and where appropriate ergonomic features;	Covered in respect of the following: 8.1 Electric shock 9.1 Mechanical Hazards 10 Radiation (all types) 11.1 Excessive temperatures 11.2 Fire prevention

No.	Essential Requirement	Coverage
		11.4 Flammable anaesthetics 11.5 Flammable agent 11.6.3 Spillage 11.8 Interruption of power supply 12.4 Hazardous output 13.1 Hazardous situations 13.2 Single Fault condition 15.3 Mechanical strength 15.4 Components and general assembly 15.5.3 Construction of transformers 16.3 Power supply 16.5 Separation devices 16.6 Leakage currents 16.8 Interruption of power supply
	- risks connected with reasonably foreseeable environmental conditions, such as magnetic fields, external electrical influences, electrostatic discharge, pressure, temperature or variations in pressure and acceleration;	Not covered See for EMC EN 60601-1-2 as referenced in Annex ZA See for acceleration EN 60601-1-11 and EN 60601-1-12 as referenced in Annex ZA Covered in respect of the following: pressure, temperature: test in 5.3 according to manufacturers' specification in 7.9.3.1
	- the risks of reciprocal interference with other devices normally used in the investigations or for the treatment given;	Not covered See for EMC EN 60601-1-2 as referenced in annex ZA
	- risks arising where maintenance or calibration are not possible (as with implants), from ageing of materials used or loss of accuracy of any measuring or control mechanism.	Not covered
9.3	Devices must be designed and manufactured in such a way as to minimize the risks of fire or explosion during normal use and in single fault condition.	Covered in respect of the following: Normal use 9.7.5 Pressure vessels, Single fault condition: 11.2 Fire prevention 11.3 Fire enclosures 11.4 Flammable anaesthetics Annex G ignition of flammable anaesthetic mixtures
	Particular attention must be paid to devices whose intended use includes exposure to flammable substances or to substances which could cause combustion.	Covered in respect of the following: 11.4 Flammable anaesthetics Annex G ignition of flammable anaesthetic mixtures
10	Devices with a measuring function	
10.1	Devices with a measuring function must be designed and manufactured in such	Not covered

No.	Essential Requirement	Coverage
	a way as to provide sufficient accuracy and stability within appropriate limits of accuracy and taking account of the intended purpose of the device.	See particular standards EN 60601-2-xx See 12.1 in respect of risks associated with accuracy of controls and instruments
	The limits of accuracy must be indicated by the manufacturer.	Covered by 7.9.3.1 technical description
10.2	The measurement, monitoring and display scale must be designed in line with ergonomic principles, taking account of the intended purpose of the device.	Not covered see EN IEC 60601-1-6 and EN IEC 62366
10.3	The measurements made by devices with a measuring function must be expressed in legal units conforming to the provisions of Council Directive 80/181/EEC.	Covered in respect of the following: 7.4.3 Units of measurement cmH ₂ O is not included in 80/181/EEC
11	Protection against radiation	General Guidance note 2 and 3 shall be observed
11.1	General	
11.1.1	Devices shall be designed and manufactured in such a way that exposure of patients, users and other persons to radiation shall be reduced as far as possible compatible with the intended purpose, whilst not restricting the application of appropriate specified levels for therapeutic and diagnostic purposes.	For unintended radiation, covered in respect to the following: 10.1.1 (ionizing radiation), 10.3 (microwave), 10.4 (lasers). For intended radiation, covered in respect to the following: 10.3 (microwave), 10.4 (lasers). Other types of radiation of these devices and other devices not covered. For devices intended to produce radiation see EN 60601-1-3 for diagnostic x-radiation. For other radiation see particular standards EN 60601-2-xx.
11.2	Intended radiation	
11.2.1	Where devices are designed to emit hazardous levels of radiation necessary for a specific medical purpose the benefit of which is considered to outweigh the risks inherent in the emission, it must be possible for the user to control the emissions. Such devices shall be designed and manufactured to ensure reproducibility and tolerance of relevant variable parameters.	1 st and 2 nd sentence covered in respect of the following: 10.3, Microwave 10.4 Lasers First sentence covered by subclauses 15.4.6, Actuating parts of controls and 15.4.7 hand or foot switches See particular standards EN 60601-2-xx See EN 60601-1-3 for diagnostic x-radiation
11.2.2	Where devices are intended to emit	Not covered.

No.	Essential Requirement	Coverage
	potentially hazardous, visible and/or invisible radiation, they must be fitted, where practicable, with visual displays and/or audible warnings of such emissions.	
11.3	Unintended radiation	
11.3.1	Devices shall be designed and manufactured in such a way that exposure of patients, users and other persons to the emission of unintended, stray or scattered radiation is reduced as far as possible.	Covered in respect to the following: 10.1.1 (ionizing radiation), 10.3 (microwave), 10.4 (lasers). Other types of radiation of these devices and other devices not covered.
11.4	Instructions	
11.4.1	The operating instructions for devices emitting radiation must give detailed information as to the nature of the emitted radiation, means of protecting the patient and the user and on ways of avoiding misuse and of eliminating the risks inherent in installation.	Covered in respect of information relating to the nature of the emitted radiation: 7.9.2.17 – ME equipment emitting radiation
11.5	Ionizing radiation	
11.5.1	Devices intended to emit ionizing radiation must be designed and manufactured in such a way as to ensure that, where practicable, the quantity, geometry and quality of radiation emitted can be varied and controlled taking into account the intended use.	Not covered For diagnostic x-radiation see EN 60601-1-3. For other devices see particular standards EN 60601-2-xx
11.5.2	Devices emitting ionizing radiation intended for diagnostic radiology shall be designed and manufactured in such a way as to achieve appropriate image and/or output quality for the intended medical purpose whilst minimizing radiation exposure of the patient and user.	Not covered For diagnostic x-radiation see EN 60601-1-3. For other devices see particular standards EN 60601-2-xx
11.5.3	Devices emitting ionizing radiation, intended for therapeutic radiology shall be designed and manufactured in such a way as to enable reliable monitoring and control of the delivered dose, the beam type and energy and where appropriate the quality of radiation.	Not covered
12	Requirements for medical devices connected to or equipped with an energy source	General Guidance note 2 and 3 shall be observed
12.1	Devices incorporating electronic programmable systems must be designed to ensure the repeatability, reliability and performance of these systems according to the intended use.	Covered by 14 Programmable electrical medical systems (PEMS)

No.	Essential Requirement	Coverage
	In the event of a single fault condition (in the system) appropriate means should be adopted to eliminate or reduce as far as possible consequent risks.	
12.1a	For devices which incorporate software or which are medical software in themselves, the software must be validated according to the state of the art taking into account the principles of development lifecycle, risk management, validation and verification.	Covered in respect of devices which incorporate SW by 14 Programmable electrical medical systems (PEMS)
12.2	Devices where the safety of the patients depends on an internal power supply must be equipped with a means of determining the state of the power supply.	Not covered
12.3	Devices where the safety of the patients depends on an external power supply must include an alarm system to signal any power failure.	Not covered
12.4	Devices intended to monitor one or more clinical parameters of a patient must be equipped with appropriate alarm systems to alert the user of situations which could lead to death or severe deterioration of the patient's state of health.	Not covered
12.5	Devices must be designed and manufactured in such a way as to minimize the risks of creating electromagnetic fields which could impair the operation of other devices or equipment in the usual environment.	Not covered EMC: see EN 60601-1-2 as referenced in annex ZA
12.6	Protection against electrical risks	
12.6.1	Devices must be designed and manufactured in such a way as to avoid, as far as possible, the risk of accidental electric shocks during normal use and in single fault condition, provided the devices are installed correctly.	Covered in respect of the following: 6.2 Protection against electric shock 7.2.10 Applied parts 7.9 Accompanying documents 8 Protection against electrical hazard 13.1 Specific hazardous situation 13.2 Single fault conditions 16.6 Leakage currents
12.7	Protection against mechanical and thermal risks	
12.7.1	Devices must be designed and manufactured in such a way as to	Covered in respect of the following: 9.1 Mechanical Hazard

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	protect the patient and user against mechanical risks connected with, for example, resistance, stability and moving parts.	15.3 Mechanical strength
12.7.2	Devices must be designed and manufactured in such a way as to reduce to the lowest possible level the risks arising from vibration generated by the devices, taking account of technical progress and of the means available for limiting vibrations, particularly at source, unless the vibrations are part of the specified performance.	Covered in respect of the following: 9.6 Acoustic energy and vibration 9.8.1 Support systems
12.7.3	Devices must be designed and manufactured in such a way as to reduce to the lowest possible level the risks arising from the noise emitted, taking account of technical progress and of the means available to reduce noise, particularly at source, unless the noise emitted is part of the specified performance.	Covered in respect of 9.6 Acoustic energy and vibration
12.7.4	Terminals and connectors to the electricity, gas or hydraulic and pneumatic energy supplies which the user has to handle must be designed and constructed in such a way as to minimize all possible risks.	Covered in respect of the following: Electrical Risks: 8.1 Fundamental rule of protection against electric shock 8.2 Connection to power sources 8.4 Limitation of voltage current or energy 8.5 Separation of parts 8.6 Functional earthing 8.7 Leakage current 8.11.3 Power supply cords Gas or Hydraulic and Pneumatic: 9.7 Pressure vessels and parts
12.7.5	Accessible parts of the devices (excluding the parts or areas intended to supply heat or reach given temperatures) and their surroundings must not attain potentially dangerous temperatures under normal use.	Covered by 11.1 Excessive temperatures
12.8	Protection against the risks posed to the patient by energy supplies or substances	
12.8.1	Devices for supplying the patient with energy or substances must be designed and constructed in such a way that the flow-rate can be set and maintained accurately enough to guarantee the safety of the patient and of the user.	Covered in respect of the following: 15.4.2 Temperature and overload control devices 15.4.4 Indicators for standby and output 15.4.6 Actuating parts of controls 15.4.7 Cord-connected hand-held and foot-operated control devices

No.	Essential Requirement	Coverage
12.8.2	Devices must be fitted with the means of preventing and/or indicating any inadequacies in the flow-rate which could pose a danger.	Covered in respect of the following: 15.4.1 Construction of connectors 15.4.2 Temperature and overload control devices 15.4.4 indicators for standby and output 15.4.5 Pre-set controls 15.4.6 Actuating parts of controls 15.4.7 Cord-connected hand-held and foot-operated control devices
	Devices must incorporate suitable means to prevent, as far as possible, the accidental release of dangerous levels of energy from an energy and/or substance source.	Covered in respect of the following: Energy Source: 12.4 Protection against hazardous output 14 In respect of programmable electrical medical systems (PEMS) 15.4.1 Construction of connectors 15.4.2 Temperature and overload control devices 15.4.4 Indicators for standby and output 15.4.5 Pre-set controls 15.4.6 Actuating parts of controls 15.4.7 Cord-connected hand-held and foot-operated control devices Substance Source: 9.7 Pressure vessels and parts subject to pneumatic and hydraulic pressure 12.4 Protection against hazardous output 14 In respect of programmable electrical medical systems (PEMS) 15.4.4 Indicators for standby and output 15.4.5 Pre-set controls 15.4.6 Actuating parts of controls
12.9	The function of the controls and indicators must be clearly specified on the devices	Covered in respect of the following: 7.4 Marking of controls and instruments
	Where a device bears instructions required for its operation or indicates operating or adjustment parameters by means of a visual system, such information must be understandable to the user and, as appropriate, the patient.	Covered in respect of the following: 7.5 Safety signs 7.9.1 General requirements for accompanying documents
13	Information supplied by the manufacturer	
13.1	Each device must be accompanied by the information needed to use it safely and properly, taking account of the training and knowledge of the potential	Covered in respect of the following: 7.2.2 Identification 7.2.4 Accessories

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	users, and to identify the manufacturer. This information comprises the details on the label and the data in the instructions for use.	7.2.5 Power from other equipment 7.9 Accompanying documents
	As far as practicable and appropriate, the information needed to use the device safely must be set out on the device itself and/or on the packaging for each unit or, where appropriate, on the sales packaging. If individual packaging of each unit is not practicable, the information must be set out in the leaflet supplied with one or more devices.	Covered in respect of the following: 7.2.3 Consult accompanying documents 7.9 Accompanying documents
	Instructions for use must be included in the packaging for every device. By way of exception, no such instructions for use are needed for devices in Class I or IIa if they can be used safely without any such instructions.	Covered in respect of the following: 7.9.1 Accompanying documents, general 7.9.2 Instructions for use
13.2	Where appropriate, this information should take the form of symbols. Any symbol or identification color used must conform to the harmonized standards. In areas for which no standards exist, the symbols and colors must be described in the documentation supplied with the device.	Covered in respect of the following: 7.6 Symbols Annex D Symbols on marking – informative annex for information only
13.3	The label must bear the following particulars: (a) the name or trade name and address of the manufacturer. For devices imported into the Community, in view of their distribution in the Community, the label, or the outer packaging, or instructions for use, shall contain in addition the name and address of the authorized representative where the manufacturer does not have a registered place of business in the Community; (b) the details strictly necessary to identify the device and the contents of the packaging especially for the users; (c) where appropriate, the word 'STERILE'; (d) where appropriate, the batch code, preceded by the word 'LOT', or the serial number; (e) where appropriate, an indication of the date by which the device should be used, in safety,	Covered in respect of the following: a) 7.2.2 Identification (partially covered: in order to comply with this ER, name and address must be used). Std. does not address the specifics of imported devices (authorized representative). b) 7.2.2 Identification (limited to details related to the identification of the device) c) 7.2.17 Protective packaging d) 7.2.2 Identification, 7.2.4 Accessories (the std. does not require to use the word LOT which has to be added) e) 7.2.2 Identification (std. does not specify the format, however, the note directs to a standard that specifies the format) f) 7.2.1 Marking (std. allows three options, manufacturer needs to limit himself on just one) g) Not covered h) Not covered i) 7.2.17 Protective packaging j) 7.2 Marking on the outside of equipment and parts 7.3 Marking on the inside of equipment and parts 7.5 Safety signs k) Covered

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	expressed as the year and month; (f) where appropriate, an indication that the device is for single use. A manufacturer's indication of single use must be consistent across the Community; (g) if the device is custom-made, the words 'custom-made device'; (h) if the device is intended for clinical investigations, the words 'exclusively for clinical investigations'; (i) any special storage and/or handling conditions; (j) any special operating instructions; (k) any warnings and/or precautions to take; (l) year of manufacture for active devices other than those covered by (e). This indication may be included in the batch or serial number; (m) where applicable, method of sterilization;	7.2.2 Identification 7.2.20 Removable protective means 7.3 Marking on the inside of equipment and parts l) 7.2.2 Identification m) 7.2.17 Protective packaging
	(n) in the case of a device within the meaning of Article 1(4a), an indication that the device contains a human blood derivative.	n) Not covered
13.4	If the intended purpose of the device is not obvious to the user, the manufacturer must clearly state it on the label and in the instructions for use.	Not covered
13.5	Wherever reasonable and practicable, the devices and detachable components must be identified, where appropriate in terms of batches, to allow all appropriate action to detect any potential risk posed by the devices and detachable components.	Covered 7.2.2 Identification
13.6	Where appropriate, the instructions for use must contain the following particulars: (a) the details referred to in Section 13.3, with the exception of (d) and (e); (b) the performances referred to in Section 3 and any undesirable side-effects; (c) if the device must be installed with or connected to other medical devices or equipment in order to operate as required for its intended purpose, sufficient details of its characteristics to identify the	a) Details referred to in Section 13.3, with the exception of (d) and (e);: 13.3 a) Instructions for Use: authorized representative: not covered Instructions for Use: 7.9.2 Instructions for use 13.3 b) Instructions for Use: 7.9.1 General on accompanying documents (for electronic Instructions for Use adhere to EU legislation 2007/12) 13.3 c) Instructions for Use: 7.9.2.18 Equipment and accessories supplied sterile (partly covered, the word "sterile" is not required by the standard) 13.3 d) Exempted for Instructions for Use.

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	correct devices or equipment to use in order to obtain a safe combination; (d) all the information needed to verify whether the device is properly installed and can operate correctly and safely, plus details of the nature and frequency of the maintenance and calibration needed to ensure that the devices operate properly and safely at all times; (e) where appropriate, information to avoid certain risks in connection with implantation of the device; (f) information regarding the risks of reciprocal interference posed by the presence of the device during specific investigations or treatment; (g) the necessary instructions in the event of damage to the sterile packaging and, where appropriate, details of appropriate methods of re-sterilization;	13.3 e) Exempted for Instructions for Use 13.3 f) Instructions for Use: not covered 13.3 g) Instructions for Use: not covered 13.3 h) Instructions for Use: not covered 13.3 i) Instructions for Use: Covered in respect of the following: 7.9.2.2 Warning and safety notices 7.9.2.18 Equipment and accessories supplied sterile 7.9.3.1 General on Technical description 9.4.4.a Grips and other handling devices Remark: handling is assumed to include installation, but to be different from operating use 13.3 k) Instructions for Use: Covered in respect of the following: 7.9.2.2 Warning and safety notices, first sentence 13.3 l) Instructions for Use: not covered 13.3 m) Instructions for Use: not covered 13.3 n) Instructions for Use: not covered ----- b) Performances referred to in Section 3 Not covered c) If the device must be installed with or connected to other medical devices Covered in respect of the following: 7.9.1, General on accompanying documents 7.9.2.1 General on instructions for use 7.9.2.14 Accessories, supplementary equipment, used material 7.9.3, Technical description 14 Programmable electrical medical systems (PEMS) d) Covered in respect of the following: 7.9.2.9 Operating instructions 7.9.2.13 Maintenance e) Not covered f) Not covered g) Covered in respect of the following: 7.2.17 Protective packaging 7.9.2.18 ME equipment and accessories supplied sterile
	(h) if the device is reusable, information on the appropriate processes to allow reuse, including cleaning, disinfection, packaging and, where appropriate, the method of sterilization of the device to be re-sterilized, and any	h) Covered in respect of 7.9.2.12 Cleaning, disinfection and sterilization i) Covered in respect of 7.9 Accompanying documents

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	restriction on the number of reuses. Where devices are supplied with the intention that they be sterilized before use, the instructions for cleaning and sterilization must be such that, if correctly followed, the device will still comply with the requirements in Section I; If the device bears an indication that the device is for single use, information on known characteristics and technical factors known to the manufacturer that could pose a risk if the device were to be re-used. If in accordance with Section 13.1 no instructions for use are needed, the information must be made available to the user upon request; (i) details of any further treatment or handling needed before the device can be used (for example sterilization, final assembly, etc.); (j) in the case of devices emitting radiation for medical purposes, details of the nature, type, intensity and distribution of this radiation. The instructions for use must also include details allowing the medical staff to brief the patient on any contra-indications and any precautions to be taken. These details should cover in particular: (k) precautions to be taken in the event of changes in the performance of the device; (l) precautions to be taken as regards exposure, in reasonably foreseeable environmental conditions, to magnetic fields, external electrical influences, electrostatic discharge, pressure or variations in pressure, acceleration, thermal ignition sources, etc.; (m) adequate information regarding the medicinal product or products which the device in question is designed to administer, including any limitations in the choice of substances to be delivered; (n) precautions to be taken against any special, unusual risks related	j) Covered in respect of 7.9.2.17 ME equipment emitting radiation k) Not covered l) Not covered m) Not covered n) Not covered o) Not covered p) Not covered q) Not covered

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	to the disposal of the device; (o) medicinal substances, or human blood derivatives incorporated into the device as an integral part in accordance with Section 7.4; (p) degree of accuracy claimed for devices with a measuring function; (q) date of issue or the latest revision of the instructions for use.	

FOREWORD

This amendment has been prepared by subcommittee 62A: Common aspects of electrical equipment used in medical practice, of IEC technical committee 62: Electrical equipment in medical practice.

The text of this amendment is based on the following documents:

FDIS	Report on voting
62A/805/FDIS	62A/820/RVD

Full information on the voting for the approval of this amendment can be found in the report on voting indicated in the above table.

The committee has decided that the contents of this amendment and the base publication will remain unchanged until the stability date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

IMPORTANT – The 'colour inside' logo on the cover page of this publication indicates that it contains colours which are considered to be useful for the correct understanding of its contents. Users should therefore print this document using a colour printer.

INTRODUCTION TO THE AMENDMENT

The third edition of IEC 60601-1 was published in 2005. At the time of publication, there were 94 National Committee comments on the 2nd CDV and the FDIS that were deferred to a future amendment/revision. Each of their deferred comments was captured in an Issue Sheet by the SC 62A secretariat. By the time of the Auckland meeting in April 2008, the Subcommittees had developed two Interpretation Sheets and the SC 62A secretariat has received an additional 15 issues from National Committees and other interested parties.

At the Auckland meeting, IEC/TC 62 approved a project to develop the 1st amendment to IEC 60601-1:2005 based on the issues outstanding at the time. The TC approved developing the 1st amendment with a view to addressing outstanding issues, including but not limited to:

- those listed in 62A/593/DC and 62A/602/INF;
- the way in which risk management has been introduced into IEC 60601-1:2005; and
- the way the concept of essential performance is used in IEC 60601-1:2005.

Since the Auckland meeting, the secretariat has received 73 additional issues from National Committees or other interested parties for a total of 182 Issues Sheets. This amendment is intended to address those issues.

FOREWORD

Replace the paragraph beginning "This third edition cancels..." with the following:

This third edition cancels and replaces the second edition published in 1988, its Amendment 1 (1991) and Amendment 2 (1995), the second edition of IEC 60601-1-1 published in 2000 and the first edition of IEC 60601-1-4 published in 1996 and its Amendment 1 (1999). This edition constitutes a technical revision. This edition has been significantly restructured. Requirements in the electrical section have been further aligned with those for information technology equipment covered by IEC 60950-1 and a requirement for including a RISK MANAGEMENT PROCESS has been added. For an expanded description of this revision, see Annex A.3.

Add the following note at the end of the Foreword:

NOTE The attention of National Committees is drawn to the fact that equipment manufacturers and testing organizations may need a transitional period following publication of a new, amended or revised IEC or ISO publication in which to make products in accordance with the new requirements and to equip themselves for conducting new or revised tests. It is the recommendation of the committee that the content of this publication be adopted for mandatory implementation nationally not earlier than 3 years from the date of publication.

INTRODUCTION

In the second dash of the existing ninth paragraph, replace the final sentence with:

Compliance with this edition of IEC 60601-1 requires that the MANUFACTURER have in place a RISK MANAGEMENT PROCESS complying with parts of ISO 14971 (see 4.2).

After the last paragraph of the introduction, insert the following new paragraph:

Amendment 1 to this standard is intended to address:

- issues identified by National Committees and other interested parties since the publication of IEC 60601-1:2005;
- the way in which RISK MANAGEMENT has been introduced into IEC 60601-1:2005; and
- the way the concept of ESSENTIAL PERFORMANCE is used in IEC 60601-1:2005.

1 Scope, object and related standards

1.1 * Scope

Renumber the note as Note 1.

Delete the fourth paragraph.

Replace the fifth paragraph with:

The IEC 60601 series does not apply to:

- in vitro diagnostic equipment that does not fall within the definition of ME EQUIPMENT, which is covered by the IEC 61010 series [61];
- implantable parts of active implantable medical devices covered by the ISO 14708 series [69]; or
- medical gas pipeline systems covered by ISO 7396-1 [68].

NOTE 2 ISO 7396-1 applies the requirement of IEC 60601-1-8 to certain monitoring and ALARM SIGNALS.

1.3 * Collateral standards

Replace Note 3 with:

NOTE 3 Collateral standards in the IEC 60601 family are numbered IEC 60601-1-xx. The IEC maintains a catalogue of valid International Standards. Users of this standard should consult this catalogue at "<http://webstore.iec.ch>" to determine which collateral standards have been published.

1.4 * Particular standards

Replace the note with:

NOTE Particular standards in the IEC 60601 family that are developed by IEC committees are numbered IEC 60601-2-xx. In addition, particular standards developed by joint projects between ISO and IEC can be numbered either IEC 80601-2-xx or ISO 80601-2-xx depending on which committee administered the project. IEC and ISO maintain catalogues of valid International Standards. Users of this standard should consult these catalogues at "<http://webstore.iec.ch>" and "<http://www.iso.org/iso/store.htm>" to determine which particular standards have been published.

2 * Normative references

Update the following normative references:

IEC 60065:2001, *Audio, video and similar electronic apparatus – Safety requirements* ¹⁾
Amendment 1:2005
Amendment 2:2010

IEC 60068-2-2:2007, *Environmental testing – Part 2-2: Tests – Test B: Dry heat*

IEC 60227-1:2007, *Polyvinyl chloride insulated cables of rated voltages up to and including 450/750 V – Part 1: General requirements*

IEC 60245-1:2003, *Rubber insulated cables – Rated voltages up to and including 450/750 V – Part 1: General requirements* ²⁾
Amendment 1:2007

IEC 60335-1:2010, *Household and similar electrical appliances – Safety – Part 1: General requirements*

IEC 60417, *Graphical symbols for use on equipment*. Available from: <<http://www.graphical-symbols.info/equipment>>

IEC 60601-1-3, *Medical electrical equipment – Part 1-3: General requirements for basic safety and essential performance: Collateral standard: Radiation protection in diagnostic X-ray equipment*

IEC 60664-1:2007, *Insulation coordination for equipment within low-voltage systems – Part 1: Principles, requirements and tests*

IEC 60730-1:2010, *Automatic electrical controls for household and similar use – Part 1: General requirements*

IEC 60825-1:2007, *Safety of laser products – Part 1: Equipment classification and requirements*

1) There exists a consolidated edition 7.2 including IEC 60065:2001 and its Amendment 1 (2005) and Amendment 2 (2010).

2) There exists a consolidated edition 4.1 including IEC 60245-1:2003 and its Amendment 1 (2007).

IEC 60851-3:2009, *Winding wires – Test methods – Part 3: Mechanical properties*

IEC 60851-5:2008, *Winding wires – Test methods – Part 5: Electrical properties*

IEC 61058-1:2000, *Switches for appliances – Part 1: General requirements*³⁾

Amendment 1:2001

Amendment 2:2007

ISO 7010:2011, *Graphical symbols – Safety colours and safety signs –Registered safety signs*

ISO 14971:2007, *Medical devices – Application of risk management to medical devices*

Replace the existing references to ISO 11135, ISO 11137, ISO 13852 and ISO 15223 by the following:

ISO 11135-1:2007, *Sterilization of health care products – Ethylene oxide – Part 1: Requirements for development, validation and routine control of a sterilization process for medical devices*

ISO 11137-1:2006, *Sterilization of health care products – Radiation – Part 1: Requirements for development, validation and routine control of a sterilization process for medical devices*

ISO 13857:2008, *Safety of machinery – Safety distances to prevent hazard zones being reached by the upper and lower limbs*

ISO 15223-1:2012, *Medical devices – Symbols to be used with medical device labels, labelling and information to be supplied – Part 1: General requirements*

Delete the following normative references:

IEC 60878:2003, *Graphical symbols for electrical equipment in medical practice*

IEC 61558-1:1997, *Safety of power transformers, power supply units and similar – Part 1: General requirements and tests*
Amendment 1:1998

ISO 31 (all parts), *Quantities and units*

ISO 1000, *SI units and recommendations for the use of their multiples and of certain other units*

ISO 11134, *Sterilization of health care products – Requirements for validation and routine control – Industrial moist heat sterilization*

Add the following new normative references:

IEC 62133, *Secondary cells and batteries containing alkaline or other non-acid electrolytes – Safety requirements for portable sealed secondary cells, and for batteries made from them, for use in portable applications*

IEC 62304:2006, *Medical device software – Software lifecycle processes*

³⁾ There exists a consolidated edition 3.2, including IEC 61058-1:2000 and its Amendment 1 (2001) and Amendment 2 (2007)

ISO 17665-1:2006, *Sterilization of health care products – Moist heat – Part 1: Requirements for the development, validation and routine control of a sterilization process for medical devices*

ISO 80000-1:2009, *Quantities and units – Part 1: General*

3 * Terminology and definitions

Add a new Note 3 and renumber the existing Note 3 to Note 4:

NOTE 3 When the term "safety" is used in this document in roman or italic type, it does not mean "safety" as defined in ISO 14971, but rather is used to mean "the state of being protected from or guarded against hurt or injury; freedom from danger".

3.5

AIR CLEARANCE

Replace the existing note with the following:

NOTE Adapted from IEC 60664-1:2007, definition 3.2.

Figure 2 – Example of the defined terminals and conductors

In the key, replace "Ⓜ MAINS CONDUCTOR" by "Ⓜ MAINS CONNECTOR".

3.15

CLEARLY LEGIBLE

Add an asterisk before the term and replace the existing note by the following:

NOTE See the test in 7.1.2.

3.25

EARTH LEAKAGE CURRENT

Replace the existing definition with the following:

current flowing from the MAINS PART through or across the insulation into the PROTECTIVE EARTH CONDUCTOR or a functional earthed connection according to 8.6.9

3.27

*** ESSENTIAL PERFORMANCE**

Replace the existing definition with the following:

performance of a clinical function, other than that related to BASIC SAFETY, where loss or degradation beyond the limits specified by the MANUFACTURER results in an unacceptable RISK

3.28

EXPECTED SERVICE LIFE

Replace the existing definition with the following:

time period specified by the MANUFACTURER during which the ME EQUIPMENT or ME SYSTEM is expected to remain safe for use (i.e. maintain BASIC SAFETY and ESSENTIAL PERFORMANCE)

NOTE Maintenance can be necessary during the EXPECTED SERVICE LIFE.

3.30

FIXED

Add the following note after the examples:

NOTE See the taxonomy in the rationale for definition 3.63.

3.37

HAND-HELD

Replace the existing definition with the following:

term referring to equipment that, once installed and placed into service, is intended to be supported by the hand

NOTE 1 Equipment can refer to ACCESSORIES or equipment parts.

NOTE 2 See the taxonomy in the rationale for definition 3.63.

3.38

HARM

Replace the source citation with the following:

[ISO 14971:2007, definition 2.2, modified]

3.39

HAZARD

Replace the existing source citation with the following:

[ISO 14971:2007, definition 2.3]

3.40

HAZARDOUS SITUATION

Replace the existing source citation with the following:

[ISO 14971:2007, definition 2.4]

3.43

INSULATION CO-ORDINATION

Add a note to the existing definition as follows:

NOTE This includes insulation types, CREEPAGE DISTANCES, AIR CLEARANCES, distance through insulation, coatings, encapsulation, environmental aspects, etc.

3.44

* INTENDED USE

INTENDED PURPOSE

Delete the asterisk in front of the term and replace the existing definition and note with the following:

use for which a product, PROCESS or service is intended according to the specifications, instructions and information provided by the MANUFACTURER

[ISO 14971:2007, definition 2.5]

NOTE INTENDED USE should not be confused with NORMAL USE. While both include the concept of use as intended by the MANUFACTURER, INTENDED USE focuses on the medical purpose while NORMAL USE incorporates not only the medical purpose, but maintenance, transport, etc. as well.

3.49

* MAINS PART

Replace the existing definition and the notes with the following:

part of electrical equipment forming a circuit that is intended to be connected to the SUPPLY MAINS

NOTE 1 The MAINS PART includes all conductive parts that are not separated from the SUPPLY MAINS by at least one MEANS OF PROTECTION.

NOTE 2 For the purpose of this definition, the PROTECTIVE EARTH CONDUCTOR is not regarded as a part of the MAINS PART (see Figure 2 and Figure 3).

3.55

MANUFACTURER

Replace existing Note 4 with:

NOTE 4 Adapted from ISO 14971:2007, definition 2.8.

3.65

MOBILE

Replace the existing definition with the following:

term referring to TRANSPORTABLE equipment that, once installed and placed into service, is intended to be moved from one location to another while supported by its own wheels or equivalent means

NOTE See the taxonomy in the rationale for definition 3.63.

3.67

MULTIPLE SOCKET-OUTLET

MSO

In the existing definition, replace "cables or cords or ME EQUIPMENT for" with "cables, cords or ME EQUIPMENT providing".

3.71

NORMAL USE

In the existing note, delete the term "service" in the last line of the sentence.

3.72

OBJECTIVE EVIDENCE

Replace the existing definition with the following:

data supporting the existence or verity of something

NOTE Objective evidence can be obtained through observation, measurement, testing or other means.

[ISO 14971:2007, definition 2.10]

3.76

PATIENT

Add a note to the definition as follows:

NOTE A PATIENT can be an OPERATOR.

3.81

PEAK WORKING VOLTAGE

Replace the existing source citation with the following:

[IEC 60950-1:2005, definition 1.2.9.8, modified]

3.85

PORTABLE

Replace the existing definition with the following:

term referring to TRANSPORTABLE equipment that, once installed and placed into service, is intended to be moved from one location to another while being carried by one or more persons

NOTE 1 Equipment can refer to ACCESSORIES or equipment parts.

NOTE 2 See the taxonomy in the rationale for definition 3.63.

3.88

PROCEDURE

Replace the existing definition with the following:

specified way to carry out an activity or a PROCESS

[ISO 14971:2007, definition 2.12]

3.89

PROCESS

Replace the existing definition with the following:

set of interrelated or interacting activities which transforms inputs into outputs

[ISO 14971:2007, definition 2.13]

3.98

RECORD

Replace the existing definition with the following:

document stating results achieved or providing evidence of activities performed

[ISO 14971:2007, definition 2.14]

3.100

RESIDUAL RISK

Replace the existing definition with the following:

RISK remaining after RISK CONTROL measures have been taken

[ISO 14971:2007, definition 2.15]

3.102

RISK

Replace the existing source citation with the following:

[ISO 14971:2007, definition 2.16]

3.103

RISK ANALYSIS

Replace the existing source citation with the following:

[ISO 14971:2007, definition 2.17]

3.104

RISK ASSESSMENT

Replace the existing source citation with the following:

[ISO 14971:2007, definition 2.18]

3.105**RISK CONTROL**

Replace the existing definition with the following:

PROCESS in which decisions are made and measures implemented by which RISKS are reduced to, or maintained within, specified levels

[ISO 14971:2007, definition 2.19]

3.106**RISK EVALUATION**

Replace the existing definition with the following:

PROCESS of comparing the estimated RISK against given RISK criteria to determine the acceptability of the RISK

[ISO 14971:2007, definition 2.21]

3.107**RISK MANAGEMENT**

Replace the existing definition with the following:

systematic application of management policies, PROCEDURES and practices to the tasks of analyzing, evaluating and controlling RISK

[ISO 14971:2007, definition 2.22]

NOTE For the purposes of this standard, RISK MANAGEMENT does not include planning for or monitoring of production and post-production information; whereas this is required for compliance with ISO 14971 (see 4.2.2).

3.108**RISK MANAGEMENT FILE**

Replace the existing definition with the following:

set of RECORDS and other documents that are produced by RISK MANAGEMENT

[ISO 14971:2007, definition 2.23]

NOTE All safety related information including MANUFACTURER'S calculations, test results, etc. is considered to be part of the RISK MANAGEMENT FILE. See also 4.2.

3.114**SEVERITY**

Replace the existing source citation with the following:

[ISO 14971:2007, definition 2.25]

3.116**SINGLE FAULT CONDITION**

Replace the existing definition with the following:

condition of ME EQUIPMENT in which a single means for reducing a RISK is defective or a single abnormal condition is present

NOTE See 4.7 and 13.2.

3.118**STATIONARY**

Replace the existing definition with the following:

term referring to equipment that, once installed and placed into service, is not intended to be moved from one place to another

NOTE See the taxonomy in the rationale for definition 3.63.

3.130

TRANSPORTABLE

Replace the existing definition with the following:

term referring to equipment that, once installed and placed into service, is intended to be moved from one place to another whether or not connected to a supply and without an appreciable restriction of range

EXAMPLES MOBILE equipment, PORTABLE equipment and BODY-WORN equipment

NOTE See the taxonomy in the rationale for definition 3.63.

3.132

TYPE B APPLIED PART

Delete the reference to "DB" in two places in Note 1.

3.133

TYPE BF APPLIED PART

Delete the reference to "DB" in two places in Note 1.

3.134

TYPE CF APPLIED PART

Delete the reference to "DB" in two places in Note 1 and insert Note 3 as follows:

NOTE 3 See the rationale for the definition of DIRECT CARDIAC APPLICATION (3.22) concerning the applied parts that need to be TYPE CF APPLIED PARTS.

3.136

USABILITY

Replace the existing definition with the following:

characteristic of the OPERATOR interface that establishes effectiveness, efficiency, ease of OPERATOR learning and OPERATOR satisfaction

[IEC 62366:2007, definition 3.17, modified]

3.137

USABILITY ENGINEERING

Replace the existing definition with the following:

application of knowledge about human behaviour, abilities, limitations, and other characteristics to the design of tools, devices, systems, tasks, jobs, and environments to achieve adequate USABILITY

[IEC 62366:2007, definition 3.18]

3.138

VERIFICATION

Replace the existing definition with the following:

confirmation, through the provision of OBJECTIVE EVIDENCE, that specified requirements have been fulfilled

NOTE 1 The term "verified" is used to designate the corresponding status.

NOTE 2 Confirmation can comprise activities such as:

- performing alternative calculations;
- comparing a new design specification with a similar proven design specification;
- undertaking tests and demonstrations;
- reviewing documents prior to issue.

[ISO 14971:2007, definition 2.28]

3.139

WORKING VOLTAGE

Replace the existing source citation with the following:

[IEC 60950-1:2005, definition 1.2.9.6]

Add the following new definitions:

3.140

AIR KERMA

K

quotient of dE_{tr} by dm , where dE_{tr} is the sum of the initial kinetic energies of all the charged particles liberated by uncharged particles in a mass dm of air, thus

$$K = \frac{dE_{tr}}{dm}$$

Unit: J kg⁻¹

The special name for the unit of AIR KERMA is gray (Gy) (ICRU 60)

[IEC 60601-1-3:2008, definition 3.4]

3.141

ALARM CONDITION

state of the ALARM SYSTEM when it has determined that a potential or actual HAZARDOUS SITUATION exists for which OPERATOR awareness or response is required

NOTE 1 An ALARM CONDITION can be invalid, i.e. a false positive ALARM CONDITION.

NOTE 2 An ALARM CONDITION can be missed, i.e. a false negative ALARM CONDITION.

[IEC 60601-1-8:2006 / Amendment 1 (2012), definition 3.1]

3.142

ALARM SIGNAL

type of signal generated by the ALARM SYSTEM to indicate the presence (or occurrence) of an ALARM CONDITION

[IEC 60601-1-8:2006, definition 3.9]

3.143

ALARM SYSTEM

parts of ME EQUIPMENT or a ME SYSTEM that detect ALARM CONDITIONS and, as appropriate, generate ALARM SIGNALS

[IEC 60601-1-8:2006, definition 3.11]

3.144

BODY-WORN

term referring to TRANSPORTABLE equipment whose INTENDED USE includes operation while being worn by a PATIENT or attached to a PATIENT'S clothing

NOTE 1 TRANSPORTABLE equipment can be both BODY-WORN and HAND-HELD.

[IEC 60601-1-11:2010, definition 3.1]

NOTE 2 See the taxonomy in the rationale for definition 3.63.

3.145

IT-NETWORK

INFORMATION TECHNOLOGY NETWORK

a system or systems composed of communicating nodes and transmission links to provide physically linked or wireless transmission between two or more specified communication nodes

[IEC 80001-1:2010, definition 2.12]

3.146

PRIMARY OPERATING FUNCTION

function that involves OPERATOR interaction that is frequently used or related to the safety of the ME EQUIPMENT

[IEC 62366:2007, definition 3.14, modified]

3.147

USABILITY ENGINEERING FILE

set of RECORDS and other documents that are produced by the USABILITY ENGINEERING PROCESS

[IEC 62366:2007, definition 3.19]

4 General requirements

4.2 * RISK MANAGEMENT PROCESS for ME EQUIPMENT or ME SYSTEMS

Replace the existing text of the subclause with the following:

4.2.1 Introduction to RISK MANAGEMENT

Subclause 4.2 specifies the RISK MANAGEMENT PROCESS that is required for compliance with this standard. This RISK MANAGEMENT PROCESS is intended to serve the following purposes:

- a) To identify whether the normative requirements specified in Clauses 5 to 17 of this standard, together with the requirements of applicable collateral and particular standards, address all the HAZARDS associated with the particular ME EQUIPMENT or ME SYSTEM under consideration.
- b) To identify the way in which some particular tests specified in this standard should be applied to a particular ME EQUIPMENT or ME SYSTEM.
- c) To identify whether particular HAZARDS or HAZARDOUS SITUATIONS for which this standard does not provide specific acceptance criteria result in any RISKS for a particular ME EQUIPMENT or ME SYSTEM, and, if so, to establish acceptable RISK levels and evaluate the RESIDUAL RISKS.
- d) To evaluate the acceptability of alternative RISK CONTROL strategies by comparing RESIDUAL RISK with that achieved by applying the full requirements of this standard.

Although the RISK MANAGEMENT PROCESS specified in this standard is required to comply with the relevant requirements of ISO 14971, it is not as extensive as and does not include all of the elements required for compliance with ISO 14971. For example, the RISK MANAGEMENT PROCESS required for compliance with this standard does not include the production and post-production monitoring required in ISO 14971. Furthermore, verification of compliance with the RISK MANAGEMENT requirements of this standard can be accomplished by examination of the

RECORDS and other documentation required by this standard and assessment of the processes cited in this standard and does not require auditing of the RISK MANAGEMENT PROCESS.

4.2.2 General requirement for RISK MANAGEMENT

A RISK MANAGEMENT PROCESS complying with ISO 14971 shall be performed. For compliance with this standard, all elements of the ISO 14971:2007 RISK MANAGEMENT PROCESS shall be applied except:

- the planning for and execution of production and post-production monitoring (subclause 3.1, fourth dash, subclause 3.4, item f), and Clause 9 of ISO 14971:2007), and
- periodic reviews of the suitability of the RISK MANAGEMENT PROCESS (subclause 3.2, fourth dash, of ISO 14971:2007).

When applying any of the requirements of ISO 14971:

- the term “medical device” shall assume the same meaning as ME EQUIPMENT or ME SYSTEM; and
- the term “fault conditions” referred to in ISO 14971 shall include, but shall not be limited to, SINGLE FAULT CONDITIONS identified in this standard.

NOTE 1 This standard specifies requirements that are generally applicable to RISKS associated with the ME EQUIPMENT or ME SYSTEMS. Those requirements are intended to facilitate the RISK MANAGEMENT PROCESS. The RISK MANAGEMENT PROCESS is aimed at identifying not only those HAZARDS addressed by this standard, but all HAZARDS, their associated RISKS and RISK CONTROL measures.

NOTE 2 Conditions or faults that can give rise to HAZARDOUS SITUATIONS are identified in the clauses of this standard. In these cases, it will often be necessary to carry out a RISK MANAGEMENT PROCESS to determine what the actual HAZARDOUS SITUATIONS are and the tests that need to be done to show that the identified HAZARDOUS SITUATIONS do not arise in the specified circumstances.

NOTE 3 It is recognized that the MANUFACTURER might not be able to follow all the PROCESSES identified in this standard for each constituent component of the ME EQUIPMENT or ME SYSTEM, such as proprietary components, subsystems of non-medical origin, and legacy devices. In this case, the MANUFACTURER would need to take special account of the need for additional RISK CONTROL measures.

NOTE 4 Where requirements of this standard refer to freedom from unacceptable RISK, acceptability or unacceptability of this RISK is determined by the MANUFACTURER in accordance with the MANUFACTURER'S policy for determining acceptable RISK.

NOTE 5 Not all the RISKS associated with the ME EQUIPMENT or ME SYSTEM are subject to specific requirements of this standard (see 1.1).

Compliance is checked by:

- *inspection of the MANUFACTURER'S policy for determining criteria for RISK acceptability;*
- *inspection of the RISK MANAGEMENT plan for the particular ME EQUIPMENT or ME SYSTEM under consideration; and*
- *confirming the MANUFACTURER has prepared a RISK MANAGEMENT FILE containing the RISK MANAGEMENT RECORDS and other documentation required by this standard for the particular ME EQUIPMENT or ME SYSTEM under consideration.*

NOTE 6 It is helpful to keep an index containing references or pointers to the RISK MANAGEMENT RECORDS and other documentation required by this standard.

4.2.3 Evaluating RISK

4.2.3.1 HAZARDS identified in the IEC 60601-series

The requirements of this standard shall be applied in the following way when evaluating RISK:

- a) Where this standard or its collateral or particular standards specify requirements addressing particular HAZARDS or HAZARDOUS SITUATIONS, together with specific acceptance criteria, compliance with these requirements is presumed to establish that the RESIDUAL

RISKS have been reduced to acceptable levels unless there is OBJECTIVE EVIDENCE to the contrary.

EXAMPLE 1 Subclause 8.5.1.2, MEANS OF PATIENT PROTECTION (MOPP)

EXAMPLE 2 Subclause 9.4.2.1, Instability in transport position

Compliance is checked by satisfying the relevant requirements of this standard and its collateral and particular standards.

- b) Where this standard or its collateral or particular standards specify requirements addressing particular HAZARDS or HAZARDOUS SITUATIONS but do not provide specific acceptance criteria, the MANUFACTURER shall provide the acceptance criteria defined in the RISK MANAGEMENT plan. These acceptance criteria shall ensure that the RESIDUAL RISK is acceptable according to the criteria for RISK acceptability recorded in the RISK MANAGEMENT plan.

EXAMPLE 3 Subclause 9.8.3.3, Dynamic forces due to loading from persons

EXAMPLE 4 Subclause 11.6.3, Spillage on ME EQUIPMENT and ME SYSTEMS

Compliance is checked by confirming that the RECORDS in the RISK MANAGEMENT FILE demonstrate that, after applying the specific requirements of this standard, the acceptance criteria determined by the MANUFACTURER are satisfied. Only the relevant parts of the RISK MANAGEMENT FILE need to be reviewed, e.g. the MANUFACTURER'S calculations or test results, or the determination of RISK acceptability.

- c) Where this standard or its collateral or particular standards identify particular HAZARDS or HAZARDOUS SITUATIONS that have to be investigated without providing specific technical requirements:
- the MANUFACTURER shall determine whether such HAZARDS or HAZARDOUS SITUATIONS exist for the particular ME EQUIPMENT or ME SYSTEM, and
 - where such HAZARDS or HAZARDOUS SITUATIONS exist for the particular ME EQUIPMENT or ME SYSTEM, the MANUFACTURER shall evaluate and (if necessary) control these RISKS following the RISK MANAGEMENT PROCESS specified in 4.2.2.

EXAMPLE 5 Subclause 10.2, Alpha, beta, gamma, neutron and other particle radiation

Compliance is checked by confirming that the RECORDS in the RISK MANAGEMENT FILE demonstrate that the RESIDUAL RISK is acceptable using the criteria for RISK acceptability recorded in the RISK MANAGEMENT plan, i.e. no unacceptable RISK remains. Only the relevant parts of the RISK MANAGEMENT FILE need to be reviewed, e.g. the MANUFACTURER'S calculations or test results, or the determination of RISK acceptability.

NOTE When ME EQUIPMENT or an ME SYSTEM has been designed in such way that for a certain type of HAZARDS no HAZARDOUS SITUATION exists, no further RISK ASSESSMENT for that HAZARD is necessary. This can be verified by tests or inspections.

4.2.3.2 HAZARDS not identified in the IEC 60601 series

For HAZARDS or HAZARDOUS SITUATIONS that are identified for the particular ME EQUIPMENT or ME SYSTEM but are not specifically addressed in this standard or its collateral or particular standards, the MANUFACTURER shall address those HAZARDS in the RISK MANAGEMENT PROCESS as specified in 4.2.2.

EXAMPLE ME EQUIPMENT or an ME SYSTEM for which there are particular RISKS but no particular standard.

Compliance is checked by inspection of the RISK MANAGEMENT FILE.

4.3 * ESSENTIAL PERFORMANCE

Replace the existing text of the subclause with the following:

During RISK ANALYSIS, the MANUFACTURER shall identify the performance of the clinical function(s) of the ME EQUIPMENT or ME SYSTEM, other than that related to BASIC SAFETY, that is necessary to achieve its INTENDED USE or that could affect the safety of the ME EQUIPMENT or ME SYSTEM.

The MANUFACTURER shall then specify performance limits between fully functional and total loss of the identified performance in both NORMAL CONDITION and SINGLE FAULT CONDITION.

The MANUFACTURER shall then evaluate the RISK from the loss or degradation of the identified performance beyond the limits specified by the MANUFACTURER. If the resulting RISK is unacceptable, then the identified performance constitutes an ESSENTIAL PERFORMANCE of the ME EQUIPMENT or ME SYSTEM.

The MANUFACTURER shall implement RISK CONTROL measures to reduce the RISK from the loss or degradation of the identified performance to an acceptable level.

NOTE 1 The performance of the RISK CONTROL measure might well become an aspect of the ESSENTIAL PERFORMANCE of the ME EQUIPMENT or ME SYSTEM. For example, the generation of the ALARM SIGNAL to indicate the interruption of the SUPPLY MAINS could be "essential" if an interruption of the SUPPLY MAINS could result in an unacceptable RISK if it went unattended.

The MANUFACTURER shall specify the methods used to verify the effectiveness of the RISK CONTROL measures. This shall include any assessment made to determine whether verification is needed.

NOTE 2 Following the principles of RISK MANAGEMENT, the MANUFACTURER is required to verify the effectiveness of each RISK CONTROL measure. This can involve demonstrating that the RISK CONTROL measure will operate in the presence of the conditions that result in the loss or degradation of the identified performance.

EXAMPLE 1 If ESSENTIAL PERFORMANCE is lost because of an interruption of the SUPPLY MAINS, an ALARM SYSTEM intended to notify the OPERATOR that the SUPPLY MAINS has been interrupted would need to have a backup power source so that the generation of the ALARM SIGNAL would not depend on the SUPPLY MAINS.

EXAMPLE 2 If the failure of a component results in the loss of ESSENTIAL PERFORMANCE, the design of the ME EQUIPMENT or ME SYSTEM would need to be such that the component failure does not compromise the effectiveness of any RISK CONTROL measures put in place to mitigate the RISK from the loss of the ESSENTIAL PERFORMANCE.

NOTE 3 Each collateral standard or particular standard in the IEC 60601 or the IEC/ISO 80601 series can list potential ESSENTIAL PERFORMANCES to guide the MANUFACTURER to identify particular ESSENTIAL PERFORMANCE in accordance with 4.3.

Compliance is checked by inspection of the RISK MANAGEMENT FILE.

Where this standard requires that ESSENTIAL PERFORMANCE is to be maintained following a particular test, compliance is checked by inspection and, if necessary, by functional test(s) that demonstrate the MANUFACTURER'S specified limits are maintained or the ME EQUIPMENT or ME SYSTEM transitions to a safe state as defined by the MANUFACTURER.

4.5 * Equivalent safety for ME EQUIPMENT or ME SYSTEMS

Replace the heading and the text of the subclause with the following:

4.5 * Alternative RISK CONTROL measures or test methods for ME EQUIPMENT or ME SYSTEMS

Where this standard specifies a particular RISK CONTROL measure or test method, an alternative RISK CONTROL measure or test method is acceptable, provided that the MANUFACTURER can demonstrate through scientific data or clinical opinion or comparative studies that the RESIDUAL RISK that results from applying the alternative RISK CONTROL measure or test method remains acceptable and is comparable to the RESIDUAL RISK that results from applying the requirements of this standard.

Comparative studies in this context mean studies comparing the effect of the alternative RISK CONTROL measure or test method with the RISK CONTROL measure or test method specified in this standard.

NOTE Alternative RISK CONTROL measures can allow for exceeding limits specified in this standard or in its collateral or particular standards if additional measures for compensation are provided.

Compliance is checked by inspection of the RISK MANAGEMENT FILE.

4.6 * ME EQUIPMENT or ME SYSTEM parts that contact the PATIENT

Replace the text of the subclause with the following:

The RISK MANAGEMENT PROCESS shall include an assessment of whether parts that can come into contact with the PATIENT but fall outside of the definition of APPLIED PARTS need to be subject to the requirements for APPLIED PARTS. For the parts concerned, the requirements for TYPE B APPLIED PART shall be applied unless the assessment identifies a need for the requirements for a TYPE BF APPLIED PART or TYPE CF APPLIED PART to apply.

If the RISK MANAGEMENT PROCESS determines that such parts are subject to the requirements for APPLIED PARTS, then all the relevant requirements and tests of this standard and of relevant collateral and particular standards shall apply, except that 7.2.10 does not apply to such parts.

Compliance is checked by inspection of the RISK MANAGEMENT FILE.

4.8 Components of ME EQUIPMENT

Add an asterisk to the title.

4.11 Power input

After the third dashed item, insert the following note:

NOTE 1 Technical details of a true r.m.s. meter can be found in IEC 62354 [66].

Replace the existing final paragraph with the following note:

NOTE 2 Supplier information can be used to supplement the above measurement as a power input specification.

5 * General requirements for testing ME EQUIPMENT

5.1 * Type tests

Replace the existing third paragraph and the note with the following:

The combination of simultaneous independent faults that could result in a HAZARDOUS SITUATION shall be documented in the RISK MANAGEMENT FILE (see also 4.7). When testing is necessary to demonstrate that BASIC SAFETY and ESSENTIAL PERFORMANCE are maintained under such simultaneous independent faults, the related testing may be limited to worst case situations.

NOTE 1 The test results might necessitate a revision of the RISK ANALYSIS.

NOTE 2 When considering use of tests described in this standard as routine tests, see Annex K of IEC/TR 62354 (production line tests) or see IEC 62353 (recurrent tests).

5.4 Other conditions

Replace existing list item a) with the following:

- a) Unless otherwise specified in this standard, ME EQUIPMENT is to be tested under the least favourable working conditions. The working conditions are specified in the ACCOMPANYING DOCUMENTS. The least favourable working conditions shall be documented for every test where they apply.

5.5 Supply voltages, type of current, nature of supply, frequency

Replace existing list item a) with the following:

- a) Where test results are influenced by deviations of the characteristics of the SUPPLY MAINS from their RATED values, the effect of such deviations is taken into account.

The values used during testing are according to 4.10.2 or according to that marked on the ME EQUIPMENT (see 7.2.6), whichever is least favourable.

NOTE 1 For the RATED value(s) given on the ME EQUIPMENT, deviations of the SUPPLY MAINS as provided in 4.10.2 are only to be tested if the deviations have a negative impact on BASIC SAFETY or ESSENTIAL PERFORMANCE.

In the existing first sentence of list item b), delete the phrase "(if marked) ± 1 Hz up to and including 100 Hz and ± 1 % above 100 Hz".

Replace existing list item c) with the following:

- c) ME EQUIPMENT designed for more than one RATED voltage, for both a.c. and d.c., or for both external power and an INTERNAL ELECTRICAL POWER SOURCE, is tested in conditions (described in 5.4) related to the least favourable voltage and nature of supply, for example, number of phases (except for single-phase supply) and type of current. It could be necessary to perform some tests more than once in order to establish which supply configuration is least favourable.

Renumber the note following list item f) as Note 2.

5.7 * Humidity preconditioning treatment

Replace the existing sixth, seventh and eighth paragraphs with the following:

The humidity preconditioning treatment is performed in a humidity cabinet containing air with a relative humidity of $93\% \pm 3\%$ where the ME EQUIPMENT or its parts under test are located. The humidity conditions at other locations in the chamber may vary by $\pm 6\%$. The temperature of the air in the cabinet, at all places where ME EQUIPMENT or its parts can be located, shall be maintained within $2\text{ }^{\circ}\text{C}$ of any convenient value T in the range of $+20\text{ }^{\circ}\text{C}$ to $+30\text{ }^{\circ}\text{C}$. Before being placed in the humidity cabinet, ME EQUIPMENT or its parts are brought to a temperature between T and $T + 4\text{ }^{\circ}\text{C}$, and kept at this temperature for at least 4 h before the humidity treatment starts.

Keep ME EQUIPMENT and its parts, where the ENCLOSURE is classified as IPX0, in the humidity cabinet for 48 h.

Keep ME EQUIPMENT and its parts, where the ENCLOSURE is designed to provide higher ingress protection against liquids, in the humidity cabinet for 168 h.

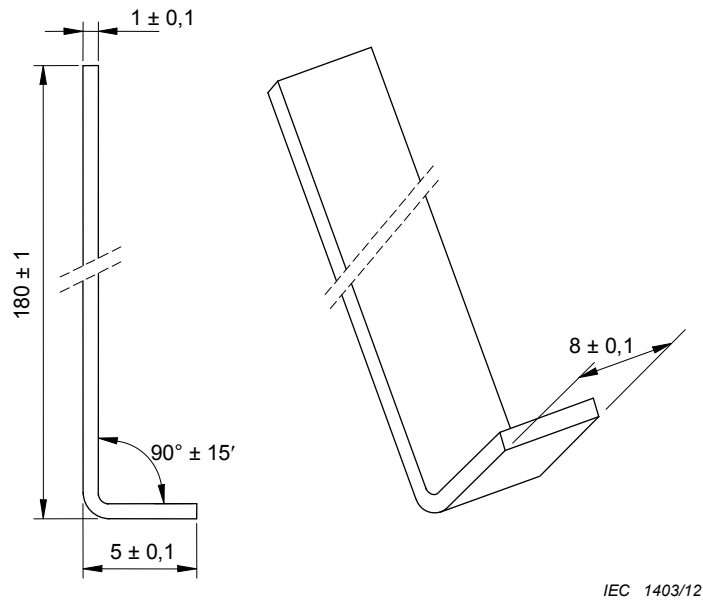
5.9 * Determination of APPLIED PARTS and ACCESSIBLE PARTS

5.9.2 ACCESSIBLE PARTS

5.9.2.2 Test hook

In the second paragraph, replace "Figure 7" by "Figure 6".

Replace existing Figure 7 with the following:



Material: steel

Figure 7 – Test hook
(see 5.9.2.2)

5.9.2.3 Actuating mechanisms

Replace the existing text of the subclause with the following:

Conductive parts of actuating mechanisms of electrical controls that are accessible after the removal of handles, knobs, levers and the like are regarded as ACCESSIBLE PARTS. Conductive parts of actuating mechanisms are not considered ACCESSIBLE PARTS if removal of handles, knobs, etc. requires the use of a TOOL.

Compliance is checked by tests according to 5.9.2.1 and 15.4.6.1.

7 ME EQUIPMENT identification, marking and documents

7.1 General

7.1.1 * USABILITY of the identification, marking and documents

Replace the existing text of the subclause with the following:

See 12.2.

7.1.2 * Legibility of markings

Replace the existing compliance paragraphs with the following:

Compliance for clear legibility is checked by the following test:

The ME EQUIPMENT or its part is positioned so that the viewpoint is the intended position of the OPERATOR. If the intended position of the OPERATOR is not specified and the position is not obvious, the viewpoint is at any point within the base of a cone subtended by an angle of 30 ° to the axis normal to the centre of the plane of the marking and at a distance of 1 m. The ambient illumination is the least favourable level in the range of 100 lx to 1 500 lx.

The observer has a visual acuity, corrected if necessary, of:

- 0 on the log Minimum Angle of Resolution (log MAR) scale or 6/6 (20/20); and
 - is able to read N6 of the Jaeger test card;
- in normal room lighting conditions (approximately 500 lx).

The observer correctly reads the marking from the viewpoint.

7.1.3 * Durability of markings

In existing list item b), replace "methylated spirit" with "ethanol 96 %".

7.2 Marking on the outside of ME EQUIPMENT or ME EQUIPMENT parts (see also Table C.1)

7.2.1 Minimum requirements for marking on ME EQUIPMENT and on interchangeable parts

Replace the existing last paragraph with the following and insert a note:

Any material, component, ACCESSORY or ME EQUIPMENT that is intended for a single use or its packaging shall be marked "Single Use Only", "Do Not Reuse" or with symbol ISO 7000-1051 (2004-01) (see Table D.1, symbol 28).

NOTE See also 7.9.2.12.

7.2.2 * Identification

Replace the existing first paragraph with the following:

ME EQUIPMENT shall be marked with:

- the name or trademark and contact information of the MANUFACTURER;
- a MODEL OR TYPE REFERENCE;
- a serial number or lot or batch identifier; and
- the date of manufacture or use by date, if applicable.

NOTE See ISO 15223-1 for symbols for MANUFACTURER, serial number, lot or batch, year of manufacture and use by date.

The serial number, lot or batch identifier, and the date of manufacture may be provided in a human readable code or through automatic identification technology such as barcodes or RFID.

Detachable components of the ME EQUIPMENT shall be marked with:

- the name or trademark of the MANUFACTURER; and
- a MODEL OR TYPE REFERENCE;

unless misidentification does not result in an unacceptable RISK.

In the final sentence of the existing second paragraph, replace "does not need to be" with "need not be".

7.2.4 * ACCESSORIES

Replace the existing text of the subclause with the following:

ACCESSORIES shall be marked with:

- the name or trade-mark and contact information of their MANUFACTURER
- a MODEL OR TYPE REFERENCE;
- a serial number or lot or batch identifier; and
- the date of manufacture or use by date, if applicable.

NOTE See ISO 15223-1 for symbols for MANUFACTURER, serial number, lot or batch, year of manufacture and use by date.

The serial number, lot or batch identifier, and the date of manufacture may be provided in a human readable code or through automatic identification technology such as barcodes or RFID.

Where no marking of the ACCESSORIES is practicable, these markings may be affixed to the individual packaging.

7.2.5 ME EQUIPMENT intended to receive power from other equipment

Replace the existing text of the subclause with the following:

If ME EQUIPMENT is intended to receive its power from other electrical equipment in an ME SYSTEM and compliance with the requirements of this standard is dependent on that other equipment, one of the following shall be provided:

- the name or trademark of the manufacturer of the other electrical equipment with a MODEL OR TYPE REFERENCE of the specified other equipment adjacent to the relevant connection point;
- placing safety sign ISO 7010-M002 (see Table D.2, safety sign 10) adjacent to the relevant connection point and listing of the required details in the instructions for use; or
- using a special connector style that is not commonly available on the market and listing of the required details in the instructions for use.

See also 7.9.2.3, 8.2.1 and 16.3.

7.2.6 Connection to the SUPPLY MAINS

In the existing second and fourth bullets, delete "DB:".

In the last line of the existing final paragraph, replace "supply connection terminals" with "SUPPLY MAINS connection".

7.2.7 Electrical input power from the SUPPLY MAINS

Replace the existing first paragraph with:

The RATED input from the SUPPLY MAINS shall be marked on the ME EQUIPMENT. The RATED input shall be given in:

- amperes or volt-amperes, or
- if the power factor exceeds 0,9, in amperes, volt-amperes or watts.

7.2.10 * APPLIED PARTS

In the existing second and third paragraphs, delete "DB:".

7.2.14 HIGH VOLTAGE TERMINAL DEVICES

In the existing paragraph, delete "DB:", and insert the following note at the end of the subclause.

NOTE CREEPAGE DISTANCES and AIR CLEARANCES between the high voltage contacts and the standard test finger are subject to the requirements in 8.9.

7.2.17 Protective packaging

In the existing second paragraph, replace "ISO 15223" with "ISO 15223-1".

Replace the existing fourth paragraph with the following

The packaging of ME EQUIPMENT or ACCESSORIES supplied sterile shall be marked as sterile and indicate the method of sterilization (see ISO 15223-1).

7.2.18 External pressure source

Replace the existing subclause with the following:

Adjacent to each input connector, the ME EQUIPMENT shall be marked with:

- the RATED maximum supply pressure from an external source, and
- the RATED flow rate if required to maintain BASIC SAFETY or ESSENTIAL PERFORMANCE.

7.2.19 FUNCTIONAL EARTH TERMINALS

In the existing paragraph, replace "(DB:2002-10)" with "(2006-08)".

Add a new subclause following 7.2.20:

7.2.21 * Mass of MOBILE ME EQUIPMENT

MOBILE ME EQUIPMENT shall be marked with its mass including its SAFE WORKING LOAD in kilograms. The marking shall be done in a way that it is obvious that it applies to the whole of the MOBILE ME EQUIPMENT when loaded with its SAFE WORKING LOAD and is separate and distinct from any markings related to maximum bin, shelf or drawer loading requirements.

7.3 Marking on the inside of ME EQUIPMENT or ME EQUIPMENT parts (see also Table C.2)

7.3.2 * HIGH VOLTAGE parts

In the existing paragraph, delete "DB:".

7.3.4 * Fuses, THERMAL CUT-OUTS and OVER-CURRENT RELEASES

Replace the existing text of the subclause with the following:

Fuses and replaceable THERMAL CUT-OUTS and OVER-CURRENT RELEASES that are accessible only by the use of a TOOL shall be identified either by specification (voltage, current, operating speed, size, breaking capacity) adjacent to the component, or by a reference to information in the ACCOMPANYING DOCUMENTS.

The nomenclature of IEC 60127-1 may be used.

7.3.5 PROTECTIVE EARTH TERMINALS

Add an asterisk before the subclause heading, and in the existing first paragraph replace "(DB:2002-10)" with "(2006-08)".

7.3.6 FUNCTIONAL EARTH TERMINALS

In the existing paragraph, replace "(DB:2002-10)" with "(2006-08)".

7.3.7 Supply terminals

In the existing first paragraph, replace "HAZARDOUS SITUATION" with "unacceptable RISK".

7.4 Marking of controls and instruments (see also Table C.3)

7.4.1 Power switches

Add an asterisk before the subclause heading and delete "DB:" in two places in the first dash and in one place in the fourth and seventh dashes.

7.4.2 Control devices

Add an asterisk before the subclause heading and delete "DB:" in two places in the first paragraph.

After the last dash of the existing second paragraph, insert the following:

A control device or switch that brings the ME EQUIPMENT into the "stand-by" condition may be indicated by use of symbol IEC 60417-5009 (2002-10) (see Table D.1, Symbol 29).

7.4.3 Units of measure

Replace the existing subclause heading as follows:

7.4.3 Units of measurement

In the existing first paragraph, replace "ISO 31" with "ISO 80000-1", and in the existing second paragraph, replace "ISO 1000" with "ISO 80000-1".

Table 1 – Units outside the SI units system that may be used on ME EQUIPMENT

In the existing footnote ^a in Table 1, replace "ISO 31" with "ISO 80000-1".

7.5 Safety signs

Add to the end of the existing first paragraph, the following:

If a safety sign with an established meaning is appropriately used, the use of the general warning sign ISO 7010:2003-W001 (see Table D.2, safety sign 2) is not required.

Add, immediately before the compliance statement:

When supplementary text is placed together with safety signs, the supplementary text shall be in a language that is acceptable to the intended OPERATOR.

NOTE 4 In some countries, more than one language is required.

7.6 Symbols

7.6.3 Symbols for controls and performance

In the existing note, append bibliography reference "[60]" to IEC 60878.

7.7 Colours of the insulation of conductors

7.7.3 Green and yellow insulation

After the existing second dash, insert the following:

NOTE In other safety standards such as IEC 60950-1, internal connections between conductive parts and the main protective earth are called protective bonding conductors.

7.9 ACCOMPANYING DOCUMENTS

7.9.1 * General (see also Table C.4)

Replace the existing first dash with the following:

- name or trade-name of the MANUFACTURER and contact information to which the RESPONSIBLE ORGANIZATION can refer;

Replace the existing third paragraph with the following:

ACCOMPANYING DOCUMENTS may be provided electronically, e.g. electronic file format on CD-ROM. If the ACCOMPANYING DOCUMENTS are provided electronically, the USABILITY ENGINEERING PROCESS shall include consideration of which information also needs to be provided as hard copy or as markings on the ME EQUIPMENT (see 12.2).

EXAMPLE Information to cover emergency operation

NOTE ACCOMPANYING DOCUMENTS provided electronically might not be acceptable in all jurisdictions.

Replace the existing compliance paragraph with the following:

Compliance is checked by inspection of the ACCOMPANYING DOCUMENTS, and, when provided electronically, as specified in IEC 60601-1-6.

7.9.2 Instructions for use (see also Table C.5)

7.9.2.1 * General

Replace the existing first paragraph with the following new text, renumbering subsequent notes in this subclause:

The instructions for use shall document:

- the use of the ME EQUIPMENT as intended by the MANUFACTURER;
- the frequently used functions;
- any known contraindication(s) to the use of the ME EQUIPMENT; and
- those parts of the ME EQUIPMENT that shall not be serviced or maintained while in use with a PATIENT.

Where the PATIENT is an intended OPERATOR, the instructions for use shall indicate:

- the PATIENT is an intended OPERATOR;
- a warning against servicing and maintenance while the ME EQUIPMENT is in use;
- which functions the PATIENT can safely use and, where applicable, which functions the PATIENT cannot safely use; and
- which maintenance the PATIENT can perform (e.g. changing batteries).

NOTE 1 For ME EQUIPMENT where the INTENDED USE includes the PATIENT partially or fully operating the ME EQUIPMENT, the PATIENT becomes an OPERATOR.

NOTE 2 For ME EQUIPMENT where the PATIENT is allowed to perform restricted maintenance, the PATIENT becomes SERVICE PERSONNEL.

The instructions for use shall indicate:

- the name or trademark and address of the MANUFACTURER;
- the MODEL OR TYPE REFERENCE.

Insert after the last paragraph:

NOTE 5 In some countries, more than one language is required.

7.9.2.7 * Isolation from the SUPPLY MAINS

In the existing text of the subclause, replace "separable plug" with "MAINS PLUG or other separable plug".

7.9.2.14 ACCESSORIES, supplementary equipment, used material

In the existing second paragraph, replace "continuous service" with "continuous duty".

7.9.2.15 Environmental protection

Replace the existing text of the subclause with the following:

The instructions for use shall provide advice on the proper disposal of waste products, residues, etc. and of the ME EQUIPMENT and ACCESSORIES at the end of their EXPECTED SERVICE LIFE.

Insert the following new subclauses ahead of the overall compliance paragraph for subclause 7.9.2:

7.9.2.17 ME EQUIPMENT emitting radiation

For ME EQUIPMENT emitting radiation for medical purposes, when appropriate, the instructions for use shall indicate the nature, type, intensity and distribution of this radiation.

7.9.2.18 ME EQUIPMENT and ACCESSORIES supplied sterile

The instructions for use for ME EQUIPMENT or ACCESSORIES supplied sterile shall indicate that they have been sterilized and indicate the method of sterilization.

The instructions for use shall indicate the necessary instructions in the event of damage to the sterile packaging, and where appropriate, details of the appropriate methods of re-sterilization (see also 7.9.2.12).

7.9.2.19 * Unique version identifier

The instructions for use shall contain a unique version identifier such as its date of issue.

EXAMPLE Month and year.

7.9.3 Technical description (see also Table C.6)

7.9.3.1 * General

Add the following dashed item to the first paragraph:

- information pertaining to ESSENTIAL PERFORMANCE and any necessary recurrent ESSENTIAL PERFORMANCE and BASIC SAFETY testing including details of the means, methods and recommended frequency.

NOTE 2 This information is directly derived from the ESSENTIAL PERFORMANCE identified by the MANUFACTURER in 4.3.

Replace the existing dash immediately ahead of Note 2 with the following and renumber Note 2 to Note 3 and Note 3 to Note 4:

- a brief description of the ME EQUIPMENT, how the ME EQUIPMENT functions and its significant physical and performance characteristics; and
- a unique version identifier such as its date of issue.

8 Protection against electrical HAZARDS from ME EQUIPMENT

8.1 Fundamental rule of protection against electric shock

In existing list item b), replace the sixth and seventh dashes as follows:

- interruption of any one power-carrying conductor between ME EQUIPMENT parts in separate ENCLOSURES, if this condition might cause permitted limits to be exceeded;
- unintended movement of a component (see 8.10.1);

8.2 Requirements related to power sources

8.2.2 Connection to an external d.c. power source

Replace the existing first paragraph and the note with the following:

If ME EQUIPMENT is specified for power supplied from an external d.c. power source, then a connection with the wrong polarity shall not lead to the HAZARDOUS SITUATIONS described in 13.1. The ME EQUIPMENT, when connection is subsequently made with the correct polarity, shall maintain BASIC SAFETY and ESSENTIAL PERFORMANCE.

Protective devices that can be reset by anyone without the use of a TOOL are acceptable provided the ME EQUIPMENT returns to NORMAL CONDITION on reset.

NOTE 1 See also 11.8.

NOTE 2 The external d.c. power source can be a SUPPLY MAINS or another item of electrical equipment. In the latter case, the combination is considered to be an ME SYSTEM as specified in 8.2.1.

8.3 Classification of APPLIED PARTS

In existing list item a), replace the existing note with the following:

NOTE See the rationale for the definition of DIRECT CARDIAC APPLICATION (3.22) concerning the APPLIED PARTS that need to be TYPE CF APPLIED PARTS. Other restrictions can apply for cardiac applications.

Delete existing list item d).

8.4 Limitation of voltage, current or energy

8.4.2 ACCESSIBLE PARTS including APPLIED PARTS

Replace the existing title and list item b) with the following:

8.4.2 ACCESSIBLE PARTS and APPLIED PARTS

- b) *. The LEAKAGE CURRENTS from, to or between ACCESSIBLE PARTS shall not exceed the limits for TOUCH CURRENT specified in 8.7.3 c) when measured as specified in 8.7.4.

In the second paragraph of existing list item c), replace "potential up to 2 V" with "potential of 2 V or more".

Replace the compliance paragraph in list item c) with the following:

Compliance is checked by inspection of the instructions for use and by measurement.

In existing list item d), replace the second dash with the following:

- internal parts that can be touched by a metal test rod with a diameter of $4 \begin{smallmatrix} 0 \\ -0,05 \end{smallmatrix}$ mm and a length of $100 \begin{smallmatrix} +0,5 \\ 0 \end{smallmatrix}$ mm, inserted through any opening in the top of an ENCLOSURE or through any opening provided for the adjustment of pre-set controls that can be adjusted by the RESPONSIBLE ORGANIZATION in NORMAL USE by using a TOOL.

8.4.3 * ME EQUIPMENT intended to be connected to a power source by a plug

In the existing sixth paragraph, replace "affect the test" with "affect the test significantly".

Add, after the sixth paragraph, the following note:

NOTE An example of a measuring arrangement that is considered acceptable is an oscilloscope and probe having an input impedance consisting of a resistance of $100 \text{ M}\Omega \pm 5 \text{ M}\Omega$ in parallel with an input capacitance of $20 \text{ pF} \pm 5 \text{ pF}$.

Add, after the last paragraph, the following new text:

Where appropriate, a d.c. input voltage equal to the peak of the RATED SUPPLY MAINS voltage may be used.

8.4.4 * Internal capacitive circuits

In the existing second paragraph, delete "DB:"

8.5 Separation of parts

8.5.1 * MEANS OF PROTECTION (MOP)

8.5.1.1 General

Replace the existing note with the following:

NOTE Coatings and other insulation that are intended as a MEANS OF PROTECTION and that comply with IEC 60950-1:2005 may be used as a MEANS OF OPERATOR PROTECTION but not automatically as a MEANS OF PATIENT PROTECTION. For MEANS OF PATIENT PROTECTION, considerations can arise as a result of the RISK MANAGEMENT PROCESS.

Add the following new final paragraph:

Compliance is checked by the test in 8.5.1.3.

8.5.1.2 MEANS OF PATIENT PROTECTION (MOPP)

Replace the existing title with the following:

8.5.1.2 * MEANS OF PATIENT PROTECTION (MOPP)

Replace the existing fourth paragraph with the following:

A Y capacitor (Y1 or Y2 only) complying with IEC 60384-14 is considered equivalent to one MEANS OF PATIENT PROTECTION. Where two capacitors are used in series, they shall be identical in type (either both Y1 or both Y2) and shall have the same NOMINAL capacitance. The capacitor(s) shall meet the dielectric strength for the type of protection for which they are being used (i.e. one or two MEANS OF PATIENT PROTECTION).

Where the working voltage across a barrier forming a MEANS OF PATIENT PROTECTION is less than 42,4 V peak a.c. or 60 V d.c., a single Y1 capacitor is acceptable for two MEANS OF PATIENT PROTECTION

Compliance is checked by the test in 8.5.1.3.

8.5.1.3 MEANS OF OPERATOR PROTECTION (MOOP)

Replace the existing fourth paragraph with the following:

A Y capacitor (Y1 or Y2 only) complying with IEC 60384-14 is considered equivalent to one MEANS OF OPERATOR PROTECTION. Where two capacitors are used in series, they shall be identical in type (either both Y1 or both Y2) and shall have the same NOMINAL capacitance. The capacitor(s) shall meet the dielectric strength for the type of protection for which they are being used (i.e. one or two MEANS OF OPERATOR PROTECTION). A Y1 capacitor can be used for two MEANS OF OPERATOR PROTECTION.

Replace the first paragraph of the compliance statement and the note with the following:

Compliance is checked by examination of the physical and electrical configuration of the ME EQUIPMENT to identify points at which insulation, CREEPAGE DISTANCES, AIR CLEARANCES, impedances of components or PROTECTIVE EARTH CONNECTIONS prevent ACCESSIBLE PARTS or APPLIED PARTS from exceeding the limits specified in 8.4.

NOTE Such points typically include insulation between parts different from earth potential and ACCESSIBLE PARTS or APPLIED PARTS but can also include, for example, insulation between a floating circuit and earth or other circuits. A survey of insulation paths is found in Annex J.

Replace the last paragraph of the compliance statement with:

The voltage, current or energy that can appear between any ACCESSIBLE PART or APPLIED PART and any other ACCESSIBLE PART, APPLIED PART or earth in NORMAL CONDITION and in SINGLE FAULT CONDITION is determined by inspection or calculation or, where necessary, by measurement in the relevant conditions.

8.5.2 Separation of PATIENT CONNECTIONS

8.5.2.3 * PATIENT leads

Replace the title, and in the existing first paragraph, replace the first sentence and the first dash with the following:

8.5.2.3 * PATIENT leads or PATIENT cables

Any connector for electrical connections on a PATIENT lead or PATIENT cable that:

- is at the end of the lead or cable that is remote from the PATIENT; and

Delete the full stop at the end of the second dash of the first paragraph.

8.5.5 DEFIBRILLATION-PROOF APPLIED PARTS

8.5.5.1 * Defibrillation protection

Delete the existing Note 1 and renumber the subsequent notes.

In existing list item a), delete the "or" at the end of the third dash, replace the full stop at the end of the fourth dash with "; or", and insert a fifth dash as follows:

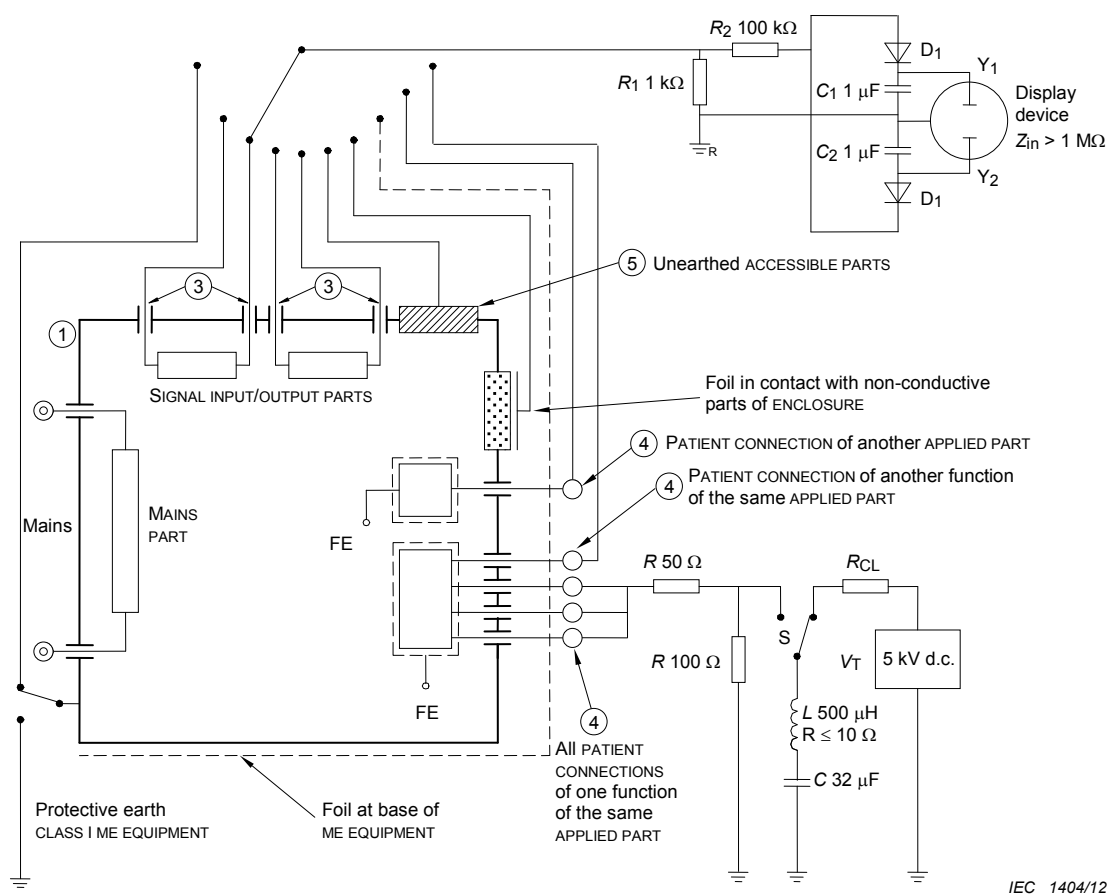
- any unused or disconnected connections of the APPLIED PART under test or any function of the same APPLIED PART. ME EQUIPMENT that is completely BODY-WORN (e.g. a Holter monitor) is exempt from this requirement.

- **Common-mode test**

Add the following sentence to the end of the first paragraph.

If an APPLIED PART has multiple functions, the test voltage is applied to all the PATIENT CONNECTIONS of one function at a time with the other functions disconnected.

Replace existing Figure 9 with the following:



For legends, see Table 5.

Components

- V_T Test voltage
 S Switch for applying the test voltage
 R_1, R_2 Tolerance at $\pm 2\%$, not less than 2 kV
 R_{CL} Current limiting resistor
 D_1, D_2 Small signal silicon diodes
 Other components toleranced at $\pm 5\%$

Figure 9 – Application of test voltage to bridged PATIENT CONNECTIONS for DEFIBRILLATION-PROOF APPLIED PARTS
 (see 8.5.5.1)

8.5.5.2 Energy reduction test

Replace the third sentence of the existing second compliance paragraph with the following:

The test voltage is applied to each PATIENT CONNECTION or APPLIED PART in turn with all the remaining PATIENT CONNECTIONS of the same APPLIED PART being connected to earth (differential mode). Other DEFIBRILLATION PROOF APPLIED PARTS, if any, are tested separately, in turn.

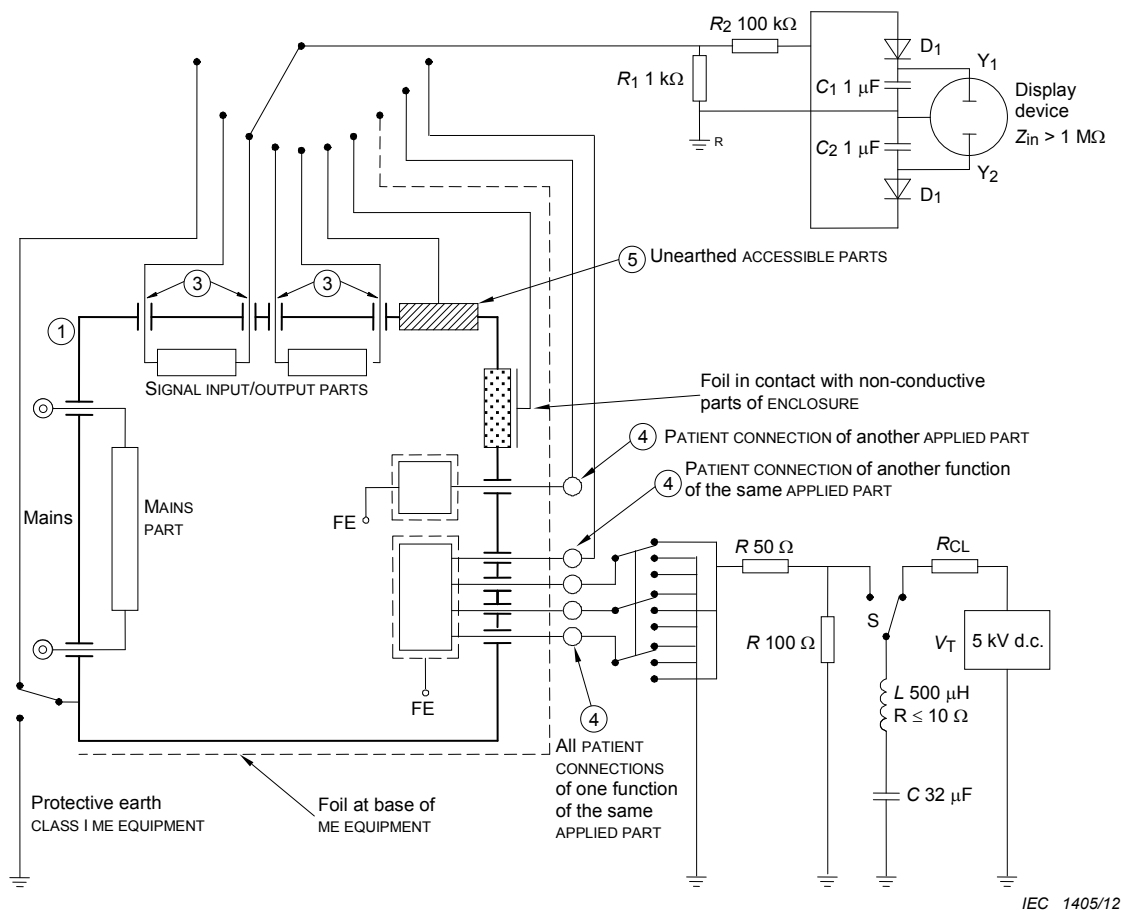
Replace the existing list item a) with the following:

- a) Connect the APPLIED PART or PATIENT CONNECTION to the test circuit. The parts described in 8.5.5.1 a) are connected to earth.

Insert a new list item f) as follows:

- f) Repeat the test with V_T reversed.

Move existing Figure 10 to immediately follow Figure 9 and replace it with the following:



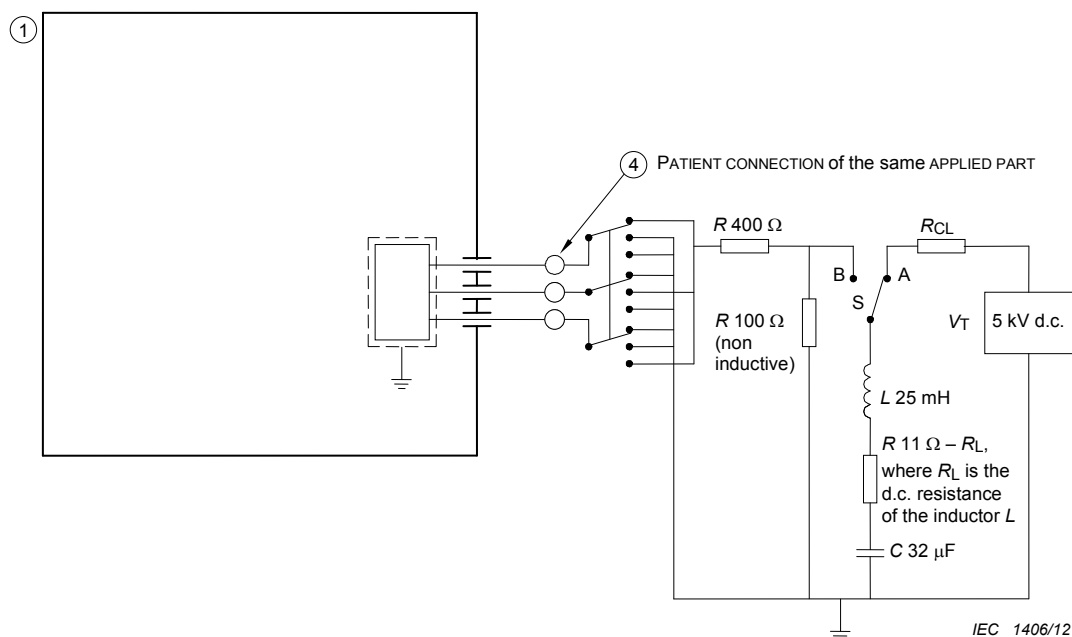
For legends, see Table 5.

Components

- V_T Test voltage
 S Switch for applying the test voltage
 R_1, R_2 Tolerance at $\pm 2\%$, not less than 2 kV
 R_{CL} Current limiting resistor
 D_1, D_2 Small signal silicon diodes
 Other components toleranced at $\pm 5\%$

Figure 10 – Application of test voltage to individual PATIENT CONNECTIONS for DEFIBRILLATION-PROOF APPLIED PARTS
 (see 8.5.5.1)

Replace the existing Figure 11 with the following:



For legends, see Table 5

Components

S Switch for applying the test voltage

A, B Switch positions

R_{CL} Current limiting resistor

Components toleranced at $\pm 5\%$

Figure 11 – Application of test voltage to test the delivered defibrillation energy
(see 8.5.5.2)

8.6 * Protective earthing, functional earthing and potential equalization of ME EQUIPMENT

8.6.4 Impedance and current-carrying capability

In existing list item a) immediately before the compliance statement, insert the following:

Additionally, the impedance between the protective earth pin in the MAINS PLUG of any DETACHABLE POWER SUPPLY CORD supplied or specified by the MANUFACTURER, when attached to the ME EQUIPMENT, and any part of the ME EQUIPMENT that is PROTECTIVELY EARTHED shall not exceed 200 mΩ, except as allowed by 8.6.4 b).

Where a DETACHABLE POWER SUPPLY CORD is neither supplied nor specified, testing shall be carried out using a 3 m long cord of appropriate cross sectional area based on 8.11.3.3 and Table 17.

In existing list item a) immediate after the second paragraph of the compliance statement, insert the following:

Alternatively, d.c. may be used for this test.

8.6.7 * POTENTIAL EQUALIZATION CONDUCTOR

Replace the existing second dash with the following:

- Accidental disconnection shall be avoided in NORMAL USE.

In the existing fifth dash, delete "DB:".

8.6.9 * CLASS II ME EQUIPMENT

At the end of the existing first paragraph, insert the following:

In such a case, the ACCOMPANYING DOCUMENTS shall state that the third conductor in the POWER SUPPLY CORD is only a functional earth.

Replace the existing second paragraph with the following:

The insulation between internal screens, including internal wiring connected to them and ACCESSIBLE PARTS, shall provide two MEANS OF PROTECTION.

8.7 LEAKAGE CURRENTS and PATIENT AUXILIARY CURRENTS

8.7.1 General requirements

After the first dash in existing list item b), insert the following:

- after any required sterilization PROCEDURE (see 11.6.7);

8.7.3 * Allowable values

Renumber the existing note in list item d) to Note 1 and insert Note 2 as follows:

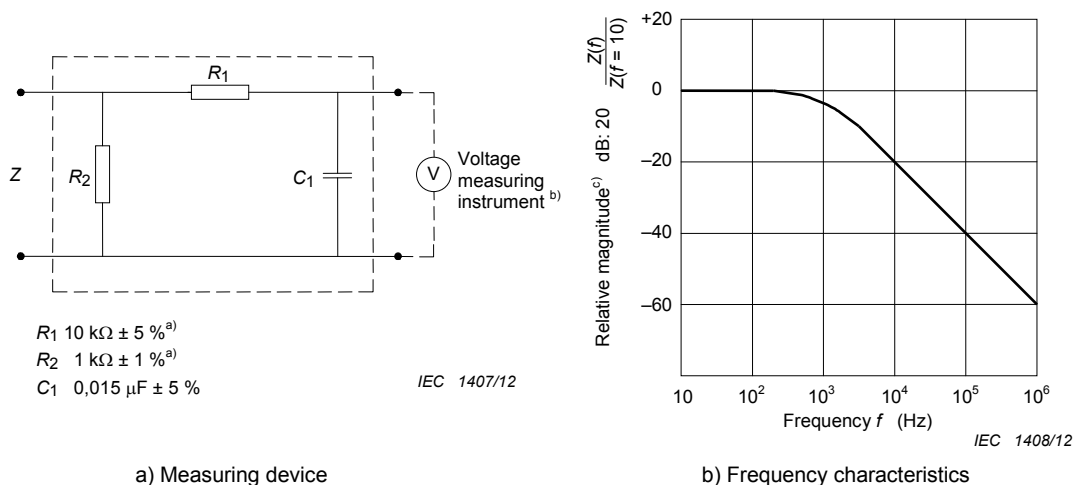
NOTE 2 The EARTH LEAKAGE CURRENT in NORMAL CONDITION can become TOUCH CURRENT in SINGLE FAULT CONDITION unless the ME EQUIPMENT is PERMANENTLY INSTALLED or the protective earth is not accessible from outside the ME EQUIPMENT.

Insert a new item f) as follows:

- f) *. The allowable values of LEAKAGE CURRENTS that can flow in a FUNCTIONAL EARTH CONDUCTOR in a non-PERMANENTLY INSTALLED ME EQUIPMENT are 5 mA in NORMAL CONDITION and 10 mA in SINGLE FAULT CONDITION.

NOTE 3 Subclause 16.6.1 requires that when ME EQUIPMENT is included in an ME SYSTEM, the TOUCH CURRENTS from any part of the ME SYSTEM cannot exceed 100 μ A in NORMAL CONDITION and 500 μ A in SINGLE FAULT CONDITION.

Replace existing Figure 12 with the following:



NOTE The network and voltage measuring instrument above are replaced by the symbol  in the following figures.

a) Non-inductive components

b) Resistance $\geq$ 1 M Ω and capacitance $\leq$ 150 pF

c) $Z(f)$ is the transfer impedance of the network, i.e. V_{out}/I_{in} , for a current of frequency f .

Figure 12 – Example of a measuring device and its frequency characteristics
(see 8.7.3)

8.7.4 Measurements

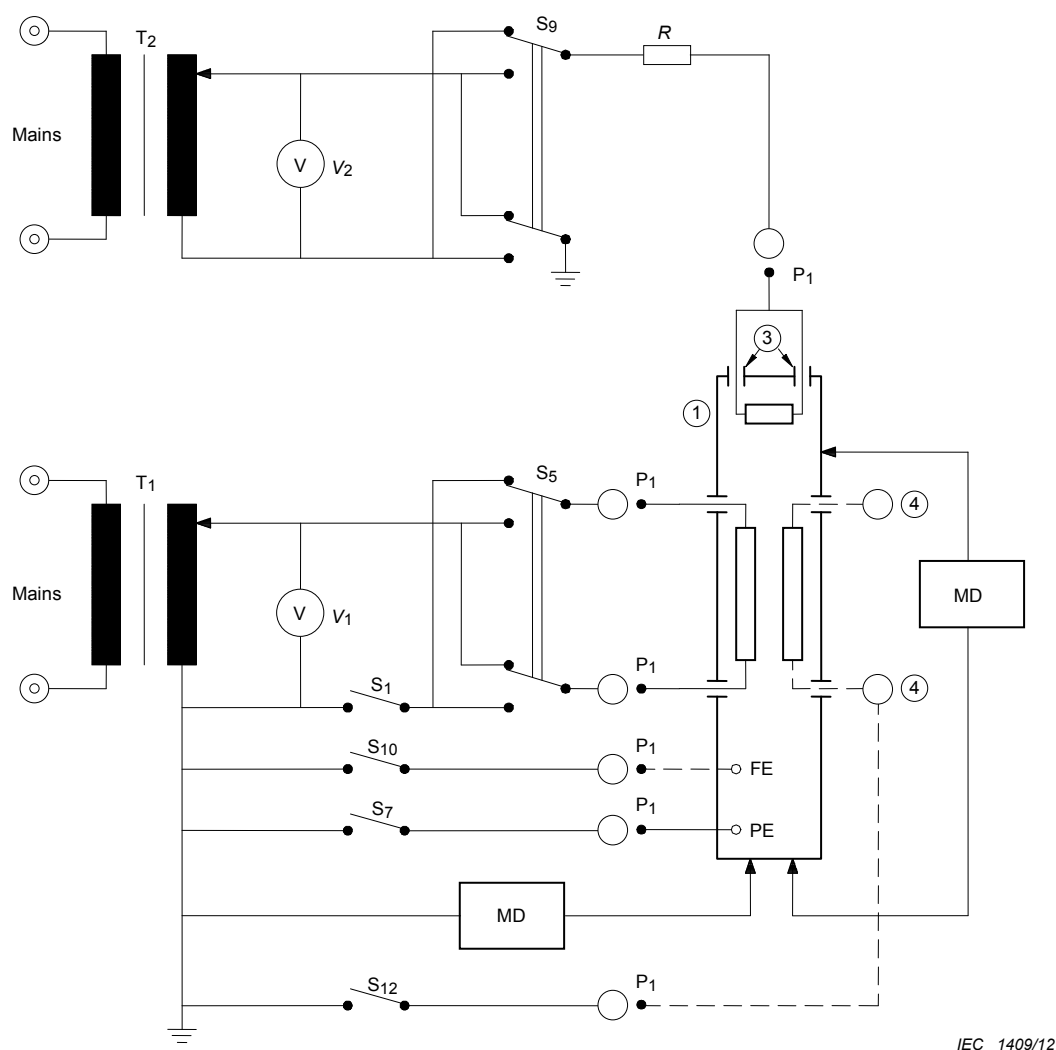
8.7.4.1 General

In existing list item b), replace "any HAZARDOUS SITUATION" with "any HAZARDOUS SITUATION described in 13.1".

Figure 13 – Measuring circuit for the EARTH LEAKAGE CURRENT of CLASS I ME EQUIPMENT, with or without APPLIED PART (see 8.7.4.5)

In the existing title of the figure, delete the word "the".

Replace existing Figure 14 with the following:



For legends, see Table 5.

Key

Measure (with S_7 closed if CLASS I ME EQUIPMENT) under all possible combinations of positions of S_1 , S_5 , S_9 , S_{10} , and S_{12} .

S_1 open is a SINGLE FAULT CONDITION.

CLASS I ME EQUIPMENT only:

Measure with S_7 open (SINGLE FAULT CONDITION) and with S_1 closed under all possible combinations of S_5 , S_9 , S_{10} and S_{12} .

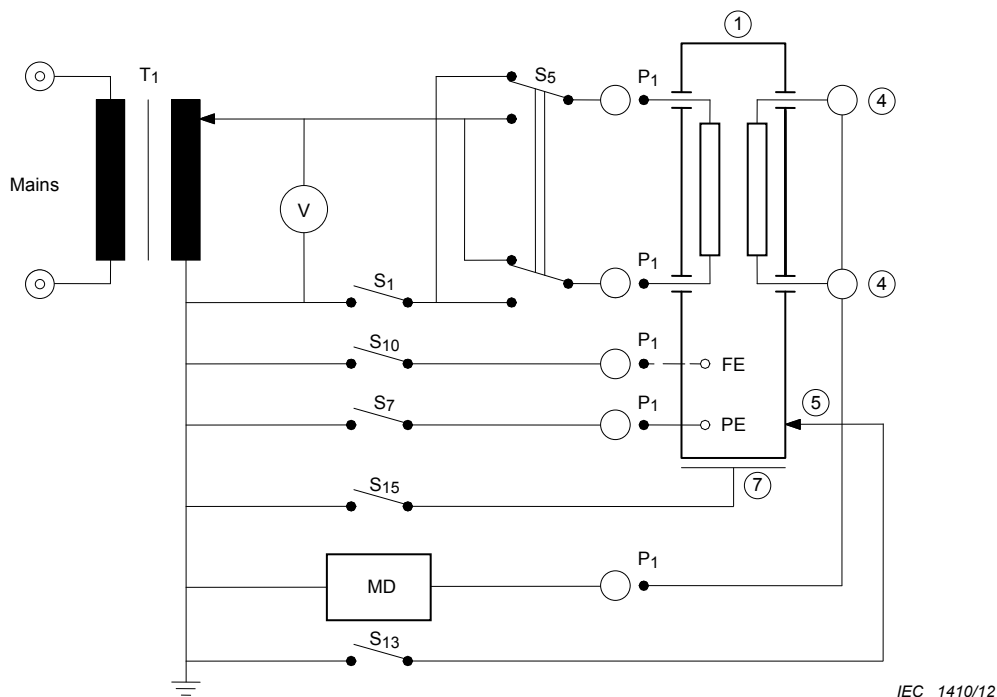
For CLASS II ME EQUIPMENT, the PROTECTIVE EARTH CONNECTION and S_7 are not used.

Transformer T₂ is used if required (see 8.1 a))

Example with the measuring supply circuit of Figure F.1.

Figure 14 – Measuring circuit for TOUCH CURRENT
(see 8.7.4.6)

Replace existing Figure 15 with the following:



For legends, see Table 5.

Key

Measure (with S_7 closed if CLASS I ME EQUIPMENT)
under all possible combinations of positions of S_1 ,
 S_5 , S_{10} , S_{13} and S_{15} :

S_1 open is a SINGLE FAULT CONDITION.

CLASS I ME EQUIPMENT only:

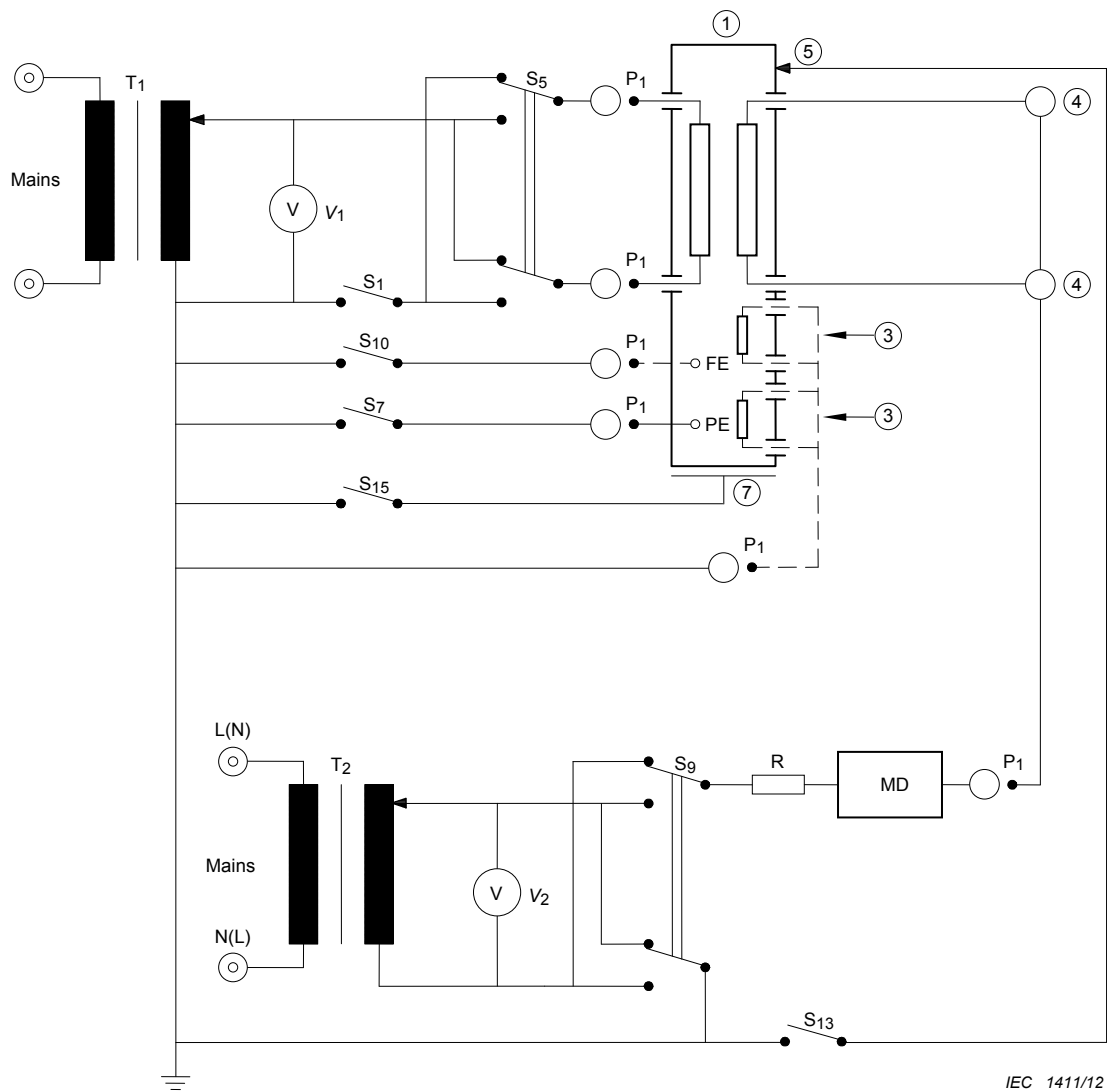
Measure with S_7 open (SINGLE FAULT CONDITION) and with S_1 closed under all possible combinations of S_5 , S_{10} , S_{13} and S_{15} .

For CLASS II ME EQUIPMENT, the PROTECTIVE EARTH CONNECTION and S_7 are not used.

Example with the measuring supply circuit of Figure F.1.

**Figure 15 – Measuring circuit for PATIENT LEAKAGE CURRENT from the
PATIENT CONNECTION to earth
(see 8.7.4.7 a))**

Replace existing Figure 16 with the following:



For legends, see Table 5.

Key

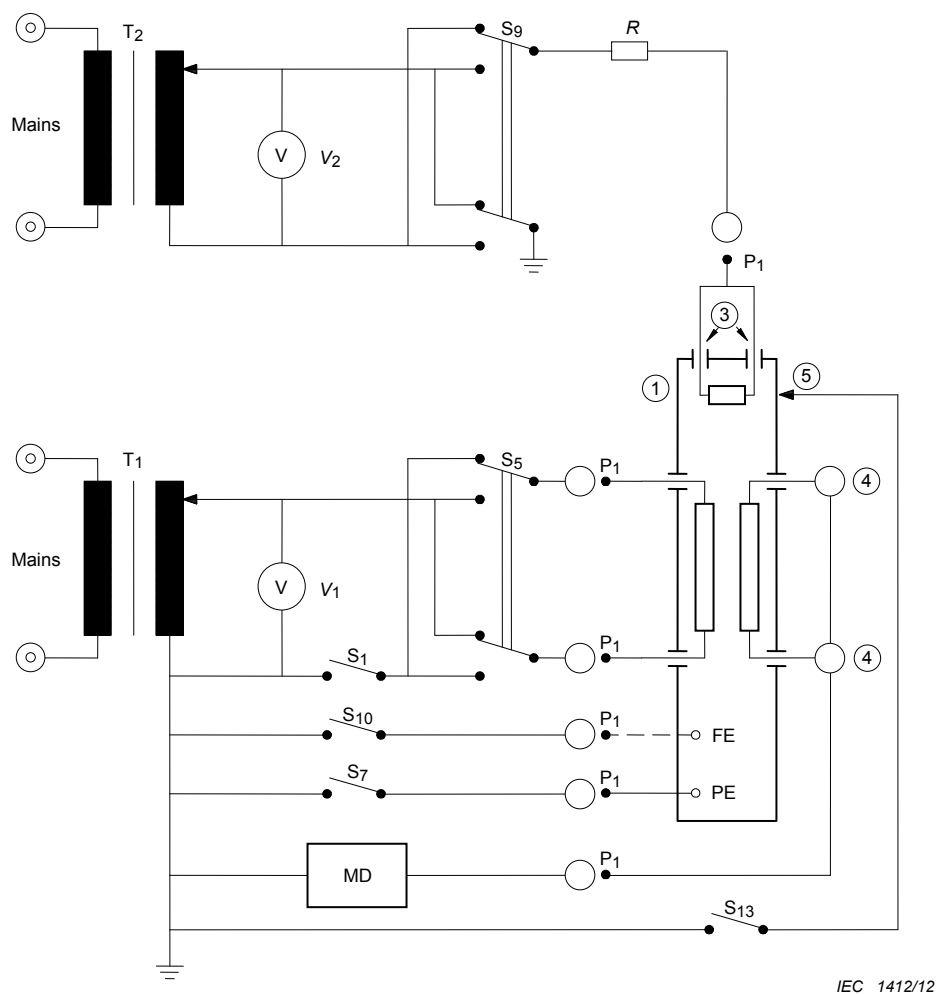
Measure (with S_7 closed, if CLASS I ME EQUIPMENT) WITH S_1 closed under all possible combinations of positions of S_5 , S_9 , S_{10} and S_{13} .

For CLASS II ME EQUIPMENT, the PROTECTIVE EARTH CONNECTION and S_7 are not used.

Example with the measuring supply circuit of Figure F.1.

Figure 16 – Measuring circuit for PATIENT LEAKAGE CURRENT via the PATIENT CONNECTION(s) of an F-TYPE APPLIED PART to earth caused by an external voltage on the PATIENT CONNECTION(s)
(see 8.7.4.7 b))

Replace existing Figure 17 with the following:



For legends, see Table 5.

Key

Measure (with S_7 closed, if CLASS I ME EQUIPMENT) under all possible combinations of positions of S_1 , S_5 , S_9 , S_{10} and S_{13} (S_1 open is SINGLE FAULT CONDITION).

CLASS I ME EQUIPMENT only:

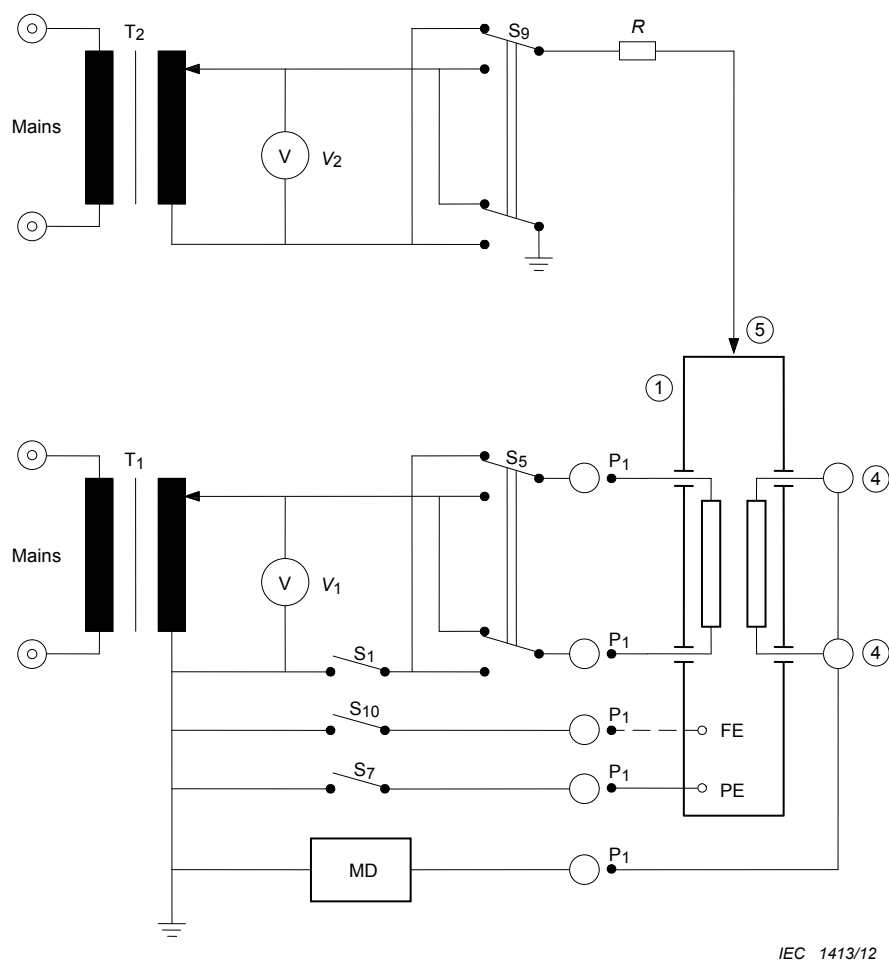
Measure with S_7 open (SINGLE FAULT CONDITION) and with S_1 closed under all possible combinations of S_5 , S_9 , S_{10} and S_{13} .

For CLASS II ME EQUIPMENT, the PROTECTIVE EARTH CONNECTION and S_7 are not used.

Example with the measuring supply circuit of Figure F.1.

Figure 17 – Measuring circuit for PATIENT LEAKAGE CURRENT from PATIENT CONNECTION(S) to earth caused by an external voltage on a SIGNAL INPUT/OUTPUT PART
(see 8.7.4.7 c))

Replace existing Figure 18 with the following:



IEC 1413/12

For legends, see Table 5.

Key

Measure with S_1 closed (and with S_7 closed, if CLASS I ME EQUIPMENT) under all possible combinations of positions of S_5 , S_9 and S_{10}

For CLASS II ME EQUIPMENT, the PROTECTIVE EARTH CONNECTION and S_7 are not used.

Example with the measuring supply circuit of Figure F.1.

Figure 18 – Measuring circuit for PATIENT LEAKAGE CURRENT from PATIENT CONNECTION(S) to earth caused by an external voltage on a metal ACCESSIBLE PART that is not PROTECTIVELY EARTHED (see 8.7.4.7 d))

Figure 19 – Measuring circuit for the PATIENT AUXILIARY CURRENT (see 8.7.4.8

In the existing title of the figure, delete the word "the".

Replace the existing second entry in the figure key with the following:

S_1 open is a SINGLE FAULT CONDITION.

Figure 20 – Measuring circuit for total PATIENT LEAKAGE CURRENT with all PATIENT CONNECTIONS of all APPLIED PARTS of the same type (TYPE B APPLIED PARTS, TYPE BF APPLIED PARTS or TYPE CF APPLIED PARTS) connected together(see 8.7.4.7 h))

Replace, in the current title of the figure, the phrase " "the total patient leakage current" with "total patient leakage current".

**Table 5 – Legends of symbols for Figure 9 to Figure 11, Figure 13 to Figure 20
Figure A.15, Annexes E and F**

In the existing legend (2), replace "(see 5.5 g) and" with "(See 5.5 f) and".

Replace the existing legend (5) with the following:

(5)	Metal ACCESSIBLE PART NOT PROTECTIVELY EARTHED In case of a non-conductive ENCLOSURE and PATIENT LEAKAGE CURRENT measurement, connection is replaced by a metal foil of maximum 20 cm × 10 cm in intimate contact with the ENCLOSURE or relevant parts of the ENCLOSURE and connected to the reference earth.
-----	--

After legend (6) insert the following new legend:

(7)	Metal plate under a non-conductive ENCLOSURE with dimensions at least equal to the plan projection of the ENCLOSURE connected to the reference earth
-----	--

Replace legend T_1 T_2 as follows

T_1, T_2	Single- or polyphase isolation transformers with sufficient power rating and adjustable output voltage (See also the rationale for 8.7.4.2 and 8.7.4.3.)
------------	--

Below legend S_{14} , insert the following new legend:

S_{15}	Switch for connecting to earth a metal plate under a non-conductive ENCLOSURE
----------	---

Replace legend R with the following:

R	Impedance to protect the circuitry and the person performing the test, but low enough to accept currents higher than the allowable values of the LEAKAGE CURRENT to be measured (optional)
-----	--

Replace legend $\frac{R}{\equiv}$ with the following:

$\frac{\perp}{\equiv}$	Reference earth (for LEAKAGE CURRENT and PATIENT AUXILIARY CURRENT measurements and for testing of DEFIBRILLATION-PROOF APPLIED PARTS, not connected to protective earth of the SUPPLY MAINS)
------------------------	---

8.7.4.5 * Measurement of the EARTH LEAKAGE CURRENT

Replace the title and the existing list item a) with the following:

8.7.4.5 * Measurement of the EARTH LEAKAGE CURRENT and current in functional earth connection

a) CLASS I ME EQUIPMENT is tested according to Figure 13. CLASS II ME EQUIPMENT with a functional earth connection according to 8.6.9 is tested as if it were CLASS I ME EQUIPMENT.

8.7.4.7 Measurement of the PATIENT LEAKAGE CURRENT

Replace existing list item g) with the following:

g) The PATIENT LEAKAGE CURRENT is measured (see also Annex E):

- for TYPE B APPLIED PARTS, from all PATIENT CONNECTIONS connected directly together.

- for TYPE BF APPLIED PARTS, from and to all PATIENT CONNECTIONS of a single function either connected directly together or loaded as in NORMAL USE.
- for TYPE CF APPLIED PARTS, from and to every PATIENT CONNECTION in turn.

In the existing second paragraph of list item g), replace "PATIENT leads and electrodes" with "PATIENT leads or PATIENT cables and electrodes".

8.8 Insulation

8.8.1 * General

Replace the first paragraph and its two dashes with:

Only insulation that is relied upon as a MEANS OF PROTECTION, including REINFORCED INSULATION, shall be subject to testing.

Table 6 – Test voltages for solid insulation forming a MEANS OF PROTECTION

Add the following new table notes:

NOTE 1 For a barrier according to:

- Figure J.6, use the column MEANS OF PATIENT PROTECTION – Protection from SECONDARY CIRCUITS – Two MOPP.
- 8.5.2.1 and Figure J.7, use the column MEANS OF PATIENT PROTECTION – Protection from MAINS PART – One MOPP.

NOTE 2 See the rationale for 8.8.3.

8.8.4 Insulation other than wire insulation

8.8.4.1 * Mechanical strength and resistance to heat

Replace the existing second paragraph with the following:

Compliance is checked by inspection of the ME EQUIPMENT and the design documentation, and, if necessary, inspection of the RISK MANAGEMENT FILE in conjunction with the following tests:

- *resistance to moisture, etc. (see 11.6);*
- *dielectric strength (see 8.8.3);*
- *mechanical strength (see 15.3).*

8.9 * CREEPAGE DISTANCES and AIR CLEARANCES

8.9.1 * Values

8.9.1.1 General

Replace the existing text of the subclause with the following:

CREEPAGE DISTANCES and AIR CLEARANCES of ME EQUIPMENT shall be equal to or greater than the values of:

- for insulation between parts of opposite polarity of the MAINS PART on the SUPPLY MAINS side of any mains fuse or OVER-CURRENT RELEASE, one MEANS OF OPERATOR PROTECTION in accordance with Table 13, Table 14 and Table 16; and
- for insulation providing at least a MEANS OF PROTECTION, in accordance with Table 12 to Table 16 (inclusive) except as specified in 8.9.1.2 to 8.9.1.15. See also 8.9.2 to 8.9.4.

8.9.1.2 CREEPAGE DISTANCES and AIR CLEARANCES complying with IEC 60950-1

In the existing text, replace "Table 11 to Table 16 (inclusive)" with "Table 12 to Table 16 (inclusive)".

8.9.1.4 Minimum CREEPAGE DISTANCE

In the existing text, replace "Table 11 to Table 16 (inclusive)" with "Table 12 to Table 16 (inclusive)".

8.9.1.6 * Interpolation

In the existing text, replace "Table 11 to Table 16 (inclusive)" with "Table 12 to Table 16 (inclusive)".

8.9.1.5 ME EQUIPMENT RATED for high altitudes

Table 8 – Multiplication factors for AIR CLEARANCES for altitudes up to 5 000 m

In existing table Note 3, replace "IEC 60664-1:1993 as amended" with "IEC 60664-1:2007".

8.9.1.8 Pollution degree classification

After the last paragraph of the subclause, insert the following:

Annex M specifies measures that may be used to reduce the pollution degree.

8.9.1.15 * CREEPAGE DISTANCES and AIR CLEARANCES for DEFIBRILLATION-PROOF APPLIED PARTS

In the existing note, delete "Table 11 and" and replace "the clearance is related" with "the AIR CLEARANCE is related".

Delete the existing Table 11 and replace the title with "Table 11 – Not used".

Table 14 – Additional AIR CLEARANCES for insulation in MAINS PARTS with PEAK WORKING VOLTAGES exceeding the peak value of the NOMINAL MAINS VOLTAGE ^a (see 8.9.1.10)

Add the following table note:

NOTE For voltage values above the PEAK WORKING VOLTAGE values given in the table, linear extrapolation is permitted.

Table 16 – Minimum CREEPAGE DISTANCES providing MEANS OF OPERATOR PROTECTION ^a

Replace existing Table 16 with the following:

WORKING VOLTAGE V r.m.s or d.c.	Spacing for one MEANS OF OPERATOR PROTECTION						
	Pollution degree 1	Pollution degree 2			Pollution degree 3		
	Material group	Material group			Material group		
	I, II, IIIa, IIIb	I	II	IIIa or IIIb	I	II	IIIa or IIIb
25	Use the AIR CLEARANCE from the appropriate table	0,5	0,5	0,5	1,3	1,3	1,3
50		0,6	0,9	1,2	1,5	1,7	1,9
100		0,7	1,0	1,4	1,8	2,0	2,2
125		0,8	1,1	1,5	1,9	2,1	2,4
150		0,8	1,1	1,6	2,0	2,2	2,5
200		1,0	1,4	2,0	2,5	2,8	3,2
250		1,3	1,8	2,5	3,2	3,6	4,0
300		1,6	2,2	3,2	4,0	4,5	5,0
400		2,0	2,8	4,0	5,0	5,6	6,3
600		3,2	4,5	6,3	8,0	9,6	10,0
800		4,0	5,6	8,0	10,0	11,0	12,5
1 000		5,0	7,1	10,0	12,5	14,0	16,0
NOTE 1 Minimum CREEPAGE DISTANCES for two MEANS OF OPERATOR PROTECTION are obtained by doubling the values in this table.							
NOTE 2 A CREEPAGE DISTANCE cannot be less than the required air clearance. See 8.9.1.4.							
NOTE 3 For WORKING VOLTAGE values greater than 1 000 V, refer to Table A.2.							
^a CREEPAGE DISTANCES within this table apply to all situations.							

8.9.2 * Application

In existing list item a), replace "result in a HAZARDOUS SITUATION" with "result in a HAZARDOUS SITUATION described in 13.1".

8.9.4 * Measurement of CREEPAGE DISTANCES and AIR CLEARANCES

Replace the existing first paragraph with the following:

Compliance is checked by measurement taking into account the rules in Figure 22 to Figure 31 (inclusive). In each figure, the dashed line (— — —) represents AIR CLEARANCE and the shaded bar (▨) represents CREEPAGE DISTANCE.

The minimum spacing (X) for grooves transverse to the CREEPAGE DISTANCE in Figure 23 to Figure 25 and Figure 27 to Figure 31 (inclusive) that are considered a MEANS OF OPERATOR PROTECTION may be adjusted based on pollution degree if the minimum AIR CLEARANCE is 3 mm or more. The minimum width of the groove is:

- 0,25 mm for pollution degree 1
- 1,0 mm for pollution degree 2
- 1,5 mm for pollution degree 3

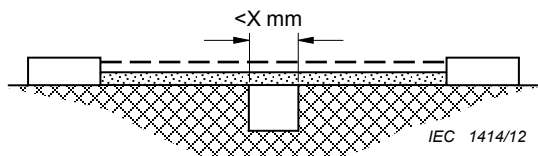
If the specified minimum AIR CLEARANCE is less than 3 mm, the minimum spacing (X) for grooves transverse to the CREEPAGE DISTANCE is the lesser of:

- the relevant value specified in the previous paragraph, or
- one third of the specified minimum AIR CLEARANCE.

The minimum spacing (X) for a groove transverse to a CREEPAGE DISTANCE that is considered a MEANS OF PATIENT PROTECTION is 1 mm for pollution degree 1 and pollution degree 2, and 1,5 mm for pollution degree 3.

In the existing seventh paragraph, replace "Table 11 to Table 16 (inclusive)" with "Table 12 to Table 16 (inclusive)".

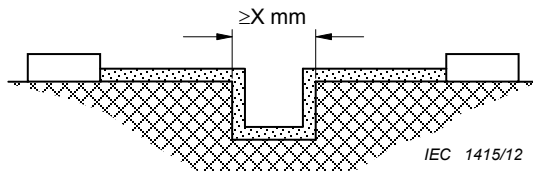
Replace existing Figures 23, 24 and 25 with the following:



Condition: Path under consideration includes a parallel- or converging-sided groove of any depth with a width less than X mm.

Rule: CREEPAGE DISTANCE and AIR CLEARANCE are measured directly across the groove as shown.

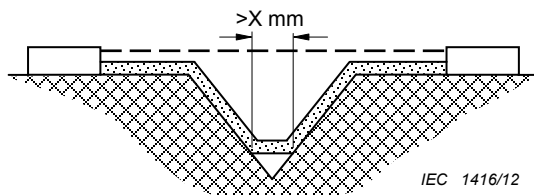
Figure 23 – CREEPAGE DISTANCE and AIR CLEARANCE – Example 2



Condition: Path under consideration includes a parallel-sided groove of any depth and equal to or more than X mm.

Rule: AIR CLEARANCE is the "line of sight" distance. CREEPAGE DISTANCE path follows the contour of the groove.

Figure 24 – CREEPAGE DISTANCE and AIR CLEARANCE – Example 3



Condition: Path under consideration includes a V-shaped groove with a width greater than X mm and an internal angle of less than 80° .

Rule: AIR CLEARANCE is the "line of sight" distance. CREEPAGE DISTANCE path follows the contour of the groove but "short circuits" the bottom of the groove by an X mm link.

Figure 25 – CREEPAGE DISTANCE and AIR CLEARANCE – Example 4

Replace existing Figures 27, 28, 29, 30 and 31 with the following:

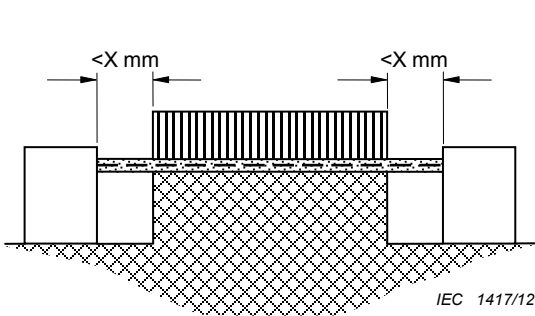


Figure 27 – CREEPAGE DISTANCE and AIR CLEARANCE – Example 6

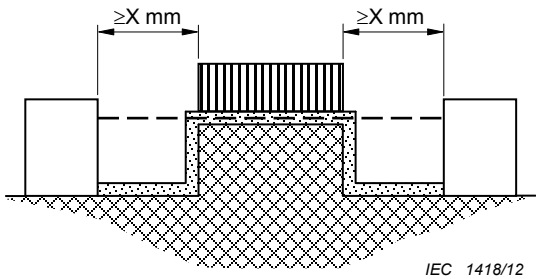


Figure 28 – CREEPAGE DISTANCE and AIR CLEARANCE – Example 7

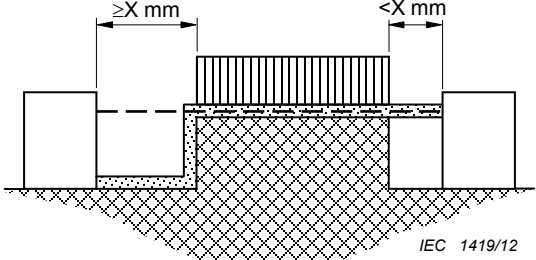


Figure 29 – CREEPAGE DISTANCE and AIR CLEARANCE – Example 8

Condition: Path under consideration includes an uncemented joint (see 8.9.3) with grooves less than X mm wide on each side.

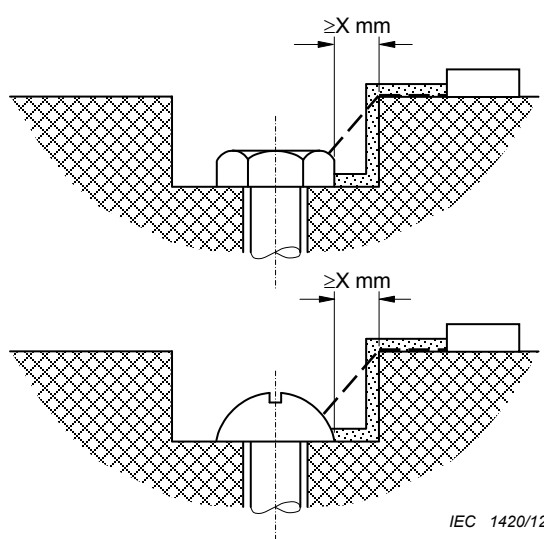
Rule: CREEPAGE DISTANCE and AIR CLEARANCE path are the “line of sight” distance shown.

Condition: Path under consideration includes an uncemented joint (see 8.9.3) with grooves equal to or more than X mm wide on each side.

Rule: AIR CLEARANCE is the “line of sight” distance. CREEPAGE DISTANCE path follows the contour of the groove.

Condition: Path under consideration includes an uncemented joint (see 8.9.3) with a groove on one side less than X mm wide and the groove on the other side equal to or more than X mm wide.

Rule: AIR CLEARANCE and CREEPAGE DISTANCE are as shown.

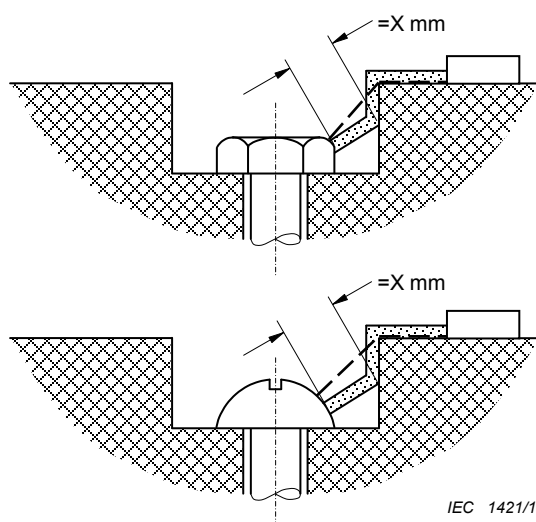


IEC 1420/12

Figure 30 – CREEPAGE DISTANCE and AIR CLEARANCE – Example 9

Condition: Gap between head of screw and wall of recess wide enough to be taken into account.

Rule: The AIR CLEARANCE is the shortest distance to any point on the head of the screw. CREEPAGE DISTANCE path follows the surface.



IEC 1421/12

Figure 31 – CREEPAGE DISTANCE and AIR CLEARANCE – Example 10

Condition: Gap between head of screw and wall of recess too narrow to be taken into account.

Rule: Measurement of CREEPAGE DISTANCE is from screw to wall at any point where the distance is equal to X mm. The AIR CLEARANCE is the shortest distance to any point on the head of the screw.

8.10 Components and wiring

8.10.2 * Fixing of wiring

In the existing first paragraph, replace "touching circuit points resulting in a HAZARDOUS SITUATION" with "touching parts resulting in a HAZARDOUS SITUATION described in 13.1".

In the existing third paragraph, replace "result in a HAZARDOUS SITUATION" with "result in a HAZARDOUS SITUATION described in 13.1".

Replace the compliance paragraph with the following:

Compliance is checked by inspection of the ME EQUIPMENT.

8.10.4 * Cord-connected HAND-HELD parts and cord-connected foot-operated control devices (see also 15.4.7)

8.10.4.2 Connection cords

Replace the existing first paragraph with the following:

The connection and anchorage at both ends of a flexible cord to a HAND-HELD or foot-operated control device of ME EQUIPMENT shall comply with the requirements specified for POWER SUPPLY CORDS in 8.11.3, if breaking free or shorting between the conductors could result in a HAZARDOUS SITUATION described in 13.1. This requirement also applies to other HAND-HELD parts if disturbance or breaking of one or more of the connections could result in a HAZARDOUS SITUATION described in 13.1.

8.10.5 * Mechanical protection of wiring

In existing list items a) and b), replace "could result in a HAZARDOUS SITUATION" with "could result in a HAZARDOUS SITUATION described in 13.1".

Replace the existing compliance paragraph with the following:

Compliance is checked by inspection and, where appropriate, by manual test.

8.11 MAINS PARTS, components and layout

8.11.1 Isolation from the SUPPLY MAINS

In existing list item a), after the second paragraph, insert the following:

For PERMANENTLY INSTALLED ME EQUIPMENT, the means provided to isolate its circuits electrically from the SUPPLY MAINS shall be capable of being locked in the off position if:

- reconnection would result in a HAZARDOUS SITUATION; or
- any OPERATOR including SERVICE PERSONNEL is unable to view the means of isolation from their intended position.

The locking mechanism may be in a SUPPLY MAINS switch provided by the RESPONSIBLE ORGANIZATION.

The requirements for the isolation device shall be specified in the ACCOMPANYING DOCUMENTS.

Replace existing list items e) and f) with the following:

- e) The actuator of a SUPPLY MAINS switch that is used to comply with 8.11.1 a) shall comply with IEC 60447.
- f) In non-PERMANENTLY INSTALLED ME EQUIPMENT that has no SUPPLY MAINS switch, a suitable plug device used to isolate ME EQUIPMENT from the SUPPLY MAINS shall be considered as complying with the requirements of 8.11.1 a). An APPLIANCE COUPLER or a flexible cord with a MAINS PLUG may be used.

8.11.5 Mains fuses and OVER-CURRENT RELEASES

Replace the existing final sentence of the second dash with the following:

The effect of short-circuit fault conditions in other circuits shall be VERIFIED before eliminating fuses or OVER-CURRENT RELEASES.

Replace the existing first paragraph following the note and the compliance paragraph with the following:

Justification for omission of fuses or OVER-CURRENT RELEASES shall be documented.

Compliance is checked by inspection of the ME EQUIPMENT and the MANUFACTURER'S documentation.

8.11.6 Internal wiring of the MAINS PART

Replace existing list item a) with the following:

- a) Internal wiring in a MAINS PART between the MAINS TERMINAL DEVICE or the APPLIANCE INLET and the protective devices shall have a cross-sectional area not less than the minimum required for the POWER SUPPLY CORD as specified in 8.11.3.3.

9 * Protection against MECHANICAL HAZARDS of ME EQUIPMENT and ME SYSTEMS

9.2 * HAZARDS associated with moving parts

Replace the existing title with the following:

9.2 * MECHANICAL HAZARDS associated with moving parts

9.2.1 * General

In the existing second paragraph, replace "protective measures" with "RISK CONTROL measures".

Replace the existing third paragraph with the following:

The RESIDUAL RISK associated with moving parts is considered acceptable if exposure is needed for the ME EQUIPMENT to perform its intended function, and RISK CONTROL measures have been implemented (e.g. warnings).

Renumber the note to Note 1 and insert the following:

NOTE 2 See ISO 14971:2007, subclauses 6.2 and 6.5.

9.2.2 Trapping zone

9.2.2.1 General

In the third dash of the existing first paragraph, replace "protective measures" with "other RISK CONTROL measures".

In the existing second paragraph, replace "protective measures" with "RISK CONTROL measures".

9.2.2.3 Safe distances

In the third line of the existing paragraph, replace "ISO 13852" with ISO 13857:2008".

9.2.2.4 * GUARDS and protective measures

Replace the existing title with the following:

9.2.2.4 * GUARDS and other RISK CONTROL measures

Table 20 – Acceptable gaps

In the existing table footnote ^a, replace "ISO 13852:1996" with "ISO 13857:2008".

9.2.2.4.1 Access to TRAPPING ZONES

In the existing first paragraph, replace "and protective measures:" with "or other RISK CONTROL measures (e.g. electro-mechanical):".

Immediately after the last dashed item in the list, insert the following note:

NOTE RISK CONTROL measures (e.g. electro-mechanical) addressed by this subclause are intended to include collision detection or collision avoidance systems, such as those employing light barrier(s) and similar feedback control(s).

9.2.2.4.3 Movable GUARDS

Replace the existing compliance paragraph with the following.

Compliance is checked by inspection of the ME EQUIPMENT and by conducting any applicable tests.

9.2.2.4.4 Protective measures

Replace the existing title with the following:

9.2.2.4.4 Other RISK CONTROL measures

Replace, in the first paragraph, the second and third dashes, and the compliance paragraph as follows.

Other RISK CONTROL measures (e.g. electro-mechanical) shall be designed and incorporated into the control system so that:

- once the ME EQUIPMENT has started to move, if the TRAPPING ZONE is reached, system movement shall stop; and
- if the RISK CONTROL measure is defeated in a SINGLE FAULT CONDITION, a second RISK CONTROL measure shall be provided, such as one or more emergency stopping device(s) (see 9.2.4), or the ME EQUIPMENT shall otherwise be SINGLE FAULT SAFE (see 4.7).

Compliance is checked by the following as necessary:

- *inspection of the ME EQUIPMENT;*
- *examination of the construction and circuits;*
- *conducting any applicable tests including, if necessary, tests under SINGLE FAULT CONDITION.*

9.2.2.5 * Continuous activation

Replace the existing first sentence of the first paragraph with the following:

Where it is impractical to make the TRAPPING ZONE inaccessible, continuous activation may be used as a RISK CONTROL measure.

A TRAPPING ZONE is not considered to present a MECHANICAL HAZARD if:

Replace existing list item c) and the associated compliance paragraph with the following:

- c) the continuous activation system is defeated in a SINGLE FAULT CONDITION, then a second RISK CONTROL measure shall be provided, such as one or more emergency stopping device(s) (see 9.2.4), or the ME EQUIPMENT shall otherwise be SINGLE FAULT SAFE (see 4.7).

Compliance is checked by the following as necessary:

- *inspection of the ME EQUIPMENT;*
- *examination of the construction and circuits;*
- *conducting any applicable tests including, if necessary, tests under SINGLE FAULT CONDITION.*

9.2.2.6 * Speed of movement(s)

Replace the existing text of the subclause with the following:

The speed of movement(s) that position parts of the ME EQUIPMENT or PATIENT, where contact with the ME EQUIPMENT could result in an unacceptable RISK, shall be limited so that the OPERATOR will have adequate control of the movement.

The overtravel (stopping distance) of such movement, occurring after operation of a control to stop the movement, shall not result in an unacceptable RISK.

Compliance is checked by the following as necessary:

- *inspection of the overtravel (stopping distance) calculations and evaluation;*
- *any functional tests.*

NOTE The overtravel (stopping distance) calculations and evaluation are part of the RISK MANAGEMENT FILE.

9.2.3 * Other HAZARDS associated with moving parts

Replace the existing title with the following.

9.2.3 * Other MECHANICAL HAZARDS associated with moving parts

9.2.3.1 Unintended movement

Replace the existing text of the subclause with the following:

Controls shall be so positioned, recessed, or protected by other means so that they cannot be accidentally actuated, unless for the intended PATIENT, the USABILITY ENGINEERING PROCESS concludes otherwise (e.g. PATIENT with special needs) or activation does not result in an unacceptable RISK.

Compliance is checked by inspection of ME EQUIPMENT, and, if the control is part of the PRIMARY OPERATING FUNCTIONS, inspection of the USABILITY ENGINEERING FILE.

9.2.3.2 Overtravel

Replace the existing title and text of the subclause with the following:

9.2.3.2 Overtravel end stops

Overtravel past range limits of ME EQUIPMENT parts shall be prevented. End stops or other stopping means shall be provided to act as the ultimate travel limiting measure.

Such means shall have the mechanical strength to withstand the intended loading in NORMAL USE and reasonably foreseeable misuse.

Compliance is checked by inspection of the ME EQUIPMENT and the following test:

The ME EQUIPMENT is

- *loaded with the SAFE WORKING LOAD,*
- *unloaded, or*

– loaded to any intermediate level that is likely to provide the most severe test result.

The moving part is to be driven against each end stop or other mechanical means for the number of cycles, operating speed and test conditions as specified in Table 33. The end stops or other mechanical means required shall be capable of performing their intended function upon completion of the test.

Table 33 – Test conditions for overtravel end stop test

Construction	Number of cycles	Test condition
1. Motor driven: No range limit system provided ^a	6 000	Run at maximum speed
2. Motor driven: Non-independent range limit system or systems provided ^{a, b}	50	Defeat all switches simultaneously, and run at maximum speed
3. Motor driven: Two or more independent range limit systems ^{a, b}	1	Defeat all switches simultaneously, and run at maximum speed
4. Manually driven or manually driven, power assisted	50	Run at any speed, including reasonably foreseeable misuse
^a A range limit system consists of all components required to stop motion, for example, it may consist of (1) a limit switch, (2) sensing circuits, and (3) related mechanical actuating mechanism. ^b To qualify as an independent range limit system, each system shall, in addition to the criteria in footnote ^a , comply with both of the following: - the system is capable of de-energizing the motor(s) directly; that is, the switch or motor controller circuit interrupts the motor's rotor or stator current, or both, and - the system provides a means by which a malfunction of one range limit system is made obvious to the OPERATOR. This may be an audible, visual or otherwise discernible indicator.		

9.2.4 * Emergency stopping devices

In existing list item e), replace the second occurrence of "HAZARD" with "MECHANICAL HAZARD".

In existing list item i), delete "DB:".

Replace the existing compliance paragraph with the following:

Compliance is checked by inspection of the ME EQUIPMENT and the RISK MANAGEMENT FILE, and by functional tests.

9.2.5 * Release of PATIENT

In the first paragraph, replace "a protective measure" with "a RISK CONTROL measure".

Replace the existing compliance paragraph with the following:

Compliance is checked by inspection of the ME EQUIPMENT and the RISK MANAGEMENT FILE, and by functional tests.

9.3 * HAZARD associated with surfaces, corners and edges

Replace the existing title and text of the subclause with the following:

9.3 * MECHANICAL HAZARD associated with surfaces, corners and edges

Rough surfaces, sharp corners and edges of ME EQUIPMENT that could cause injury or damage shall be avoided or covered.

In particular, attention shall be paid to flange or frame edges and the removal of burrs.

Compliance is checked by inspection of the ME EQUIPMENT.

NOTE 1 If inspection is insufficient to determine sharpness of edges or burrs, the rationale for 9.3 provides a reference to an optional functional test.

NOTE 2 A sharp edge MECHANICAL HAZARD could cut wire insulation which could lead to an electrical HAZARD. This requirement is intended to cover all these HAZARDS.

9.4 * Instability HAZARDS

9.4.1 General

Replace the existing first paragraph and the note with the following:

ME EQUIPMENT and its parts, other than FIXED ME EQUIPMENT, intended to be placed on a surface such as a floor or a table in NORMAL USE shall not overbalance (tip over) or move unexpectedly.

NOTE HAND-HELD parts of FIXED ME EQUIPMENT are intended to be tested.

9.4.2 * Instability – overbalance

9.4.2.1 Instability in transport position

Add a new note following the first paragraph:

NOTE The meaning of transport in this subclause is moving ME EQUIPMENT from room to room during NORMAL USE.

9.4.2.2 Instability excluding transport

Replace the existing subclause title with the following:

9.4.2.2 Instability excluding transport position

9.4.2.3 Instability from horizontal and vertical forces

Replace the existing text of the subclause with the following:

- a) * ME EQUIPMENT or its parts having a mass of 25 kg or more other than FIXED ME EQUIPMENT that is intended to be used on the floor shall be permanently marked with a CLEARLY LEGIBLE warning of this RISK, e.g. by use of safety sign ISO 7010-P017 (see Table D.2, safety sign 5), or it shall not overbalance due to being pushed, leaned, rested upon etc.

If marking is provided because the ME EQUIPMENT overbalances, the marking shall be visible during NORMAL USE, but not on surfaces for which pushing is associated with NORMAL USE (e.g. surfaces with handles).

Compliance is checked by inspection of the marking provided or the following test:

Prior to the test, the ME EQUIPMENT is prepared as described in 9.4.2.2. The ME EQUIPMENT is placed on a horizontal plane and a force equal to 15 % of its weight, but not more than 150 N, is applied in any direction, except a direction having an upward component. Unless otherwise marked, the force is applied at any point of the ME EQUIPMENT but not exceeding 1,5 m from the floor. The ME EQUIPMENT is prevented from sliding on the floor by a horizontal obstruction, not exceeding 20 mm height, which is fastened flat on the floor. If the application of the test force results in lateral movement of the ME EQUIPMENT, increase the height of the obstruction to the minimum extent necessary to prevent lateral movement. ME EQUIPMENT without marking shall not overbalance.

- b) ME EQUIPMENT or its parts, other than FIXED ME EQUIPMENT, that is intended to be used on the floor or on a table shall be permanently marked with a CLEARLY LEGIBLE warning of this RISK, e.g. by use of safety signs ISO 7010-P018 or ISO 7010-P019 as appropriate (see Table D.2, safety signs 6 and 7), or it shall not overbalance due to being sat or stepped upon.

NOTE Requirements for PATIENT support surfaces are found in 9.8.3.

If marking is provided because the ME EQUIPMENT overbalances, the marking shall be visible during potential stepping or sitting misuse.

Compliance is checked by inspection of the marking provided or the following test:

Prior to the test the ME EQUIPMENT is prepared as described in 9.4.2.2. The ME EQUIPMENT is placed on a horizontal plane and a constant downward force of 800 N is applied at the point of maximum moment to any working surface, excluding PATIENT support surfaces, offering an obvious foothold or sitting surface of a minimum 20 cm by 20 cm area, and at a height not exceeding 1 m from the floor. ME EQUIPMENT without marking shall not overbalance.

9.4.2.4 * Castors and wheels

9.4.2.4.3 Movement over a threshold

Add an asterisk to the title and replace the existing text of the subclause with the following:

MOBILE ME EQUIPMENT exceeding 45 kg shall be able to pass over a 10 mm threshold. Passing over a 10 mm threshold shall not result in overbalancing.

Compliance is checked by the following test:

The ME EQUIPMENT is configured in transport position with any SAFE WORKING LOAD in place as indicated in the ACCOMPANYING DOCUMENTS. The ME EQUIPMENT is moved as in NORMAL USE 10 times in forward direction over (up and down) a solid vertical plane obstruction that is affixed flat on the floor. The obstruction shall have a rectangular cross section of 10 mm $\pm$ 0,5 mm high and at least 80 mm wide with a radius of 2 mm $\pm$ 0,1 mm at the top edges. The method of passing over the obstruction has to be applied in accordance with the instructions in the ACCOMPANYING DOCUMENTS, or if no instructions are given, by the following test.

All wheels and castors are to impact the obstruction at a speed of 0,8 m/s $\pm$ 0,1 m/s for manual MOBILE ME EQUIPMENT, or, for motor driven MOBILE ME EQUIPMENT, the maximum speed capable of being maintained. Manual MOBILE ME EQUIPMENT is propelled by a force acting at its handle.

It is unacceptable for ME EQUIPMENT to be unable to go over (up) the obstruction (due to small wheel diameter, for example). Overbalancing constitutes a failure. BASIC SAFETY and ESSENTIAL PERFORMANCE shall be maintained.

NOTE Examples of damage that can affect BASIC SAFETY include the reduction of CREEPAGE DISTANCES and AIR CLEARANCES below those specified in 8.9, access to parts that exceed limits in 8.4, or access to moving parts that could cause HARM.

Assessment criteria that can be useful in determining if this test has resulted in a loss of BASIC SAFETY include:

- those in Clause 8 and 11.6;
- the dielectric strength test as specified in 8.8.3 to evaluate the integrity of solid insulation providing a MEANS OF PROTECTION; and
- measurement of CREEPAGE DISTANCES or AIR CLEARANCES to compare the values with the minimum distances specified in 8.9. Small chips that do not adversely affect the protection against electric shock or moisture can normally be ignored.

9.4.3 Instability from unwanted lateral movement (including sliding)

Add an asterisk before the subclause title.

9.4.3.1 Instability in transport

Replace the existing subclause title with the following:

9.4.3.1 Instability in transport position

Replace existing first paragraph of list item c) with the following:

- c) * MOBILE ME EQUIPMENT shall be provided with wheel locks or with a braking system to prevent unwanted movement on an incline of 10° when in its transport position.

Delete the last sentence of the existing second compliance paragraph.

9.4.3.2 Instability excluding transport

Replace the existing title and the first paragraph of list item a) with the following:

9.4.3.2 Instability excluding transport position

- a) MOBILE ME EQUIPMENT shall be provided with wheel locks or with a braking system to prevent unwanted movement on an incline of 5° when in any position excluding transport position.

Delete the last sentence of the existing second compliance paragraph.

Replace the existing first paragraph of list item b) with the following:

- b) MOBILE ME EQUIPMENT shall be provided with wheel locks or with a braking system to prevent unwanted movement from lateral forces.

Replace the existing second compliance paragraph of list item b) with the following:

Prior to the test, the ME EQUIPMENT is prepared as described in 9.4.2.2. The ME EQUIPMENT is placed on a horizontal plane with the SAFE WORKING LOAD in place, and the locking device (e.g. brakes) activated. If castors are incorporated, they are positioned in their worst-case position. A force equal to 15 % of the weight of the unit, but not more than 150 N, is applied in any direction, except a direction having an upwards component, at the highest point of the ME EQUIPMENT that does not lead to overbalancing but not exceeding 1,5 m from the floor. Following the initial elastic movement, initial creepage and initial pivoting of castors, any further movement of the ME EQUIPMENT greater than 50 mm (in relation to the horizontal plane) constitutes a failure.

9.4.4 Grips and other handling devices

In existing list item a), replace "HAZARDS" with "unacceptable RISK".

9.6 Acoustic energy (including infra- and ultrasound) and vibration

9.6.1 * General

Replace the existing compliance paragraph with the following:

Compliance is checked by the tests in 9.6.2 and 9.6.3, and, if necessary, by inspection of the RISK MANAGEMENT FILE (taking into account the audibility of auditory ALARM SIGNALS and PATIENT sensitivity).

9.6.2 * Acoustic energy

9.6.2.1 Audible acoustic energy

In the existing first paragraph, place "alarm signals" in SMALL CAPITALS.

In existing second dash, replace "140 dB un-weighted sound pressure level" with "140 dBC (peak) sound pressure level".

In existing Note 2, place "alarm signals" in SMALL CAPITALS.

In existing Note 3, replace "80 dB(A)" with "80 dBA".

Add a new list item e) as follows:

- e) *When sound measurements in a test room are not feasible (e.g. for a large PERMANENTLY INSTALLED ME EQUIPMENT), measurements may be done in situ.*

9.7 * Pressure vessels and parts subject to pneumatic and hydraulic pressure

9.7.4 Pressure rating of ME EQUIPMENT parts

Replace the existing compliance paragraph with the following:

Compliance is checked by inspection of the MANUFACTURER'S data for the component, inspection of the ME EQUIPMENT, and, where necessary, by functional test.

9.7.5 * Pressure vessels

Replace the last paragraph of the existing compliance statement with the following:

Where unmarked pressure vessels and pipes (e.g. those with no national certification) cannot be hydraulically tested, integrity is verified by other suitable tests, e.g. pneumatic using suitable media, at the same test pressure as for the hydraulic test.

9.7.6 Pressure-control device

Replace the existing compliance paragraph with the following:

Compliance is checked by inspection of the MANUFACTURER'S data for the component, inspection of the ME EQUIPMENT, and, where necessary, by functional test.

9.8 * HAZARDS associated with support systems

Replace the existing title with the following:

9.8 * MECHANICAL HAZARDS associated with support systems

9.8.1 General

In the existing third dash, replace "HAZARDS" with "MECHANICAL HAZARDS".

9.8.2 TENSILE SAFETY FACTOR

Add an asterisk before the subclause title.

Replace the existing third paragraph with the following:

If testing is needed to demonstrate compliance with 9.8.1 or 9.8.2, a test load equal to the TOTAL LOAD times the required TENSILE SAFETY FACTOR is gradually applied to the support assembly under test. The support assembly under test is to be in equilibrium after 1 min, or otherwise not result in an unacceptable RISK.

9.8.3 * Strength of PATIENT or OPERATOR support or suspension systems

9.8.3.1 General

Replace the existing first paragraph with the following:

ME EQUIPMENT parts serving for support or immobilization of PATIENTS shall be designed and manufactured so there is no unacceptable RISK of physical injuries or of accidental loosening of fixings.

Replace the existing compliance paragraph with the following:

Compliance is checked by inspection of the ME EQUIPMENT (including markings), the ACCOMPANYING DOCUMENTS, the MANUFACTURER'S data for the component, the RISK MANAGEMENT FILE, and, where necessary, functional test.

9.8.3.2 * Static forces due to loading from persons

Change the existing note following the first paragraph to "Note 1".

In existing list item a), replace the compliance paragraphs with the following:

Compliance is checked by inspection of the ME EQUIPMENT, the specifications of materials used and the processing specifications for these materials, and the following test:

Prior to performing these tests, the PATIENT support/suspension system is positioned horizontally in its most disadvantageous position in NORMAL USE.

A mass equal to two times 135 kg or two times the intended person load, whichever is greater, is applied to the foot rest over an area of 0,1 m² for 1 min. After the test, a foot rest and its fixings that shows any damage or permanent deflection from normal greater than 5° constitutes a failure. BASIC SAFETY and ESSENTIAL PERFORMANCE shall be maintained.

NOTE 2 Examples of damage that can affect BASIC SAFETY include the reduction of CREEPAGE DISTANCES and AIR CLEARANCES below those specified in 8.9, access to parts that exceed limits in 8.4, or access to moving parts that could cause HARM.

Assessment criteria that can be useful in determining if this test has resulted in a loss of BASIC SAFETY include:

- those in Clause 8 and 11.6;
- the dielectric strength test as specified in 8.8.3 to evaluate the integrity of solid insulation providing a MEANS OF PROTECTION; and
- measurement of CREEPAGE DISTANCES or AIR CLEARANCES to compare the values with the minimum distances specified in 8.9. Small chips that do not adversely affect the protection against electric shock or moisture can normally be ignored.

In existing list item b), replace the compliance paragraphs with the following:

Compliance is checked by inspection of the ME EQUIPMENT, the specifications of materials used and the processing specifications for these materials, and the following test:

Prior to performing these tests, the PATIENT support/suspension system is positioned horizontally in its most disadvantageous position in NORMAL USE.

A mass of 60 % of the part of the SAFE WORKING LOAD representing the PATIENT or OPERATOR, as defined in the instructions for use, or a minimum 80 kg, is placed on the support/suspension system with the centre of the load 60 mm from the outer edge of the support/suspension system for at least one minute. Any permanent deflection of the

support/suspension system from normal greater than 5° constitutes a failure. BASIC SAFETY and ESSENTIAL PERFORMANCE shall be maintained.

NOTE 3 Examples of damage that can affect BASIC SAFETY include the reduction of CREEPAGE DISTANCES and AIR CLEARANCES below those specified in 8.9, access to parts that exceed limits in 8.4, or access to moving parts that could cause HARM.

Assessment criteria that can be useful in determining if this test has resulted in a loss of BASIC SAFETY include:

- those in Clause 8 and 11.6;
- the dielectric strength test as specified in 8.8.3 to evaluate the integrity of solid insulation providing a MEANS OF PROTECTION; and
- measurement of CREEPAGE DISTANCES or AIR CLEARANCES to compare the values with the minimum distances specified in 8.9. Small chips that do not adversely affect the protection against electric shock or moisture can normally be ignored.

9.8.3.3 * Dynamic forces due to loading from persons

Replace the text of the subclause with the following:

Where dynamic forces (due to sitting down, standing up, PATIENT handling PROCESS or the like) can be exerted on ME EQUIPMENT parts intended to support or suspend a PATIENT or OPERATOR in NORMAL USE, the ME EQUIPMENT shall maintain BASIC SAFETY and ESSENTIAL PERFORMANCE.

Compliance is checked by the following test:

Prior to performing this test, the PATIENT support/suspension system is positioned horizontally in its most disadvantageous position in NORMAL USE.

For the area of support/suspension where a PATIENT or OPERATOR can sit, the top reservoir of the body upper-carriage module described in Figure 33 is loaded with the appropriate mass to obtain the SAFE WORKING LOAD representing the PATIENT or OPERATOR as defined in the ACCOMPANYING DOCUMENTS. The body upper-carriage module with appropriate mass is dropped from a distance of 150 mm above the seat area. BASIC SAFETY and ESSENTIAL PERFORMANCE shall be maintained.

NOTE Examples of damage that can affect BASIC SAFETY include the reduction of CREEPAGE DISTANCES and AIR CLEARANCES below those specified in 8.9, access to parts that exceed limits in 8.4, or access to moving parts which could cause HARM.

Assessment criteria that can be useful in determining if this test has resulted in a loss of BASIC SAFETY include:

- those in Clause 8 and 11.6;
- the dielectric strength test as specified in 8.8.3 to evaluate the integrity of solid providing a MEANS OF PROTECTION; and
- measurement of CREEPAGE DISTANCES or AIR CLEARANCES to compare the values with the minimum distances specified in 8.9. Small chips that do not adversely affect the protection against electric shock or moisture can normally be ignored.

9.8.4 * Systems with MECHANICAL PROTECTIVE DEVICES

9.8.4.1 General

In list item b), replace the existing compliance paragraph with the following:

Compliance is checked by the following as necessary:

- *inspection of the overtravel (stopping distance from the time of engagement of the MECHANICAL PROTECTIVE DEVICE to the time of no further movement) calculations and evaluation;*
- *any functional tests.*

NOTE The overtravel (stopping distance from the time of engagement of the MECHANICAL PROTECTIVE DEVICE to the time of no further movement) calculations and evaluation are part of the RISK MANAGEMENT FILE.

Replace the existing figure note and title of Figure 33 with the following:

NOTE The top reservoir of the body upper-carriage module is formed of wood, metal or a similar material. The reservoir is intended to retain the appropriate human body mass, typically with high density material (e.g. lead). The bottom portion is foam. The resiliency or spring factor of the foam (ILD or IFD ratings) is not specified, as with a large mass being dropped, the foam properties are likely inconsequential. The foam is cylindrical, rather than spherical.

Figure 33 – Body upper-carriage module

9.8.4.3 MECHANICAL PROTECTIVE DEVICE intended for single activation

In the existing third dash, replace "safety sign 7010-W001" with "safety sign ISO 7010-W001".

Replace the existing first dashed item of the compliance paragraph with the following:

- *by inspection of the ME EQUIPMENT, the ACCOMPANYING DOCUMENTS and the specifications of materials used and the processing specifications for these materials;*

9.8.5 Systems without MECHANICAL PROTECTIVE DEVICES

Replace the existing compliance paragraph with the following:

Compliance is checked by inspection of the ME EQUIPMENT, the design documentation and the RISK MANAGEMENT FILE.

10 * Protection against unwanted and excessive radiation HAZARDS

10.1 X-Radiation

10.1.1 * ME EQUIPMENT not intended to produce diagnostic or therapeutic X-radiation

Replace the existing first paragraph with the following:

For ME EQUIPMENT not intended to produce X-radiation for diagnostic or therapeutic purposes, but that might produce ionizing radiation, the AIR KERMA rate shall not exceed 5 $\mu\text{Gy/h}$ at a distance of 5 cm from a surface of the ME EQUIPMENT taking account of the background radiation.

If the INTENDED USE of the ME EQUIPMENT requires a permanent proximity to a PATIENT, the resulting annual exposure should be made acceptable taking into account the irradiated body part and national regulations and/or international recommendations.

Replace the existing fifth paragraph of the compliance statement and Note 3 with the following:

Any measurement exceeding 5 $\mu\text{Gy/h}$ adjusted for the level of background radiation constitutes a failure.

NOTE 3 This test PROCEDURE is equivalent to that in Annex H of IEC 60950-1: 2005.

10.1.2 ME EQUIPMENT intended to produce diagnostic or therapeutic X-radiation

Replace the existing text of the subclause with the following:

Unintended X-radiation from ME EQUIPMENT designed to produce diagnostic or therapeutic X-radiation shall be reduced as far as possible by application of applicable particular and collateral standards, or in the absence of these standards by application of the RISK MANAGEMENT PROCESS.

For intended X-radiation, also see 12.4.5.2 and 12.4.5.3.

Compliance is checked by application of applicable particular and collateral standards or inspection of the RISK MANAGEMENT FILE.

10.3 * Microwave radiation

Replace the existing text of the subclause with the following:

The power density of unintended microwave radiation at frequencies between 1 GHz and 100 GHz shall not exceed 10 W/m^2 at any point 50 mm away from a surface of the ME EQUIPMENT under reference test conditions. This requirement does not apply to parts of the apparatus where microwave radiation is propagated intentionally, for example, at waveguide output ports.

NOTE This requirement is equivalent to that in subclause 12.4 of IEC 61010-1:2001.

Compliance is checked by review of MANUFACTURER'S calculations and, if necessary, the following test:

The ME EQUIPMENT is operated at the most unfavourable RATED MAINS VOLTAGE and with any control adjusted so as to give maximum microwave radiation whilst maintaining the ME EQUIPMENT in NORMAL USE.

Internal pre-set controls not intended to be adjusted during the expected SERVICE LIFE of the ME EQUIPMENT are not considered.

Measurements are made at a distance of 50 mm from any surface to which OPERATORS other than SERVICE PERSONNEL:

- can gain access without the use of a TOOL;*
- are deliberately provided with the means of access; or*
- are instructed to enter regardless of whether or not a TOOL is needed to gain access.*

Any measurement exceeding 10 W/m^2 under reference test conditions constitutes a failure.

10.4 * Lasers and light emitting diodes (LEDs)

Replace the title and existing text of the subclause with the following:

10.4 * Lasers

For lasers that produce or amplify electromagnetic radiation in the wavelength range from 180 nm to 1 mm, the relevant requirements of IEC 60825-1:2007 shall apply. If laser light barriers or similar products are used within equipment, they shall comply with the requirements of IEC 60825-1:2007.

NOTE For laser equipment that is intended for use on humans or animals for surgical, therapeutic, medical diagnostic, cosmetic, or veterinary applications and classified as a Class 3b or Class 4 laser product as defined in IEC 60825-1, see also IEC 60601-2-22 [59].

Compliance is checked by following the relevant PROCEDURES of IEC 60825-1:2007.

11 Protection against excessive temperatures and other HAZARDS

11.1 * Excessive temperatures in ME EQUIPMENT

11.1.2 * Temperature of APPLIED PARTS

11.1.2.2 * APPLIED PARTS not intended to supply heat to a PATIENT

Replace the existing text of the subclause with the following:

The limits of Table 24 shall apply in both NORMAL CONDITION and SINGLE FAULT CONDITION. If the surface temperature of an APPLIED PART exceeds 41 °C:

- the maximum temperature shall be disclosed in the instructions for use;
- the conditions for safe contact, e.g. duration or condition of the PATIENT, shall be disclosed; and
- the clinical effects with respect to characteristics such as body surface, maturity of PATIENTS, medications being taken or surface pressure shall be determined and documented in the RISK MANAGEMENT FILE.

Where 41°C is not exceeded, no justification is required.

If analyses documented in the RISK MANAGEMENT FILE demonstrate that APPLIED PART temperatures cannot be affected by operation of the ME EQUIPMENT including in SINGLE FAULT CONDITIONS, measurement of APPLIED PART temperature according to 11.1.3 is not required.

Surfaces of APPLIED PARTS that are cooled below ambient temperatures can also result in an unacceptable RISK and shall be evaluated as part of the RISK MANAGEMENT PROCESS.

11.2.2.1 Risk of fire in an OXYGEN RICH ENVIRONMENT

In existing list item b), 3), replace the compliance statement with the following:

Compliance is checked by visual inspection and by inspection of the documentation provided by the MANUFACTURER including the RISK MANAGEMENT FILE

11.6 Overflow, spillage, leakage, ingress of water or particulate matter, cleaning, disinfection, sterilization and compatibility with substances used with the ME EQUIPMENT

11.6.2 * Overflow in ME EQUIPMENT

Replace the existing text of the subclause with the following:

If ME EQUIPMENT incorporates a reservoir or liquid storage chamber that is liable to be overfilled or to overflow in NORMAL USE, liquid overflowing from the reservoir or chamber shall not wet any MEANS OF PROTECTION that is liable to be adversely affected by such a liquid, nor result in the loss of BASIC SAFETY or ESSENTIAL PERFORMANCE.

If the maximum fill level is indicated by marking on the ME EQUIPMENT and a warning or safety notice is given, no HAZARDOUS SITUATION (as specified in 13.1) or unacceptable RISK due to overflow shall develop if the reservoir or liquid storage chamber is filled to its maximum capacity and the TRANSPORTABLE ME EQUIPMENT is tilted through an angle of 10°, or for MOBILE ME EQUIPMENT exceeding 45 kg, if it is moved over a threshold as described in 9.4.2.4.3.

If no warning or safety notice is given regarding the maximum fill level, no HAZARDOUS SITUATION (as specified in 13.1) or unacceptable RISK due to overflow shall develop if the reservoir or liquid storage chamber is filled to 15 % above the maximum capacity and the TRANSPORTABLE ME EQUIPMENT is tilted through an angle of 10°, or in MOBILE ME EQUIPMENT exceeding 45 kg, if it is moved over a threshold as described in 9.4.2.4.3.

Compliance is checked by the following:

If warnings or safety notices regarding overfilling are marked on the TRANSPORTABLE ME EQUIPMENT, the liquid reservoir is filled to the indicated maximum level.

If there is no warning or safety notice regarding overfilling marked on the TRANSPORTABLE ME EQUIPMENT, the liquid reservoir is filled completely and subsequently a further quantity equal to 15 % of the capacity of the reservoir is added poured in steadily over a period of 1 min.

TRANSPORTABLE ME EQUIPMENT is subsequently tilted through an angle of 10° in the least favourable direction(s) (if necessary with refilling) starting from the position of NORMAL USE.

MOBILE ME EQUIPMENT exceeding 45 kg is moved over a threshold as described in 9.4.2.4.3.

After these PROCEDURES, the ME EQUIPMENT is to pass the appropriate dielectric strength and LEAKAGE CURRENT tests and is to show no signs of wetting of uninsulated electrical parts or electrical insulation of parts that could result in the loss of BASIC SAFETY or ESSENTIAL PERFORMANCE in NORMAL CONDITION or in combination with a SINGLE FAULT CONDITION (based on a visual inspection).

11.6.3 * Spillage on ME EQUIPMENT and ME SYSTEM

Replace the existing title and text of the subclause with the following:

11.6.3 * Spillage on ME EQUIPMENT and ME SYSTEMS

ME EQUIPMENT and ME SYSTEMS requiring the handling of liquids in NORMAL USE, including ME EQUIPMENT or ME SYSTEMS used in an environment where the PROCESS has determined that spillage on the ME EQUIPMENT is likely to occur, shall be so constructed that spillage does not wet parts that are likely to result in the loss of BASIC SAFETY or ESSENTIAL PERFORMANCE.

Compliance is checked by inspection of the RISK MANAGEMENT FILE and by the following test:

The ME EQUIPMENT is positioned according to 5.4 a). A quantity of liquid is poured steadily on a point on the top of the ME EQUIPMENT. The type of liquid, volume, duration of the spill, and location (point) are determined through RISK ANALYSIS. Test conditions that simulate the worst case for spillage shall be documented in the RISK MANAGEMENT FILE.

After these PROCEDURES, the ME EQUIPMENT is to pass the appropriate dielectric strength and LEAKAGE CURRENT tests and is to show no signs of wetting of uninsulated electrical parts or electrical insulation of parts that could result in the loss of BASIC SAFETY or ESSENTIAL PERFORMANCE in NORMAL CONDITION or in combination with a SINGLE FAULT CONDITION (based on a visual inspection).

11.6.5 * Ingress of water or particulate matter into ME EQUIPMENT and ME SYSTEMS

Replace the existing final paragraph with the following:

After these PROCEDURES, the ME EQUIPMENT is to show no signs of bridging of insulation (or electrical components) that is likely to result in the loss of BASIC SAFETY or ESSENTIAL PERFORMANCE in NORMAL CONDITION or in combination with a SINGLE FAULT CONDITION (based on a visual inspection) followed by the appropriate dielectric strength and LEAKAGE CURRENT tests.

11.6.6 Cleaning and disinfection of ME EQUIPMENT and ME SYSTEMS

Replace the existing second paragraph with the following:

The MANUFACTURER shall evaluate the effects of multiple cleanings/disinfections as indicated in the instructions for use during the EXPECTED SERVICE LIFE of the ME EQUIPMENT, ME SYSTEM, their parts and ACCESSORIES and assure that that these PROCESSES do not result in the loss of BASIC SAFETY or ESSENTIAL PERFORMANCE.

In the first line of the compliance statement, replace "affected" with "effected".

11.6.7 Sterilization of ME EQUIPMENT and ME SYSTEMS

In the existing first paragraph, replace "ISO 11134, ISO 11135-1 or ISO 11137-1" with "ISO 11135-1, ISO 11137-1 or ISO 17665-1".

11.6.8 * Compatibility with substances used with the ME EQUIPMENT

Replace the existing first paragraph with the following:

When applicable, the MANUFACTURER shall address in the RISK MANAGEMENT PROCESS the RISKS associated with compatibility with substances used with the ME EQUIPMENT. Such RISKS may be addressed through the application of appropriate ISO or IEC standards (giving the presumption of acceptable RISK according to 4.2) such as ISO 15001 [70] for components that contain oxygen at pressures greater than 50 kPa or through the MANUFACTURER'S own testing and RISK CONTROL measures.

11.8 * Interruption of the power supply / SUPPLY MAINS to ME EQUIPMENT

Replace the existing first paragraph with the following:

ME EQUIPMENT shall be so designed that an interruption and restoration of the power supply shall not result in the loss of BASIC SAFETY or ESSENTIAL PERFORMANCE.

12 * Accuracy of controls and instruments and protection against hazardous outputs

12.2 USABILITY

Replace the existing title and text of the subclause with the following:

12.2 USABILITY OF ME EQUIPMENT

The MANUFACTURER shall address the RISK(S) of poor USABILITY, including those associated with identification, marking and documents, through a USABILITY ENGINEERING PROCESS complying with IEC 60601-1-6.

Compliance is checked as specified in IEC 60601-1-6.

12.3 Alarm systems

Replace the existing title and text of the subclause with the following:

12.3 ALARM SYSTEMS

If the MANUFACTURER has implemented an ALARM SYSTEM, this ALARM SYSTEM shall comply with IEC 60601-1-8.

Compliance is checked as specified in IEC 60601-1-8.

12.4 Protection against hazardous output

12.4.2 Indications of parameters relevant to safety

Replace the existing title, the first paragraph and the example with the following:

12.4.2 Indications relevant to safety

When applicable, the MANUFACTURER shall address in the RISK MANAGEMENT PROCESS the need to indicate any hazardous output.

EXAMPLE 1 Prior to the delivery of energy or substances to a PATIENT the energy, rate or volume is indicated quantitatively.

EXAMPLE 2 During activation of an X-ray output, a flashing yellow light is displayed.

12.4.5 Diagnostic or therapeutic radiation

12.4.5.2 Diagnostic X-ray equipment

Replace the existing text of the subclause with:

ME EQUIPMENT and ME SYSTEMS designed to produce X-radiation for diagnostic imaging purposes shall comply with IEC 60601-1-3.

Compliance is checked as specified in IEC 60601-1-3.

13 * HAZARDOUS SITUATIONS and fault conditions

Replace the existing title of Clause 13 with the following:

13 * HAZARDOUS SITUATIONS and fault conditions for ME EQUIPMENT

13.1 Specific HAZARDOUS SITUATIONS

13.1.2 * Emissions, deformation of ENCLOSURE or exceeding maximum temperature

In the existing sixth dash, replace "construction or the supply" with "construction of the supply".

After the existing sixth dash, insert the following:

- Secondary circuits meet all of the following conditions:
 - mounted on material with a flammability classification of FV1 in accordance with IEC 60695-11-10 or better;
 - they are energized at a voltage of 60 V d.c. or 42,2 V peak or less in NORMAL and SINGLE FAULT CONDITION;
 - they are limited to 100 VA or are limited to 6 000 J in SINGLE FAULT CONDITION;
 - they employ wire insulation of types PVC, TFE, PTFE, FEP, polychloroprene or polybromide.

Compliance is checked by evaluation of the design documentation.

or

- The component is a COMPONENT WITH HIGH-INTEGRITY CHARACTERISTICS as described in 4.9.
Compliance is checked by evaluation of the design documentation.

or

13.2 SINGLE FAULT CONDITIONS

13.2.7 Impairment of cooling that could result in a HAZARD

Replace the existing title with:

13.2.7 Impairment of cooling that could result in a HAZARDOUS SITUATION

13.2.13 * Overload

13.2.13.1 * General overload test conditions

Replace the existing first paragraph with the following:

After the tests of 13.2.13.2 to 13.2.13.4 (inclusive), the ME EQUIPMENT, when cooled down to within 3 °C of the temperature in the test environment, shall remain safe.

14 * PROGRAMMABLE ELECTRICAL MEDICAL SYSTEMS (PEMS)

14.1 * General

Replace the existing text of the subclause with the following:

The requirements in 14.2 to 14.12 (inclusive) shall apply to PEMS unless:

- none of the PROGRAMMABLE ELECTRONIC SUBSYSTEM (PESS) provides functionality necessary for BASIC SAFETY or ESSENTIAL PERFORMANCE; or
- the application of RISK MANAGEMENT as described in 4.2 demonstrates that the failure of any PESS does not lead to an unacceptable RISK.

The requirements in 14.13 are applicable to any PEMS intended to be incorporated into an IT-NETWORK whether or not the requirements in 14.2 to 14.12 apply.

NOTE 1 This clause requires that a PROCESS be followed throughout the PEMS DEVELOPMENT LIFE-CYCLE and that a RECORD of that PROCESS be produced. The concepts of RISK MANAGEMENT and a PEMS DEVELOPMENT LIFE-CYCLE are the basis of such a PROCESS. However, because a RISK MANAGEMENT PROCESS is already required by this standard, this clause will define the minimum elements of the PEMS DEVELOPMENT LIFE-CYCLE and only the additional elements for the PEMS that needs to be considered as part of the RISK MANAGEMENT PROCESS (see 4.2).

NOTE 2 If a RISK CONTROL measure is implemented within the PEES, it is necessary to apply Clause 14 to demonstrate that the failure of the PEES does not lead to an unacceptable RISK.

NOTE 3 It is recognized that the MANUFACTURER might not be able to follow all the PROCESSES identified in Clause 14 for each constituent component of the PEMS, such as software of unknown provenance (SOUP), subsystems of non-medical origin, and legacy devices. In this case, the MANUFACTURER should take special account of the need for additional RISK CONTROL measures. SOUP is defined in IEC 62304:2006 as a "software item that is already developed and generally available and that has not been developed for the purpose of being incorporated into the medical device (also known as 'off-the-shelf software') or software previously developed for which adequate RECORDS of the development PROCESSES are not available".

Compliance is determined by inspection of all documentation required, and when necessary, assessment of the requirements in 14.2 to 14.13 (inclusive).

NOTE 4 This assessment could be performed by internal audit.

When the requirements in 14.2 to 14.13 apply, the requirements in subclause 4.3, Clause 5, Clause 7, Clause 8 and Clause 9 of IEC 62304:2006 shall also apply to the development or modification of software for each PESS.

Compliance is determined by inspection and assessment as required by subclause 1.4 of IEC 62304:2006.

NOTE 5 The software development PROCESS required for compliance with this standard does not include the post-production monitoring and maintenance required by Clause 6 of IEC 62304:2006.

14.2 * Documentation

Delete the existing first paragraph and the note.

14.3 * RISK MANAGEMENT plan

In the existing paragraph, replace "3.5 of ISO 14971" with "4.2.2".

14.4 * PEMS DEVELOPMENT LIFE-CYCLE

Delete Note 2 and re-designate Note 1 as Note.

14.6 RISK MANAGEMENT PROCESS

14.6.1 * Identification of known and foreseeable HAZARDS

Replace the text of the subclause with the following:

When compiling the list of known or foreseeable HAZARDS, the MANUFACTURER shall consider those HAZARDS associated with software and hardware aspects of the PEMS including those associated with the incorporation of the PEMS into an IT-NETWORK, components of third-party origin and legacy subsystems.

NOTE In addition to the material given in Annex E of ISO 14971:2007, the list of possible causes for HAZARDS associated with PEMS can include:

- undesired feedback [physical and data] (possibilities include: unsolicited input, out of range or inconsistent input, and input originating from electromagnetic interference);
- unavailable data;
- lack of integrity of data;
- incorrect data;
- incorrect timing of data;
- unintended interactions within and among PESS;
- unknown aspects or quality of third-party software;
- unknown aspects or quality of third-party PESS;
- lack of data security including its effects on data privacy, and particularly vulnerability to tampering, unintended interaction with other programs and viruses;
- failure of the IT-NETWORK to provide the characteristics necessary for the PEMS to achieve its BASIC SAFETY or ESSENTIAL PERFORMANCE. See Annex H.7.2 for examples.

14.6.2 * RISK CONTROL

In the existing first paragraph, replace "Subclause 6.1 of ISO 14971" with "4.2.2".

14.8 * Architecture

Replace existing list item n) with the following:

n) the IT-NETWORK specification, if applicable.

14.9 * Design and implementation

Replace the existing second paragraph with:

Descriptive data regarding the design environment shall be documented.

In the existing note, replace "H.3" with "H.4 a)".

14.11 * PEMS VALIDATION

Replace the existing first paragraph with the following:

A PEMS VALIDATION plan shall include the validation of BASIC SAFETY and ESSENTIAL PERFORMANCE.

Methods used for PEMS VALIDATION shall be documented.

Delete the final paragraph in the subclause.

14.12 * Modification

Add the following as a second paragraph:

When software is modified, the requirements in subclause 4.3, Clause 5, Clause 7, Clause 8 and Clause 9 of IEC 62304:2006 shall also apply to the modification.

14.13 * Connection of PEMS by NETWORK/DATA COUPLING to other equipment

Replace the existing title and text of the subclause with the following:

14.13 * PEMS intended to be incorporated into an IT-NETWORK

If the PEMS is intended to be incorporated into an IT-NETWORK that is not validated by the PEMS MANUFACTURER, the MANUFACTURER shall make available instructions for implementing such connection including the following:

- a) the purpose of the PEMS's connection to an IT-NETWORK;
- b) the required characteristics of the IT-NETWORK incorporating the PEMS;
- c) the required configuration of the IT-NETWORK incorporating the PEMS;
- d) the technical specifications of the network connection of the PEMS including security specifications;
- e) the intended information flow between the PEMS, the IT-NETWORK and other devices on the IT-NETWORK, and the intended routing through the IT-NETWORK; and

NOTE 1 This can include aspects of effectiveness and data and system security as related to BASIC SAFETY and ESSENTIAL PERFORMANCE (see also Clause H.6 and IEC 80001-1:2010).

- f) a list of the HAZARDOUS SITUATIONS resulting from a failure of the IT-NETWORK to provide the characteristics required to meet the purpose of the PEMS connection to the IT-NETWORK.

NOTE 2 Connecting a PEMS to another piece of equipment for the purpose of transferring data creates a two-node IT-NETWORK. For example, connecting a PEMS to a printer creates an IT-NETWORK. If the MANUFACTURER has validated the PEMS with the printer, the resulting network would be considered within the control of the MANUFACTURER.

Compliance is checked by inspection of the instructions.

In the ACCOMPANYING DOCUMENTS, the MANUFACTURER shall instruct the RESPONSIBLE ORGANIZATION that:

- connection of the PEMS to an IT-NETWORK that includes other equipment could result in previously unidentified RISKS to PATIENTS, OPERATORS or third parties;
 - the RESPONSIBLE ORGANIZATION should identify, analyze, evaluate and control these RISKS;
- NOTE 3 IEC 80001-1:2010 provides guidance for the RESPONSIBLE ORGANIZATION to address these RISKS.
- subsequent changes to the IT-NETWORK could introduce new RISKS and require additional analysis; and
 - changes to the IT-NETWORK include:
 - changes in the IT-network configuration;
 - connection of additional items to the IT-NETWORK;
 - disconnecting items from the IT-NETWORK;
 - update of equipment connected to the IT-NETWORK; and
 - upgrade of equipment connected to the IT-NETWORK.

Compliance is checked by inspection of the ACCOMPANYING DOCUMENTS.

15 Construction of ME EQUIPMENT

15.1 * Arrangements of controls and indicators of ME EQUIPMENT

Replace the existing text of the subclause with the following:

When applicable, the MANUFACTURER shall address the RISKS associated with the arrangement of controls and indicators of ME EQUIPMENT in the USABILITY ENGINEERING PROCESS. See 12.2.

Compliance is checked as specified in IEC 60601-1-6.

15.3 Mechanical strength

15.3.1 General

Replace the existing first paragraph and the note with the following:

ME EQUIPMENT or its parts shall have adequate mechanical strength and shall not result in loss of BASIC SAFETY or ESSENTIAL PERFORMANCE due to moulding stress or when subjected to mechanical stress caused by pushing, impact, dropping, and rough handling.

NOTE Examples of damage that can affect BASIC SAFETY include the reduction of CREEPAGE DISTANCES and AIR CLEARANCES below those specified in 8.9, access to parts which exceed limits in 8.4, or access to moving parts which could cause HARM.

Assessment criteria that can be useful in determining if this test has resulted in loss of BASIC SAFETY include:

- those in Clause 8 and 11.6;
- the dielectric strength test as specified in 8.8.3 to evaluate the integrity of solid INSULATION providing a MEANS OF PROTECTION; and
- measurement of CREEPAGE DISTANCES or AIR CLEARANCES to compare the values with the minimum distances specified in 8.9. Small chips that do not adversely affect the protection against electric shock or moisture can normally be ignored.

Insert as the second major row in Table 28 the following:

BODY-WORN	Push (15.3.2)
	Impact (15.3.3)
	Drop (15.3.4.1)
	Moulding stress relief (15.3.6)

15.3.2 * Push test

Replace the existing third compliance paragraph with the following:

After the test, any damage sustained that results in an unacceptable RISK constitutes a failure.

NOTE See the compliance criteria in 15.3.1.

15.3.3 * Impact test

Replace the existing sixth compliance paragraph with the following:

After the test, any damage sustained that results in an unacceptable RISK constitutes a failure.

NOTE See the compliance criteria in 15.3.1.

15.3.4 * Drop test

15.3.4.1 HAND-HELD ME EQUIPMENT

Replace the existing text of the subclause with the following:

HAND-HELD ME EQUIPMENT, ACCESSORIES and ME EQUIPMENT parts shall not result in an unacceptable RISK as a result of a free fall.

Compliance is checked by the following test.

The sample to be tested, with any SAFE WORKING LOAD in place, is allowed to fall freely once from each of three different starting orientations encountered during NORMAL USE from the height at which the ME EQUIPMENT, ACCESSORY or ME EQUIPMENT part is used (as specified in the ACCOMPANYING DOCUMENTS), or from a height of 1 m, whichever is greater, onto a 50 mm $\pm$ 5 mm thick hardwood board (hardwood > 600 kg/m³) lying flat on a concrete or a similar rigid base.

After the test, the HAND-HELD ME EQUIPMENT, ACCESSORY or ME EQUIPMENT part shall not result in an unacceptable RISK.

15.3.4.2 * PORTABLE ME EQUIPMENT

Replace the existing text of the first paragraph with the following:

PORTABLE ME EQUIPMENT, ACCESSORIES and ME EQUIPMENT parts shall withstand the stress caused by a free fall from the height indicated in Table 29 onto a hard surface.

Replace the existing third compliance paragraph with the following:

After the test, any damage sustained that results in an unacceptable RISK constitutes a failure.

NOTE See the compliance criteria in 15.3.1.

15.3.5 * Rough handling test

Replace the existing second compliance paragraph and list items a) and b) with the following:

The sample is tested in transport position with any SAFE WORKING LOAD in place and in the most adverse condition permitted in NORMAL USE.. During the test, suitable precautions shall be taken to prevent over-balance caused by the rough handling stress/shock.

a) Ascending step shock

The sample is pushed three times in its normal direction of travel at a speed of 0,8 m/s $\pm$ 0,1 m/s, or, for motor-driven MOBILE ME EQUIPMENT, at the maximum speed capable of being maintained, against a solid hardwood plane obstruction with a face of 40 mm that is rigidly attached to an otherwise flat floor. The direction of movement is perpendicular to the face of the obstacle. The sample need not go over the 40 mm obstruction.

b) Descending step shock

The sample is pushed three times in its normal direction of travel at a speed of 0,8 m/s $\pm$ 0,1 m/s, or, for motor-driven MOBILE ME EQUIPMENT, at the maximum speed capable of being maintained, in order to fall over a vertical step having a height of 40 mm affixed flat on a rigid base (e.g. concrete). The direction of movement is perpendicular to the face of the descending step.

During performance of the descending step shock test, if a part other than the castor comes in contact with the obstruction before the castor touches the ground, the ME EQUIPMENT continues to be pushed until it has fully descended.

In existing list item c), replace "0,4 m/s $\pm$ 0,1 m/s" with "0,8 m/s $\pm$ 0,1 m/s".

Replace the existing final compliance paragraph with the following:

After each test, any damage sustained that results in an unacceptable RISK constitutes a failure.

NOTE 1 See compliance criteria in 15.3.1.

NOTE 2 Instability of ME EQUIPMENT, when the ME EQUIPMENT remains undamaged from the rough handling shock/stress, is evaluated in accordance with 9.4.

15.4 ME EQUIPMENT components and general assembly

15.4.1 Construction of connectors

Replace existing list item a) and the associated compliance paragraphs with the following:

- a) Plugs for connection of PATIENT leads or PATIENT cables shall be so designed that they cannot be connected to outlets on the same ME EQUIPMENT intended for other functions, unless it can be proven that no unacceptable RISK can result.

Compliance is checked by inspection of PATIENT leads, PATIENT cables, connectors and outlets and, if interchange of the leads, cables, connectors or outlets is possible, by inspection of the RISK MANAGEMENT FILE.

Replace the compliance paragraph associated with existing list item b) with the following:

Compliance is checked by inspection of all medical gas connectors.

15.4.2 Temperature and overload control devices

15.4.2.1 Application

Replace existing list items a), b), c) and d) with the following:

- a) THERMAL CUT-OUTS and OVER-CURRENT RELEASES with automatic resetting shall not be used in ME EQUIPMENT if their use could lead to a HAZARDOUS SITUATION described in 13.1 by such resetting.

Compliance is checked by inspection of the design documentation and the RISK MANAGEMENT FILE.

- b) THERMAL CUT-OUTS with a safety function that have to be reset by a soldering operation that can affect the operating value shall not be fitted in ME EQUIPMENT.

Compliance is checked by inspection of the design documentation.

- c) In ME EQUIPMENT, where a failure of a THERMOSTAT could lead to a HAZARDOUS SITUATION described in 13.1, an independent non-SELF-RESETTING THERMAL CUT-OUT shall additionally be provided. The temperature of operation of the additional device shall be outside that attainable at the extreme setting of the normal control device (THERMOSTAT) but shall be within the safe temperature limit for the intended function of the ME EQUIPMENT.

Compliance is checked by inspection of the design documentation and the RISK MANAGEMENT FILE.

- d) Loss of function of the ME EQUIPMENT caused by operation of a THERMAL CUT-OUT or OVER-CURRENT RELEASE shall not result in the loss of ESSENTIAL PERFORMANCE or any of the HAZARDOUS SITUATIONS described in 13.1.

Compliance is checked by inspection of the design documentation and the RISK MANAGEMENT FILE.

In the existing third paragraph of list item f), replace "IEC 60730-1:1999, clauses" with "IEC 60730-1:2010, Clauses".

15.4.3 * Batteries

15.4.3.1 Housing

Replace the existing text of the subclause with the following:

In ME EQUIPMENT, housings containing batteries from which gases can escape during charging or discharging shall be ventilated so that there is no unacceptable RISK from the accumulation of gases and possible ignition is prevented.

Battery compartments of ME EQUIPMENT shall be designed to prevent accidental short circuiting of the battery where such short circuits could result in the HAZARDOUS SITUATIONS described in 13.1.

Compliance is checked by inspection of the design documentation and, where appropriate, the RISK MANAGEMENT FILE.

15.4.3.4 Lithium batteries

Replace the existing text of the subclause with the following:

Primary lithium batteries shall comply with the requirements of IEC 60086-4. Secondary lithium batteries shall comply with the requirements of IEC 62133. See also 7.3.3.

NOTE Batteries includes both single cells and assemblies of cells, i.e. battery packages.

Compliance is checked by inspection of the battery design documentation or by performance of the tests identified in IEC 60086-4 for primary lithium batteries and IEC 62133 for secondary lithium batteries.

15.4.3.5 Excessive current and voltage protection

Add an asterisk to the title and replace the third sentence of the existing paragraph with:

Justification for omission of fuses or OVER-CURRENT RELEASES shall be documented.

Add the following as a second paragraph:

The short circuit test between the positive pole and the negative pole of an INTERNAL ELECTRICAL POWER SOURCE in the area between the INTERNAL ELECTRICAL POWER SOURCE output contacts and the subsequent protection device may be omitted if two MEANS OF OPERATOR PROTECTION are provided. Alternatively, a short-circuit test shall not result in any of the HAZARDOUS SITUATIONS in 13.1.2.

Replace the existing compliance statement with:

Compliance is checked by inspection for the presence of protective means, and if necessary, by inspection of the design documentation. Alternatively, conduct the short-circuit test and none of the HAZARDOUS SITUATIONS in 13.1.2 shall occur.

15.4.6 Actuating parts of controls of ME EQUIPMENT

15.4.6.1 Fixing, prevention of maladjustment

Replace the existing list item b) with the following:

b) Controls shall be so secured that the indication of any scale always corresponds with the position of the control.

15.4.6.2 Limitation of movement

Replace the existing first paragraph with the following:

Stops of adequate mechanical strength shall be provided on rotating or movable parts of controls of ME EQUIPMENT, where necessary to prevent an unexpected change from maximum to minimum, or vice-versa, of the controlled parameter.

Replace the existing third paragraph with the following:

If an axial pull is likely to be applied to the rotating or movable parts of controls of ME EQUIPMENT in NORMAL USE, there shall be no unexpected change of the controlled parameter.

15.4.7 Cord-connected HAND-HELD and foot-operated control devices (see also 8.10.4)

15.4.7.3 * Entry of liquids

Replace existing list item b) with the following:

- b) In ME EQUIPMENT, ENCLOSURES of foot operated control devices used in areas such as emergency rooms or operating theatres where liquids are likely to be present at floor level and that contain electrical circuits shall be classified at least IPX6 according to IEC 60529.

Compliance is determined by inspection of the ACCOMPANYING DOCUMENTS, the design documentation, and by performing the appropriate tests of IEC 60529.

15.5 * MAINS SUPPLY TRANSFORMERS of ME EQUIPMENT and transformers providing separation in accordance with 8.5

15.5.1 Overheating

15.5.1.1 * Transformers

Replace the existing fourth compliance paragraph with the following:

Components intended to prevent overheating of the transformer during short circuit and overload conditions are included as part of the tests of 15.5.1.2 and 15.5.1.3 provided that:

- *the component is one with high-integrity characteristics, and*
- *two MEANS OF OPERATOR PROTECTION are provided between the output contacts of the transformer up to the COMPONENT WITH HIGH-INTEGRITY CHARACTERISTICS.*

15.5.1.2 Short-circuit test

In the existing paragraph, replace "the 5X frequency and 5X voltage test of 15.5.2" with "the 5X frequency and 5X voltage test of 15.5.2 a) or the 2X frequency and 2X voltage test of 15.5.2 b)".

15.5.1.3 Overload test

Immediately after the first paragraph, insert the following:

The overload test may be applied after rectification.

15.5.2 * Dielectric strength

Immediately after the heading, insert the following:

This subclause is not applicable to transformers operating at a frequency above 1 kHz, which are tested in accordance with 8.8.3.

15.5.3 * Construction of transformers used to provide separation as required by 8.5

Replace the text of the subclause with the following:

Transformers of ME EQUIPMENT that form MEANS OF PROTECTION as required by 8.5 shall comply with the following:

- Means shall be provided to prevent displacement of end turns beyond the interwinding insulation.
- If a protective earthed screen has only one turn, it shall have an insulated overlap of not less than 3 mm. The width of the screen shall be at least equal to the axial winding length of the primary winding.
- The exit of the wires from the internal windings of toroidal transformers shall be provided with double sleeving complying with the requirements for two MEANS OF PROTECTION and having a total wall thickness of at least 0,3 mm, extending at least 20 mm outside the winding.
- The insulation between primary and secondary windings shall comply with 8.8.2.
- CREEPAGE DISTANCES and AIR CLEARANCES shall comply with 8.9.4 with the following exceptions:
 - Enamelled or lacquered winding wires are considered as contributing 1 mm each to the CREEPAGE DISTANCES specified in 8.9.4 for MEANS OF PATIENT PROTECTION.
 - CREEPAGE DISTANCES are measured through the joint between two parts of an insulation barrier, except when:
 - either the two parts forming the joint are bonded by heat sealing or other similar means at the place where this is of importance;
 - or
 - the joint is completely filled with adhesive at the necessary places and the adhesive bonds to the surfaces of the insulating barrier so that humidity cannot be sucked into the joint.
 - CREEPAGE DISTANCES within moulded transformers are considered not to exist if it can be shown that no gas bubbles are present and the thickness of the insulation between enamelled or lacquered primary and secondary windings is at least 1 mm for reference voltages U not exceeding 250 V and increased proportionally for higher reference voltages.

Compliance is checked by inspection of the transformer construction and measurement of required distances.

16 * ME SYSTEMS

16.3 * Power supply

After the existing first paragraph, insert the following:

If an ME SYSTEM:

- is intended to receive its power from an isolated power supply (IPS) or an uninterruptible power supply (UPS), and
- the ME SYSTEM can draw large transient currents when being switching on or off or when operating,

the MANUFACTURER shall restrict such transient currents to the allowed level according to the specification of the IPS or the UPS from which the ME SYSTEM is intended to be supplied.

If an IPS or UPS is not specified, the actual transient current level shall be disclosed in the technical description and any installation instructions.

16.6 * LEAKAGE CURRENTS

16.6.4 * Measurements

16.6.4.1 General conditions for ME SYSTEMS

Replace the existing first paragraph of list item a) with the following:

- a) *The TOUCH CURRENT, the PATIENT LEAKAGE CURRENT, the total PATIENT LEAKAGE CURRENT and the total EARTH LEAKAGE CURRENT are measured after the ME SYSTEM has been brought up to operating temperature as follows.*

16.8 Interruption of the power supply to parts of an ME SYSTEM

Replace the existing first paragraph with the following:

An ME SYSTEM shall be so designed that an interruption and restoration of the power to the ME SYSTEM as a whole, or any part of the ME SYSTEM, shall not result in the loss of BASIC SAFETY or ESSENTIAL PERFORMANCE.

16.9 ME SYSTEM connections and wiring

16.9.1 Connection terminals and connectors

Replace the existing text of the subclause with the following:

Design and construction of electrical, hydraulic, pneumatic and gas connection terminals and connectors shall be such that incorrect connection of accessible connectors, removable without the use of a TOOL, shall be prevented unless it can be proven that no unacceptable RISK can result. In particular:

- Plugs for connection of PATIENT leads or PATIENT cables shall be so designed that they cannot be connected to other outlets of the same ME SYSTEM that are likely to be located in the PATIENT ENVIRONMENT unless it can be proved that no unacceptable RISK can result.

Compliance is checked by inspection of PATIENT leads, PATIENT cables, connectors and outlets and, if interchange of the leads, cables, connectors or outlets is possible, by inspection of the RISK MANAGEMENT FILE.

- Medical gas connections on the ME SYSTEM for different gases to be operated in NORMAL USE shall not be interchangeable. See also ISO 407 [27].

Compliance is checked by inspection of all medical gas connections.

16.9.2 MAINS PARTS, components and layout

16.9.2.1 * MULTIPLE SOCKET-OUTLET

In existing list item c), replace the third dash with the following:

- PROTECTIVE EARTH TERMINALS and PROTECTIVE EARTH CONNECTIONS shall comply with 8.6.

In existing list item d), replace the first dash with the following:

- The separating transformer shall comply with this standard. Alternatively the separating transformer may comply with the requirements of IEC 61558-2-1, except that the requirements of maximum RATED output power of 1 kVA and degree of protection IPX4 do not apply.

Delete the existing Note 3.

16.9.2.2 * PROTECTIVE EARTH CONNECTIONS IN ME SYSTEMS

Insert the following new paragraph before the existing first paragraph:

For each part of an ME SYSTEM that shares a MAINS CONNECTION, the impedance and current carrying capability of the total protective earth path of an ME SYSTEM when tested as a unit shall comply with 8.6.4. The impedance between the protective earth pin in the MAINS PLUG and any part that is PROTECTIVELY EARTHED shall not exceed 200 mΩ.

Annex A –General guidance and rationale

A.1 General guidance

Replace existing list item h) with the following:

- h) conditions, particularly in operating theatres, that can present a combination of humidity, moisture or HAZARDS of fire or explosion caused by air, oxygen or nitrous oxide.

A.2 Safety of ME EQUIPMENT and ME SYSTEMS

In the last line of the existing second paragraph, replace "HAZARD" with "HAZARDOUS SITUATION".

A.4 Rationale for particular clauses and subclauses

Clause 2 – Normative references

Replace the existing fourth paragraph with the following:

Dated references are made when the requirements of a particular edition are to be used to satisfy a requirement of this standard. Subsequent amendments to, or revisions of, dated references will need to be incorporated by amendment of this standard.

Subclause 3.8 – APPLIED PART

Replace the paragraph immediately before Figure A.1 including the three dashes with the following:

Figure A.1 and Figure A.2 show an ECG monitor that includes the ECG monitor, the PATIENT cable, PATIENT leads and the ECG electrodes. In Figure A.1 and Figure A.2:

- The APPLIED PART includes the electrodes and those parts of the PATIENT leads or PATIENT cable that need to physically contact the PATIENT in NORMAL USE.
- Application of RISK MANAGEMENT might identify other parts of the PATIENT lead or PATIENT cable that have to be treated as APPLIED PARTS because of the probability they will come in contact with the PATIENT.
- The PATIENT CONNECTIONS consist of the ECG electrodes, which are all part of the same function of the APPLIED PART.

In the paragraph immediately following Figure A.3, replace the first three dashes with the following:

- The ME EQUIPMENT includes the ECG monitor; the ECG PATIENT cable, the PATIENT leads and their electrodes; and the pressure transducer and its fluid filled line.

- The APPLIED PART(s) include the ECG electrodes and those parts of the PATIENT cable and PATIENT leads that need to physically contact the PATIENT in NORMAL USE; and the fluid filled pressure monitoring line.
- Application of RISK MANAGEMENT might identify other parts of the ECG PATIENT cable, the PATIENT leads or the pressure transducer that have to be treated as APPLIED PARTS because of the probability they will come in contact with the PATIENT.

In the paragraph immediately following Figure A.4, replace the existing first and third dashes with the following:

- The ME SYSTEM includes the X-ray source assembly, the X-ray table and the wall stand, which are all items of ME EQUIPMENT. Other parts of the ME SYSTEM such as the X-ray generator and OPERATOR console are not shown.
- The application of RISK MANAGEMENT might identify some parts of the tube assembly and some other parts of the table and the wall stand as having to be treated as APPLIED PARTS because of the probability they will come in contact with the PATIENT.

In the paragraph immediately following Figure A.6, replace the first four dashes and their sub-bullets with the following:

- The ME SYSTEM includes the ECG module, PATIENT cable, PATIENT leads and electrodes, and the personal computer and any of its ACCESSORIES (not shown).
- The MANUFACTURER can choose to specify one of the following situations:
 - The ECG module and its PATIENT cable, PATIENT leads and electrodes are an item of ME EQUIPMENT; and the personal computer is not an item of ME EQUIPMENT. This would be an ME SYSTEM.
 - The ECG module and its PATIENT cable, PATIENT leads and electrodes are one item of ME EQUIPMENT; and the personal computer is a separate item of ME EQUIPMENT. This would also be an ME SYSTEM.
 - The ECG module and its PATIENT cable, PATIENT leads and electrodes together with the personal computer is a single item of ME EQUIPMENT and not an ME SYSTEM.
- The APPLIED PART includes the electrodes and those parts of the PATIENT cable or PATIENT leads that need to physically contact the PATIENT in NORMAL USE.
- Application of RISK MANAGEMENT might identify other parts of the PATIENT cable or PATIENT leads have to be treated as APPLIED PARTS because of the probability they will come in contact with the PATIENT.

Insert immediately following Figure A.7 the following new paragraph:

Subclause 4.6 would likely not apply to ACCESSIBLE PARTS of ME EQUIPMENT which are positioned according their INTENDED USE in such a way that the PATIENT could touch them only by a deliberate movement. This is based on the assumption that a PATIENT could react to a negative stimulus. Further, such PATIENTS are likely to have contact with non-ME EQUIPMENT such as bedside lights, personal computers, radios, etc.

Subclause 3.10 – BASIC SAFETY

Replace the existing first paragraph with the following:

BASIC SAFETY relates to a device not resulting in HARM incidental to its operation.

Add the following new rationale:

Subclause 3.15 – CLEARLY LEGIBLE

Vision or visual acuity can be tested by reading a Snellen eye chart at a distance of 6 m. Near vision can be tested using a Jaeger test card. By examining a large number of people, doctors

have decided what a “normal” human being should be able to see at various distances. That is the description of normal vision.

Subclause 3.27 – ESSENTIAL PERFORMANCE

Replace the existing text of the rationale for this subclause with the following:

It has long been recognized that ME EQUIPMENT or an ME SYSTEM that does not perform properly could result in unacceptable RISK for PATIENTS, OPERATORS, or others. Hence the concept of "safety" has been broadened from the BASIC SAFETY considerations in the first and second editions of this standard to include ESSENTIAL PERFORMANCE matters.

In order to achieve its INTENDED USE, the ME EQUIPMENT or ME SYSTEM needs to perform within certain limits. These limits are usually specified by the MANUFACTURER but could be specified by this standard, a collateral standard or a particular standard in the IEC 60601 family.

Examples of ESSENTIAL PERFORMANCE are:

- correct administration of a drug by a syringe pump where inaccuracy/incorrect administration would cause an unacceptable RISK to the PATIENT;
- the ability of an electrocardiograph/monitor to recover from the effects of the discharge of a defibrillator where the failure to recover could lead to an incorrect response by the medical staff that would present an unacceptable RISK to the PATIENT;
- correct operation of an ALARM SYSTEM in an intensive care or operating room monitoring system where an incorrect/missing ALARM SIGNAL could lead to an incorrect response by the medical staff that would present an unacceptable RISK to the PATIENT; or
- correct output of diagnostic information from ME EQUIPMENT that is likely to be relied upon to determine treatment, where incorrect information could lead to an inappropriate treatment that would present an unacceptable RISK to the PATIENT.

For purposes of this standard, performance related to BASIC SAFETY aspects of the ME EQUIPMENT, such as the performance of BASIC INSULATION, is not considered to be ESSENTIAL PERFORMANCE.

Particular and collateral standards in the IEC 60601 family are expected to identify specific ESSENTIAL PERFORMANCE.

Subclause 3.44 – INTENDED USE

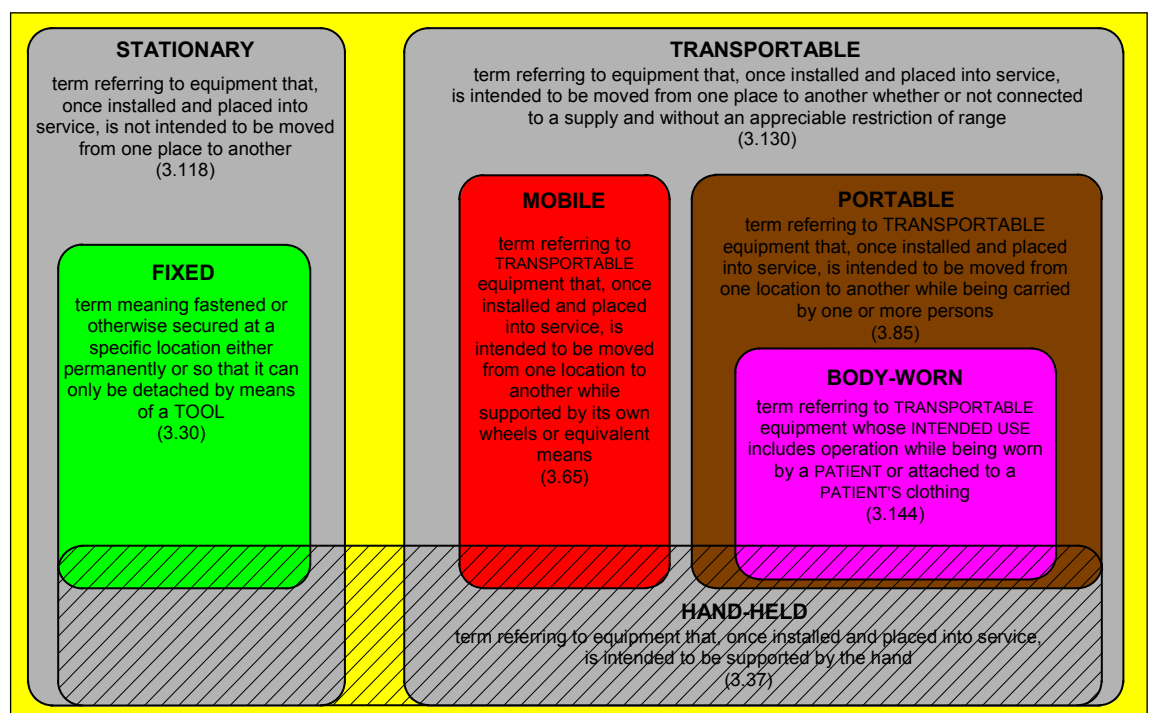
Delete the title and the entire rationale for this subclause.

Subclause 3.63 – MEDICAL ELECTRICAL EQUIPMENT

In the last line of the existing paragraph, replace "HAZARDS" with "HAZARDOUS SITUATIONS".

Insert the following as the final paragraph of the rationale:

This standard applies several terms to describe ME EQUIPMENT, ACCESSORIES and ME EQUIPMENT parts. They are FIXED (3.30), STATIONARY (3.118), TRANSPORTABLE (3.130), MOBILE (3.65), PORTABLE (3.85), HAND-HELD (3.37) and BODY-WORN (3.144). To help in understanding the relationship of these terms, the chart in Figure A.20 was developed.



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Figure A.20 – Relationship of the terms used to describe equipment, ACCESSORIES or equipment parts

Subclause 3.81 – PEAK WORKING VOLTAGE

In the existing text, replace "IEC 60950-1:2001, subclause 1.2.9.7" with "IEC 60950-1:2005, subclause 1.2.9.8".

Subclause 3.110 – SECONDARY CIRCUIT

In the existing text, replace "IEC 60950-1:2001, subclause 1.2.8.4" with "IEC 60950-1:2005, subclause 1.2.8.5".

Subclause 3.139 – WORKING VOLTAGE

In the existing text, replace "IEC 60950-1:2001, subclause 1.2.9.6" with "IEC 60950-1:2005, subclause 1.2.9.6".

Subclause 4.1 – Conditions for application to ME EQUIPMENT or ME SYSTEMS

Replace the existing text of the rationale for this subclause with the following:

The condition for application of RISK MANAGEMENT to ME EQUIPMENT and ME SYSTEMS includes reasonably foreseeable misuse. The MANUFACTURER identifies foreseeable misuse as part of the RISK ANALYSIS (see 4.2). This identification could include the results of a USABILITY ENGINEERING PROCESS.

Subclause 4.2 – RISK MANAGEMENT PROCESS for ME EQUIPMENT or ME SYSTEMS

Replace the existing first two paragraphs with the following:

A change introduced in the third edition of this standard is that, in specifying minimum BASIC SAFETY and ESSENTIAL PERFORMANCE requirements, provision is made for assessing the

adequacy of the design PROCESS where this provides an appropriate alternative to the application of laboratory testing with specific pass/fail criteria, (e.g. in assessing the safety of new technologies). Application of this principle leads to the introduction of a general requirement to carry out elements of a RISK MANAGEMENT PROCESS as part of demonstrating compliance with this standard.

The MANUFACTURER is responsible for ensuring that the design and construction of the ME EQUIPMENT renders it suitable for its INTENDED USE and that any RISKS that are associated with its use are acceptable when weighed against the benefits. ISO 14971 specifies a PROCEDURE for the MANUFACTURER to identify HAZARDS associated with the ME EQUIPMENT or ME SYSTEM and its ACCESSORIES, to estimate and evaluate the RISKS associated with those HAZARDS and to control those RISKS.

The scope of this standard is confined to type examination of ME EQUIPMENT and ME SYSTEMS; it does not extend to lifecycle monitoring. For this reason, the monitoring of production and post-production information and the planning thereof, as required by ISO 14971:2007, is excluded from the RISK MANAGEMENT PROCESS described in this standard. The requirement in ISO 14971:2007 for periodic review of the suitability of the RISK MANAGEMENT PROCESS is also excluded.

Replace the existing final paragraph with the following:

In deciding which phrase to use in a requirement the following rule has been used.

- The phrase “no unacceptable RISK” is used when the MANUFACTURER has to, or is permitted to, make a judgement about the acceptability of the RISK. This judgement is to be supported by an appropriate rationale such as experience, historical data, etc.
- The phrase “no HAZARDOUS SITUATION” is used when the MANUFACTURER has to demonstrate (e.g. by technical drawing or technical description) that there is no exposure to a HAZARD or no possibility of HARM can develop. In these cases the only determination a MANUFACTURER has to make is whether or not a HAZARDOUS SITUATION exists; this determination is made regardless of the acceptability of the RISK that could develop from that HAZARDOUS SITUATION.
- The phrase “no HAZARD” is never used.

The following provides guidance on how to use several elements of the RISK MANAGEMENT PROCESS as defined in ISO 14971 within the context of IEC 60601-1.

- **INTENDED USE:** The INTENDED USE is determined at the highest level of the ME EQUIPMENT or ME SYSTEM and should be used in identifying HAZARDS or HAZARDOUS SITUATIONS related to the use of the ME EQUIPMENT or ME SYSTEM. The INTENDED USE is not (or very rarely) used in the evaluation of specific components. Further, the INTENDED USE is the primary input when determining what aspects of the clinical functions of the ME EQUIPMENT or ME SYSTEM constitute ESSENTIAL PERFORMANCE as required by 4.3.
- **HAZARD identification:** This standard and ISO 14971 require identification of all HAZARDS and HAZARDOUS SITUATIONS associated with the ME EQUIPMENT or ME SYSTEM, regardless of whether they are identified in an international safety standard or not (see 4.2.2, Note 1, and 4.2.3.2). This holds for NORMAL USE, reasonably foreseeable misuse and SINGLE FAULT CONDITIONS. However, when HAZARDS or HAZARDOUS SITUATIONS identified in this standard cannot occur for the specific ME EQUIPMENT or ME SYSTEM, the HAZARD identification would be performed only to document that the corresponding requirements of this standard do not apply. In this case, further RISK ASSESSMENT or VERIFICATION of compliance is not required. When a requirement does apply, no HAZARD identification is necessary (because the standard has already done so). For example, no HAZARD of electric shock arises in ME EQUIPMENT having no voltages capable of producing LEAKAGE CURRENTS that exceed the applicable limits as specified in 8.4.2 c).
- **RISK EVALUATION:** RISK is evaluated at two points in the RISK MANAGEMENT PROCESS—before and then again after implementation of RISK CONTROL measures. Evaluation before implementation of RISK CONTROL measures is applicable to the requirements of this

standard only in order to demonstrate that the associated RISK is acceptable without applying a requirement. However, when this standard requires that RISKS related to a specific requirement remain acceptable or that no unacceptable RISK occurs in relation to a specific requirement, compliance is checked by inspection of the RISK MANAGEMENT FILE for the HAZARDS or HAZARDOUS SITUATIONS related to this requirement and for OBJECTIVE EVIDENCE (e.g. by MANUFACTURER'S calculations, analyses or test results) that the corresponding RISKS are judged to be acceptable based on the MANUFACTURER'S criteria for acceptability of RISK. The effectiveness of the RISK CONTROL measures is evaluated in accordance with subclause 6.3 of ISO 14971:2007.

- **Risk reduction:** When a requirement of this standard requires implementation of specific RISK CONTROL measures to assure that the associated RISK is acceptable, compliance is checked by either of the following:
 - by assuring that the specific measures have been implemented, in which case no RISK MANAGEMENT and no evaluation of RESIDUAL RISK are required because the RISKS are assumed to be reduced to an acceptable level by the presence of the specific measures (meeting the objective requirements of this standard), or
 - by inspection of the RISK MANAGEMENT FILE for the HAZARDS or HAZARDOUS SITUATIONS identified in the requirement in absence of the measures specified by the requirement and for OBJECTIVE EVIDENCE (e.g. by MANUFACTURER'S calculations, analyses or test results) that the corresponding RISKS are judged to be acceptable.

Subclauses 7.2.2 for detachable components and 7.2.17 for protective packaging are examples where special markings are required unless no unacceptable RISKS occur. So, either the markings must be provided or the RISKS in absence of those markings must be evaluated and judged acceptable. Another example is 15.4.1 a), for the design and construction of connector plugs.

- **Risk/benefit analysis:** A RISK/benefit analysis according to subclause 6.5 of ISO 14971:2007 is only required when the RESIDUAL RISK associated with a specific HAZARD or HAZARDOUS SITUATION remains unacceptable using the criteria for RISK acceptability recorded in the RISK MANAGEMENT plan and further RISK CONTROL is not practicable. Such RISK/benefit analyses are not required when the RESIDUAL RISKS are judged acceptable or when the ME EQUIPMENT or ME SYSTEM complies with specific requirements of this standard. Subclause 11.1.2 is one of the few examples where a RISK/benefit analysis is required when temperature limits are exceeded to achieve a clinical purpose (see Table 24, footnote ^b).

Subclause 4.3 – ESSENTIAL PERFORMANCE

Replace the existing text of the rationale for this subclause with the following:

During the initial RISK ANALYSIS, the MANUFACTURER identifies the performance of the clinical function(s) of the ME EQUIPMENT or ME SYSTEM that is necessary for achieving the INTENDED USE. The MANUFACTURER also identifies other qualitative and quantitative characteristics that could affect the safety of the ME EQUIPMENT or ME SYSTEM.

The performance limits specified by the MANUFACTURER could be anywhere within the full range of intended performance in NORMAL CONDITION and SINGLE FAULT CONDITION, to no performance.

The MANUFACTURER then determines if the loss or failure to perform within specified limits would result in a RISK to the PATIENT. The estimate is often made with the assumption that the performance aspect in question has been lost or degraded beyond the specified limits (i.e. P_1 in Figure A.8 is 100 %). The MANUFACTURER takes into account the probability of a HAZARDOUS SITUATION leading to HARM (P_2 in Figure A.8) as well as the SEVERITY of that HARM. In some instances P_2 could also be 100 %, in which case the RISK estimate is based entirely on the SEVERITY of the HARM. When considering the RISK associated with a level of degraded performance, which might include full loss, the analysis should include expected measures as prescribed in the ACCOMPANYING DOCUMENTS, which might return some or all performance after

a delay including service to the ME EQUIPMENT or ME SYSTEM, or a return of some or all of the PATIENT'S medical care with back-up medical care measures.

The MANUFACTURER then evaluates the RISK using their established RISK acceptance criteria. If the RISK is unacceptable, then the identified performance is ESSENTIAL PERFORMANCE because RISK reduction is required to achieve an acceptable level of RESIDUAL RISK.

Once the ESSENTIAL PERFORMANCE is identified, the MANUFACTURER puts in place RISK CONTROL measures that are appropriate to reduce the RISK to an acceptable level. When conducting the RISK CONTROL option analysis, the MANUFACTURER is to follow the priority order listed in ISO 14971, namely:

- a) inherent safety by design;
- b) protective measures;
- c) information for safety.

The RISK CONTROL measures selected need to be reasonably practicable and consistent with the generally accepted technology. For example, it might be possible to build a critical care ventilator that will continue to function in the presence of a single component failure, but, given the generally accepted technology, this is not practicable. Therefore, the MANUFACTURER might rely on a protective measure, such as an ALARM SYSTEM, to alert the OPERATOR of the failure so the OPERATOR can take appropriate and timely action to prevent the onset of HARM. The ALARM SIGNAL coupled with required OPERATOR training might be adequate RISK CONTROL measures to reduce the RISK arising from the loss or degradation of the identified performance to an appropriate level, i.e. the RESIDUAL RISK is acceptable. These RISK CONTROL measures can be integral with the consideration of ESSENTIAL PERFORMANCE.

Of course, the RISK CONTROL measure has to be sufficiently independent such that the condition that causes the loss or degradation of the identified performance does not compromise the effectiveness of the RISK CONTROL measure. This is consistent with the two MEANS OF PROTECTION philosophy employed in the IEC 60601 series. For example, an ALARM SYSTEM intended to alert the OPERATOR that the SUPPLY MAINS has been interrupted could not depend on the SUPPLY MAINS as the sole source of its power. A back-up battery could be used to power the ALARM SYSTEM if the SUPPLY MAINS is interrupted.

In RISK CONTROL, timing is everything. If there is insufficient time from the onset of the event (interruption of the SUPPLY MAINS) for the OPERATOR to register that an ALARM CONDITION has occurred and take appropriate action to prevent the onset of HARM, then the ALARM SYSTEM alone would not be an effective RISK CONTROL measure. A back-up power source to keep the ME EQUIPMENT operating for a period of time sufficient for the OPERATOR to take the necessary corrective action could also be required.

As with all aspects of the RISK MANAGEMENT PROCESS, the identification of ESSENTIAL PERFORMANCE can be iterative. The MANUFACTURER might have to revisit the process of determining what the ESSENTIAL PERFORMANCE is during the lifecycle of the ME EQUIPMENT or ME SYSTEM.

See also the rationale for 3.27.

Subclause 4.4 – EXPECTED SERVICE LIFE

Before the existing first paragraph, insert the following:

The EXPECTED SERVICE LIFE is the time period during which the ME EQUIPMENT or ME SYSTEM is expected to remain suitable for its INTENDED USE. It is also the period when all RISK CONTROL measures need to remain effective to ensure that RISKS remain acceptable.

After the existing second paragraph, insert the following:

In defining the EXPECTED SERVICE LIFE, the MANUFACTURER should assume that the RESPONSIBLE ORGANIZATION will follow the MANUFACTURER'S instructions for routine maintenance. See 7.9.2.13.

Subclause 4.5 – Equivalent safety for ME EQUIPMENT or ME SYSTEMS

Replace the existing heading and text of the rationale for this subclause with the following:

Subclause 4.5 – Alternative RISK CONTROL measures or test methods for ME EQUIPMENT or ME SYSTEMS

This subclause allows alternative means of achieving safety to those means specified in this standard. This is important as it permits a MANUFACTURER to use innovative solutions that are safe and can have other benefits, e.g. lower cost or better manufacturability.

Because it is often difficult to ascertain a sense of RISK acceptability for an individual RISK CONTROL measure or test method, the MANUFACTURER is permitted to use scientific data, clinical opinion, or comparative studies on a case-by-case basis to establish that the RESIDUAL RISK from applying an alternative remains acceptable.

Documentation in the RISK MANAGEMENT FILE should show that the RESIDUAL RISK achieved using the alternative means remains acceptable and is comparable to this standard. Comparative studies should take into account scientific studies referenced in this standard and applicable to the alternative means.

Subclause 4.6 – ME EQUIPMENT or ME SYSTEM parts that contact the PATIENT

After the existing fourth paragraph, insert the following:

Parts identified as needing to be subject to the requirements for APPLIED PARTS (except for marking) will typically contact PATIENTS less frequently than APPLIED PARTS, so the benefits of electrical separation from earth would be less. However in some cases, these parts might need to satisfy the requirements for TYPE BF APPLIED PARTS or TYPE CF APPLIED PARTS in order to provide an appropriate level of safety.

After the existing rationale for Subclause 4.7, add the following new rationale:

Subclause 4.8 – Components of ME EQUIPMENT

Resistors bridging a MEANS OF PROTECTION

A resistor or group of series connected resistors (two or more), can bridge one or two MEANS OF PROTECTION (MOOP or MOPP), providing the following is observed:

- the resistor or group of resistors has a voltage rating twice the value of the WORKING VOLTAGE across the MEANS OF PROTECTION;
- the resistor or group of resistors can withstand dielectric strength tests according to 8.8.3 and considering the expected transient voltages that the component(s) will be subjected to during NORMAL USE over the EXPECTED SERVICE LIFE;
- the resistor or group of resistors has a power rating twice that expected in NORMAL USE;
- the resistor or group of resistors have a resistance value such that TOUCH CURRENT, PATIENT LEAKAGE CURRENT and PATIENT AUXILIARY CURRENT requirements are not exceeded;
- the resistor or group of resistors have a construction that maintains the MOP separation requirements for CREEPAGE DISTANCE and AIR CLEARANCE (For a series group, separation at each component is additive making up the required separation.);
- the resistor or group of resistors are constructed such that their inductance is low;
- the resistor or group of resistors have a construction that cannot fail short-circuit; and

- when using a group of resistors they all need have the same resistance value and the voltage and power doubling requirement applies also to each component based on the expected WORKING VOLTAGE across that resistor in NORMAL USE.

Subclause 5.1 – TYPE TESTS

After the final paragraph, insert the following:

Guidance on general testing procedures that can be used with this standard is given in IEC/TR 62354 [66].

This standard uses several terms to describe fault conditions that could be encountered. They are:

SINGLE FAULT CONDITION

The SINGLE FAULT CONDITION is defined in 3.116 to describe those conditions where “a single means for reducing a RISK is defective” or “a single abnormal condition is present”. Possible SINGLE FAULT CONDITIONS are listed in this standard and can affect verifiable values like CREEPAGE DISTANCES, protective earth path impedances, LEAKAGE CURRENTS, etc.

Faults not leading to a SINGLE FAULT CONDITION

Compared with SINGLE FAULT CONDITIONS, this standard does not have verifiable requirements for faults not leading to a SINGLE FAULT CONDITION. This means this standard does not give guidance how reliable it is that faults can be prevented.

Combination of simultaneous independent faults

Because there are no reliable verifiable requirements defined in this standard for the prevention of faults, all possible simultaneous faults should be considered in accordance with 4.7.

Where a SINGLE FAULT CONDITION remains undetected (e.g. bridging of one MEANS OF OPERATOR PROTECTION), further simultaneous faults should be considered in accordance with 4.7.

NOTE For additional guidance, see IEC 60513:1994.

The worst cases of simultaneous faults are those which lead to most dangerous results. The wording of the standard is used to avoid conducting all possible simultaneous fault tests, because many of them could be covered by worst case testing.

Subclause 5.7 – Humidity preconditioning treatment

Replace the existing third paragraph with the following:

Parts sensitive to humidity, normally used in controlled environments and that do not influence BASIC SAFETY and ESSENTIAL PERFORMANCE, need not be subjected to this test. Examples are: high-density storage media in computer-based systems, disc and tape drives, etc.

Subclause 7.1.2 – Legibility of markings

After the existing third paragraph, insert the following:

The Jaeger test card is used for determining near vision. N6 corresponds to reading at a distance of 0,87 m from the face.

An ambient light level of 500 lx is recommended for conducting the visual acuity examination.

If a recommended minimum font size is to be considered, then a readable font size is a function of reading distance and the resulting visual angle of the marking image presented to the retina of the eye. Calculations of visual angle in minutes of arc are found in ANSI/AAMI HE-75:2009 [71], subclause 6.2.2.6.5, Table 6.2. Using a minimal acceptable visual angle of 12 minutes of arc, a font size of 9 would be recommended for a reading distance of one meter and a font size of 6 for a typical reading distance of 0,7 m.

Subclause 7.1.3 – Durability of markings

Replace the existing first paragraph with the following:

The rubbing test is performed with distilled water, ethanol 96 % and isopropyl alcohol.

Subclause 7.2.2 – Identification

Replace the existing first three paragraphs with the following:

This subclause is intended to apply to any detachable component when misidentification could result in a HAZARDOUS SITUATION. For example, normal consumables would probably need to be identified, but a cosmetic cover would not need to be identified.

Although a MODEL OR TYPE REFERENCE usually denotes a certain performance specification, it might not denote the exact construction, including the applied components and materials. The MODEL OR TYPE REFERENCE is supplemented by a serial number or lot or batch identifier. The serial number or lot or batch identifier can be used for other purposes such as providing traceability in case the MANUFACTURER needs to take a corrective action.

Subclause 7.2.3 – Consult ACCOMPANYING DOCUMENTS

In the existing paragraph, replace "IEC 60878 Safety 01" with "ISO 7010-M002" in two places.

Subclause 7.2.10 – APPLIED PARTS

In the existing third paragraph, replace "are non-obvious HAZARDS" with "can be non-obvious HAZARDOUS SITUATIONS", and replace "HAZARDS can include" with "These HAZARDOUS SITUATIONS can include".

After the rationale for Subclause 7.2.12, add the following new rationale:

Subclause 7.2.21 – Mass of MOBILE ME EQUIPMENT

MOBILE ME EQUIPMENT is intended to be transported from one location to another by its own wheels or equivalent means. For safety, it is important that the OPERATOR be aware of the total mass of the ME EQUIPMENT with its SAFE WORKING LOAD installed. This is important when moving the ME EQUIPMENT, and, for heavy ME EQUIPMENT, understanding any limitations on locations where the ME EQUIPMENT can be taken. The ME EQUIPMENT can have bins, shelves or drawers that might have their own maximum loading requirements. The marking on the ME EQUIPMENT needs to be sufficiently separate and distinct from any markings on the bins, shelves or draws to avoid confusing the OPERATOR as to what the marking applies.

Subclause 7.3.2 – HIGH VOLTAGE parts

Replace the existing paragraph with the following:

HIGH VOLTAGE parts present a significant electric shock HAZARD to SERVICE PERSONNEL and others who could be required to work inside the ME EQUIPMENT while it is energized. Because the parts are inside the ENCLOSURE, the RISK is perceived to be substantially less than that for HIGH VOLTAGE TERMINAL DEVICES located on the outside of the ME EQUIPMENT. Therefore, the "dangerous voltage" symbol IEC 60417-5036 (2002-10) (see Table D.1, symbol 24) is

permitted as a marking to alert SERVICE PERSONNEL and others to the potential presence of these dangerous voltages. The MANUFACTURER is permitted to use ISO 7010-W012 (see Table D.3, safety sign 3). The RISK MANAGEMENT PROCESS could determine that the safety sign is the most appropriate choice if the personnel exposed to the HAZARD have minimal training or might otherwise be unaware that HIGH VOLTAGE is present.

After the rationale for Subclause 7.3.4, add the following new rationales:

Subclause 7.3.5 – PROTECTIVE EARTH TERMINALS

This standard does not require marking of terminals for internal PROTECTIVE EARTH CONNECTIONS, but does not preclude it.

Subclause 7.4.1 – Power switches

If a marking is used for a toggle-type switch, the symbols for "on" () and "off" () [symbols 12 and 13 from Table D.1] should be used to indicate its ON and OFF positions.

For a push-button type switch, the appropriate marking will depend on the type of switch. For a push button switch with bistable positions, the symbol () [symbol 14 from Table D.1] should be used to indicate that both the ON and OFF positions are stable.

For a push button switch with a momentary ON position, the symbol () [symbol 15 from Table D.1] should be used to indicate that mains power is applied for only as long as the push button is depressed. As soon as the button is released the switch returns to the stable OFF position.

Subclause 7.4.2 – Control devices

The different positions of switches that control the functionality of the ME EQUIPMENT or a part of the ME EQUIPMENT, but do not control mains power to the ME EQUIPMENT or a part of the ME EQUIPMENT should also be marked in such a way as to clearly indicate the state of the control function. Any appropriate means including figures, letters, indicator lights, etc. can be used. If the marking includes symbols, the symbols for "on" for part of the equipment () and "off" for part of the equipment () [symbols 16 and 17 from Table D.1] can be used to indicate its ON and OFF positions.

A switch or switch position by means of which the ME EQUIPMENT or part of the ME EQUIPMENT is switched on in order to bring it into the "stand-by" condition can be indicated by the stand-by symbol () [symbol 29 from Table D.1]. This symbol is required by IEC 60950-1 to indicate the stand-by mode of the information technology equipment covered by that standard.

Subclause 7.8 – Indicator lights and controls

After the first paragraph, insert the following:

Colour alone should not be used to convey important information. A redundant means of conveying information such as shape, location, sound or marking is recommended.

Subclause 7.9.1 – General

After the first paragraph, insert the following:

The USABILITY ENGINEERING PROCESS requires the MANUFACTURER to develop an applications specification that includes the intended environments of use. The MANUFACTURER is required to supply a summary of this specification in the ACCOMPANYING DOCUMENTS. See 12.2.

Subclause 7.9.2.1 – General

In the existing first paragraph, delete the final sentence.

After the existing first paragraph, insert the following:

When considering if the PATIENT can act as an OPERATOR, the MANUFACTURER needs to take into account that the level of protection provided by this standard for the OPERATOR can be different from that for the PATIENT. For example, acceptable LEAKAGE CURRENT levels and accessible voltages values are higher for the OPERATOR than they are for the PATIENT. Instead of PATIENT LEAKAGE CURRENT, TOUCH CURRENT could be applied. Additionally the OPERATOR can have access to voltages up to 42,4 V peak a.c. or 60 V d.c.

Other conditions to consider include:

- Is the ME EQUIPMENT to be used under medical supervision or guidance?
- What is the health condition of the PATIENT?
- What treatment is being provided to the PATIENT?
- Are there special USABILITY considerations (see IEC 60601-1-6)?
- Is the ME EQUIPMENT to be used in the professional healthcare environment, the home healthcare environment (see IEC 60601-1-11 [55]) or the emergency medical service environment (see IEC 60601-1-12 [58])?
- Does the PATIENT have access to the instructions for use?
- Is adequate training of the PATIENT as an OPERATOR available?

After the rationale for Subclause 7.9.2.7, add the following new rationale:

Subclause 7.9.2.19 – Unique version identifier

MANUFACTURERS frequently issue multiple revisions of the instructions for use. RESPONSIBLE ORGANIZATIONS and OPERATORS need to be able to determine if the instructions for use they have are current. This is particularly important when dealing with PEMS software that can be updated in the environment of use. Providing a date of issuance is a widely accepted method for identifying versions of a document. The MANUFACTURER decides what level of detail is appropriate with month and year being common for printed material. However, greater granularity might be required, particularly for material distributed electronically. The MANUFACTURER could include a revision number when appropriate.

Clause 8 – Protection against electrical HAZARDS from ME EQUIPMENT

In the existing first paragraph, replace "present a HAZARD" with "result in a HAZARDOUS SITUATION".

Subclause 8.3 d)

Delete the title and text of the rationale for lettered paragraph 8.3 d).

Subclause 8.4.2 c)

In both existing paragraphs, replace "IEC 60950-1:2001" with "IEC 60950-1:2005".

Subclause 8.5.1 – MEANS OF PROTECTION

In the existing thirteenth paragraph, replace "2 500 V a.c." with "1 500 V a.c.".

Delete the existing fourteenth paragraph immediately before Figure A.12.

Replace existing Figure A.12 with:

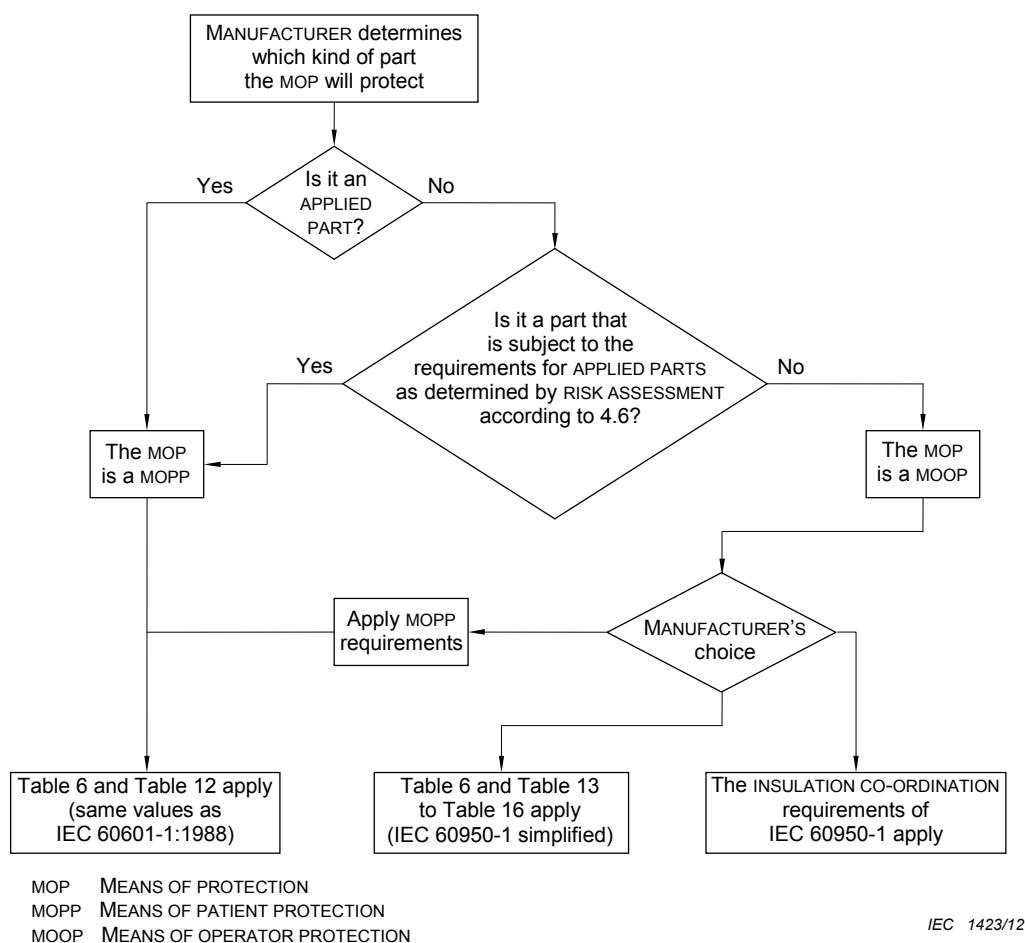


Figure A.12 – Identification of MEANS OF PATIENT PROTECTION and MEANS OF OPERATOR PROTECTION

Immediately following the existing Figure A.12, add the following new rationale:

Subclause 8.5.1.2 – MEANS OF PATIENT PROTECTION (MOPP)

When fitting Y capacitors across barriers, the dielectric strength requirement must be considered. For example, for voltages in the range of 212 V peak to 354 V peak, two MEANS OF PATIENT PROTECTION are required to be tested at 4 000 V a.c. This would then require the use of Y1 capacitors. Two Y2 capacitors in series would not withstand this voltage as they are rated at 1 500 V a.c. each. For voltages below 212 V peak, two Y2 capacitors would suffice as the dielectric strength requirement is 3 000 V a.c.

Subclause 8.5.2.2 – TYPE B APPLIED PARTS

In the existing second dash, replace "The RISK cannot arise" with "A HAZARDOUS SITUATION cannot develop".

Subclause 8.5.2.3 – PATIENT leads

Replace the existing title with the following:

Subclause 8.5.2.3 – PATIENT leads or PATIENT cables

In the existing eleventh paragraph, replace "HAZARD" with "HAZARDOUS SITUATION".

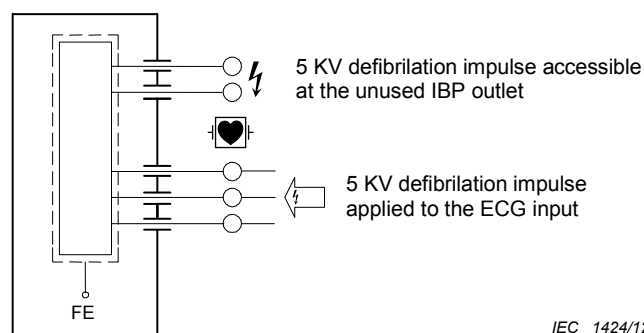
Subclause 8.5.4 – WORKING VOLTAGE

In the existing fourth paragraph, replace "10 s" with "1 s".

Subclause 8.5.5.1 – Defibrillation protection

Directly before "Rationale for impulse test voltage", insert the following:

If ME EQUIPMENT would be designed according to 8.5.2.1 and 8.5.5.1, it might have two different functions on one common APPLIED PART circuit. Consider the example in Figure A.21. In this example, the ECG and an invasive blood pressure catheter (IBP) share one common APPLIED PART circuit. In real use in the hospital, sometimes only one function (ECG) is used while the other function (IBP) is not connected. If the PATIENT needs to be defibrillated, there is a potential that a second OPERATOR would get an electric shock at unused IBP outlets during selecting of proper adjustments at the ME EQUIPMENT (e.g. at a multi-parameter patient monitor). It should be pointed out that in principle, the 5 KV could occur at every disconnected connector, which does not have adequate recessed pins. However, in practice, the probability of occurrence of HARM is much higher at the unused APPLIED PART outlets on the ME EQUIPMENT compared to any connector of adapter cables. Adapter cables will normally not be touched by the OPERATOR during defibrillation, because before defibrillation the command "stand clear" will be given. Even in the case that a disconnected adapter cable would be touched by an OPERATOR during defibrillation, it will most probably be touched at the isolated plastic parts, but not at its internal pins.



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Figure A.21 – Example of ME EQUIPMENT having two different functions on one common APPLIED PART circuit

Subclause 8.6.4 a)

After the sixth paragraph, insert the following new paragraph:

Where a long (> 3 m) DETACHABLE POWER SUPPLY CORD is supplied, specified or used, the 200 mΩ total protective earth pathway requirement still applies. Therefore, it might be necessary to use a DETACHABLE POWER SUPPLY CORD with conductors of a greater cross sectional area than the minimum required by 8.11.3.3.

Subclause 8.6.9 – CLASS II ME EQUIPMENT

After the first paragraph, insert the following new paragraph:

The current flowing through the FUNCTIONAL EARTH CONDUCTOR has to be limited in the same way as current in the PROTECTIVE EARTH CONDUCTOR. To prevent the FUNCTIONAL EARTH CONDUCTOR from becoming a PROTECTIVE EARTH CONDUCTOR, the insulation between internal

screens including internal wiring connected to them and ACCESSIBLE PARTS needs to provide two MEANS OF PROTECTION.

Subclause 8.7.3 – Allowable values, Table 3 and Table 4

Renumber existing list item d) as list item a) and renumber all subsequent items in this rationale accordingly.

Replace existing item e) with the following:

- b) The RISK is highest and approximately equal for frequencies in the 10 Hz to 200 Hz range. It is lower, by a factor of nearly 5, at d.c. and by approximately 1,5 at 1 kHz. Beyond 1 kHz, the RISK decreases rapidly [45]. However lower limits are needed for d.c. to prevent tissue necrosis with long-term application.

The values in Table 3 and Table 4 apply to currents measured with the measuring device shown in Figure 12 a), which automatically allows for the reduced sensitivity at higher frequencies. SUPPLY MAINS frequencies of 50 Hz and 60 Hz are in the range of highest RISK.

*In the first line of the existing paragraph immediately preceding the heading "**Total PATIENT LEAKAGE CURRENT for TYPE BF APPLIED PARTS**", replace "TYPE CF EQUIPMENT" with "TYPE CF APPLIED PARTS".*

*In the second line of the existing paragraph immediately following the heading "**Total PATIENT LEAKAGE CURRENT for TYPE BF APPLIED PARTS**", replace "c) above" with "k) above".*

*Immediately below the existing paragraph following the heading "**Heating effects of LEAKAGE CURRENT**", add the following new rationale:*

Subclause 8.7.3 f)

LEAKAGE CURRENTS can flow through a FUNCTIONAL EARTH CONDUCTOR. This can lead to an unexpected RISK if CLASS II equipment is combined with CLASS I equipment or with an earthed trolley, e.g. using an MSO. Although proper application of the 'rules' for assembling an ME SYSTEM would eliminate this RISK, the causal RISK can be mitigated by limiting the allowed LEAKAGE CURRENT in a FUNCTIONAL EARTH CONDUCTOR.

Subclause 8.7.4.5 – Measurement of the EARTH LEAKAGE CURRENT

Replace the existing title with the following:

Subclause 8.7.4.5 – Measurement of the EARTH LEAKAGE CURRENT and current in FUNCTIONAL EARTH CONNECTION

Subclause 8.9 – CREEPAGE DISTANCES AND AIR CLEARANCES

In the existing second paragraph of item 2), delete the last sentence.

Replace the existing final paragraph with the following:

Table A.2 contains CREEPAGE DISTANCES for WORKING VOLTAGES above 1 000 V derived from IEC 60664-1:2007, Table F.4.

Subclause 8.9.1 – Values

In the existing second dash, replace "CLEARANCES" with "AIR CLEARANCES".

Subclause 8.9.1.15 – CREEPAGE DISTANCES and AIR CLEARANCES for DEFIBRILLATION-PROOF APPLIED PARTS

In the existing paragraph, replace "IEC 60664-1, Table 2" with "IEC 60664-1:2007, Table F.2"

Subclause 8.10.2 – Fixing of wiring

In the existing first paragraph, replace "from creating a HAZARD" with "from resulting in a HAZARDOUS SITUATION".

Subclause 8.11.1 c)

In the existing paragraph, replace "Table 22 of IEC 61058-1" with "Table 22 of IEC 61058-1:2000".

Subclause 8.11.3.4 – APPLIANCE COUPLERS

Replace the existing text of the rationale for this subclause with the following:

A POWER SUPPLY CORD connected to a MAINS CONNECTOR is subject to similar stresses to a non-DETACHABLE POWER SUPPLY CORD. If it is not adequately protected from excessive bending, a HAZARDOUS SITUATION could result.

Subclause 8.11.3.6 – Cord guards

In the existing second paragraph, replace "3.29 of IEC 60950-1:2001" with "IEC 60950-1:2005, subclause 3.2.8."

Clause 9 – Protection against MECHANICAL HAZARDS of ME EQUIPMENT and ME SYSTEMS

In the existing third paragraph, second dash, replace "HAZARD" with "MECHANICAL HAZARD".

Subclause 9.2 – HAZARDS associated with moving parts

Replace the existing title with the following:

Subclause 9.2 – MECHANICAL HAZARDS associated with moving parts

In the first through the fifth dashes, replace "HAZARD(S)" with "MECHANICAL HAZARD(S)".

Subclause 9.2.2.4 – GUARDS and protective measures

Replace the existing heading with the following:

Subclause 9.2.2.4 – GUARDS and other RISK CONTROL measures

Delete the existing fourth and fifth paragraphs.

Subclause 9.2.3 – Other HAZARDS associated with moving parts

Replace the existing heading and text of the rationale for this subclause with the following:

Subclause 9.2.3 – Other MECHANICAL HAZARDS associated with moving parts

Subclause 9.2.2.1 deals with MECHANICAL HAZARDS caused by TRAPPING ZONES. Movement could result in other MECHANICAL HAZARDS, such as impact, puncture, etc.

Subclause 9.3 – HAZARD associated with surfaces, corners and edges

Replace the existing heading with:

Subclause 9.3 – MECHANICAL HAZARD associated with surfaces, corners and edges

Subclause 9.4 – Instability HAZARDS

After the second paragraph, insert the following new paragraph:

All ME EQUIPMENT is tested for overbalance with 5° tip, 10° tip, and push in 9.4.2.1, 9.4.2.2, 9.4.2.3 a) and b). In 9.4.3.1 a), b) and c) and 9.4.3.2 a) and b), only MOBILE ME EQUIPMENT is tested as these are requirements for wheels and their brakes.

Subclause 9.4.2 – Instability – overbalance<-

Replace existing Table A.3 with the following:

Table A.3 – Instability test conditions

Transport warning	Test plane angle	
	10° plane	5° plane
Transport warning not provided	Must pass in all positions	Not applicable (represented by 10° test)
Transport warning provided	Must pass in transport position (only)	Must pass in all positions except transport

After Figure A.16, add the following new rationale:

Subclause 9.4.2.3 a)

The push force of 15 % of the ME EQUIPMENT'S weight or a maximum of 150 N was selected in order to represent foreseeable misuse for typical ME EQUIPMENT, based on input from ME EQUIPMENT MANUFACTURERS. The marking option for ME EQUIPMENT that overbalances with this push force is considered something that provides medical care benefit, and with the no-pushing marking the committee presumes that the RISK would be acceptable, but the MANUFACTURER needs to make this determination based on their own analysis, which might include usability studies, including the location of the marking.

Subclause 9.4.2.4 – Castors and wheels

Replace the existing text of the rationale for this subclause with the following:

Compliance with this subclause is required not only to avoid obvious unacceptable RISK but also to ensure the substantially operative movement as an ESSENTIAL PERFORMANCE. MOBILE ME EQUIPMENT is defined as equipment intended to be moved from one location to another.

After the rationale for Subclause 9.4.2.4, add the following new rationales:

Subclause 9.4.2.4.3 – Movement over a threshold

The 10 mm threshold height came from the thought that cables are represented by an 8 mm to 10 mm height. The committee felt that elevator alignment is now closer to 10 mm (e.g. 4 mm) rather than 20 mm. Normally, hospitals would not have door threshold.

The manual speed of 0,8 m/s was selected from an anticipated range of 0,4 m/s to 1,5 m/s during NORMAL USE, based on input from ME EQUIPMENT MANUFACTURERS. A speed of 0,8 m/s was felt to be the most representative of NORMAL USE and normal walking speed. There's a balance between too slow a speed such that the ME EQUIPMENT won't go over the threshold and too fast a speed causing the ME EQUIPMENT to become unstable. A single speed was selected for the test as this was felt to be necessary for reproducibility among testers.

Some MOBILE ME EQUIPMENT is provided with castors for convenience but not as means to be wheeled across thresholds or similar obstacles. When the MANUFACTURER clearly indicates this use is not intended there should be no need for the threshold test. Examples of equipment that might not require a threshold test include trolleys and treatment tables that are fitted with castors to enable them to be moved easily aside for cleaning the floor or making space.

Subclause 9.4.3 – Instability from unwanted lateral movement (including sliding)

The braking functional verification tests allow the braking system to be assessed with compliance giving the presumption of acceptable RISK. The test criteria assume that generally any initial elastic movement and subsequent movement up to 50 mm are acceptable. In accordance with 4.2, if these movements for a specific piece of ME EQUIPMENT and its INTENDED USE seem inappropriate, the RISK should be lowered with additional RISK CONTROL measures.

The braking push force of 15 % or the maximum of 150 N was selected in order to represent reasonably foreseeable misuse for typical ME EQUIPMENT based on input from ME EQUIPMENT MANUFACTURERS.

The intent of braking requirements is to provide a generic test for general ME EQUIPMENT. Thus compliance criteria are black and white. It is acknowledged that some slow unwanted movement of low weight carts (low momentum), could be considered an acceptable RISK.

It is accepted that for many carts with 4 wheels, providing 2 brakes is a reasonable RISK CONTROL measure to mitigate unwanted movement due to inadequate braking. Placing 4 brakes on many carts will not result in greater usage of the brakes, as some of the wheels will typically be pivoted out of reach.

Subclause 4.5 is always applicable and allows the MANUFACTURER to use alternative RISK CONTROL measures, or variations to test method, or test compliance criteria, with justification.

Subclause 9.4.3.1 c) – Instability in transport

The hard flat surface used in this test should reflect the worst case of INTENDED USE. Slippage of wheels against the hard flat surface is a secondary concern. If slippage is detected, it should be determined whether a more appropriate surface should be chosen for the test or the wheel material is inappropriate.

Subclause 9.7.5 – Pressure vessels

Before the existing first paragraph, insert the following:

The Oxford dictionary defines a pressure vessel as “a container designed to hold material at high pressures.” This is the usage intended by this standard.

Subclause 9.8 – HAZARDS associated with support systems

Replace the existing heading with:

Subclause 9.8 – MECHANICAL HAZARDS associated with support systems

After the last dash in the existing second paragraph, insert the following heading:

Subclause 9.8.2 – TENSILE SAFETY FACTOR

In the twelfth paragraph, replace "HAZARD" with "MECHANICAL HAZARD".

After the final paragraph, insert the following:

A jack screw nut considered impaired by wear is usually intentionally constructed from a material softer than the jack screw. The intent is to have generic requirement provided for general ME EQUIPMENT.

Newer jack screw designs can have less friction than traditional designs, and, for example, can include multiple ball bearings. Such designs can be considered as less "impaired by wear" as dictated by vendor declarations and a MANUFACTURER'S RISK ASSESSMENT.

Subclause 4.5 is always applicable and allows the MANUFACTURER to use alternative RISK CONTROL measures, or variations to a test method or test compliance criteria, with justification.

Subclause 9.8.3.2 – Static forces due to loading from persons

After the existing fifth paragraph, insert the following:

During the static loading tests, it's intended that any temporary elastic deflection (during the test), and permanent plastic deflection (after the test) are checked for any adverse effect on BASIC SAFETY or ESSENTIAL PERFORMANCE.

For the step loading test, the committee felt that a 5° deflection would provide a measurable quantity that would represent a reasonable limit for acceptance. The 2X loading factor is a precedent from the 2nd edition of IEC 60601-1, and seems to recognize the temporary loading that steps represent during their life.

For the sitting surface loading test, no specific linear or arc limits are specified. The intent is to evaluate any temporary elastic or permanent plastic deflections for any adverse effect on BASIC SAFETY or ESSENTIAL PERFORMANCE. The 60 % sitting loading factor was considered representative of actual sitting load, when legs are dangling off the edge of the sitting surface. Loss of function is not considered unacceptable unless associated with ESSENTIAL PERFORMANCE. The intent is damage should not result in injury to persons, or loss of RISK CONTROL measures. Surfaces that are intended to support sitting and at other times full patient loads are also evaluated for compliance with the appropriate loading factor(s) in 9.8.2.

Immediately before the rationale for Subclause 10.4, add the following new rationale:

Subclause 10.3 – Microwave radiation

This subclause specifies a microwave power density limit to mitigate thermal harm to whole body tissue. The intention is that compliance might be appropriately determined with engineering judgement, in lieu of testing, when it's obvious that emissions are nowhere near the 10 W/m² power density limit. Knowledge about equipment transmission specifications and engineering judgement might be used to determine compliance. The safety threshold of 10 W/m² greatly exceeds that allowed for wireless communication uses, where SAR (W/kg) limits are applicable. For example, equipment with microprocessor clock oscillators or intentional transmission radios such as Wi-Fi™, Bluetooth®, GPS, or cellular would comply with this threshold since these meet more stringent limits for their applicable bandwidth. When this confidence does not exist, appropriate power density measurements should be conducted.

Subclause 10.4 – Lasers and light emitting diodes (LEDs)

Replace the existing title and text of the rationale for this subclause with the following:

Subclause 10.4 – Lasers

IEC 60825-1 is applicable to safety of laser products emitting laser radiation in the wavelength range 180 nm to 1 mm. A laser product can consist of a single laser with or without a separate power supply or it can incorporate one or more lasers in a complex optical, electrical, or mechanical system.

Any laser product is exempt from all further requirements of IEC 60825-1 if classification by the manufacturer of that product according to Clauses 3, 8 and 9 of IEC 60825-1 shows that the emission level does not exceed the AEL (Accessible Emission Limit) of Class 1 under all conditions of operation, maintenance, service and failure.

In previous editions of IEC 60825-1, LEDs were included in the scope of that standard, and they can be still included in other parts of the IEC 60825 series. However, with the development of lamp safety standards, optical radiation safety of LEDs in general can be more appropriately addressed by lamp safety standards. The removal of LEDs from the scope of IEC 60825-1 does not preclude other standards from including LEDs whenever they refer to lasers. IEC 62471 [67] can be applied to determine the risk group class of an LED or product incorporating one or more LEDs.

IEC 62471 gives guidance for evaluating the photobiological safety of lamps and lamp systems including luminaires. Specifically, it specifies the exposure limits, reference measurement technique and classification scheme for the evaluation and control of photobiological hazards from all electrically powered incoherent broadband sources of optical radiation, including LEDs but excluding lasers, in the wavelength range from 200 nm through 3 000 nm. IEC 62471 was originally prepared as Standard CIE S 009:2002 by the International Commission on Illumination.

Subclause 11.1 – Excessive temperatures in ME EQUIPMENT

In the existing first paragraph, replace "HAZARDS" with "HAZARDOUS SITUATIONS".

Subclause 11.1.3 – Measurements

After the existing paragraph, insert the following:

During development of the requirements of 11.1.3, the working group discussed whether it would be appropriate to reduce the maximum allowable temperature for surfaces and APPLIED PARTS at higher altitudes. In response to this question, a thorough evaluation was performed by Dr. Joachim Kohl, an expert member of the group. The following summarizes the results of that evaluation (see Figure A.22).

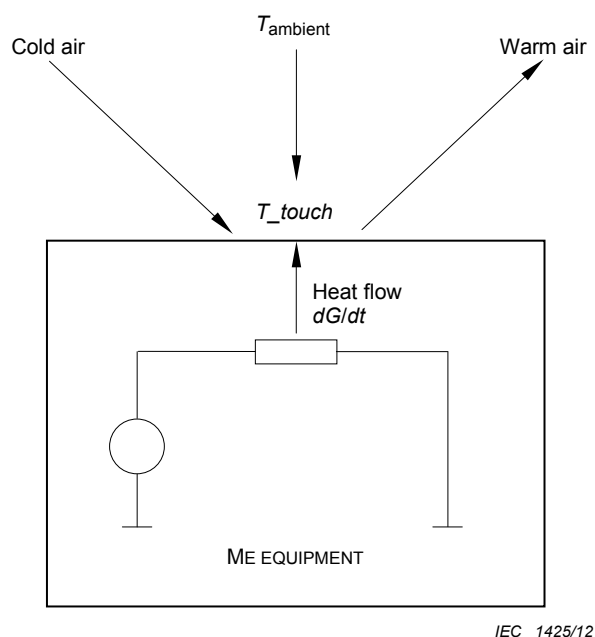


Figure A.22 – Maximum allowable temperature for surfaces and APPLIED PARTS at higher altitudes

The following variables and symbols are used in the evaluation:

T_{touch}	touch temperature at the surface of the ME EQUIPMENT
ΔT	temperature difference of the touched area to ambient
$R_{thermal}$	thermal resistance of the ME EQUIPMENT area under test to ambient
Q	heat
dQ/dt	heat flow
$P_{ambient}$	ambient air pressure
α	heat transfer coefficient = $dQ/dt / (\Delta T * area)$
L	characteristic length of the heat transfer area
$area$	cooling area of the device
λ	heat conductance
η	dynamic viscosity
ρ	density of gas
Nu	Nusselt number
Re	Reynold number
Gr	Grashof number
$\sqrt{\quad}$	square root
$\sim$	proportional

Evaluation of the effects of altitude on thermal transfer:

The standard heat transfer equation is:

$$\Delta T = R_{thermal} * dQ/dt$$

For $R_{thermal}$, one can estimate:

$$R_{thermal} \sim 1 / P_{ambient}^m$$

with the exponent $m = 0,6 \dots 0,8$, we will assume 1.

Equations about the proportionality to $1 / P_{ambient}^m$:

Therefore,

$$R_{thermal} \sim 1 / \alpha$$

where the proportionality factor contains geometric data.

Definition of Nu :

$$Nu = \alpha * L / \lambda$$

Results of theoretical investigations:

$$Nu \sim Re^m$$

The proportionality factor depends on geometry and on the kind of gas and the way of air flow, but not on temperature, pressure, density.

The exponent m is determined by geometry and the way of air flow. For free convection due to a temperature gradient Re is given by:

$$Re = \sqrt{0,4 * Gr}$$

The definition of Gr leads to:

$$Gr \sim (\rho / \eta)^2$$

so that:

$$Re \sim \rho / \eta.$$

This last relationship also holds true for forced convection.

From the general gas equation we can see:

$$\rho \sim P_{ambient}$$

λ and η do not depend on pressure (according to standard physics textbooks). Putting it all together leads to the above stated proportionality between $R_{thermal}$ and $P_{ambient}$.

Subclause 11.2.1 –Strength and rigidity required to prevent fire in ME EQUIPMENT.

In the existing first paragraph, replace "HAZARD" with "HAZARDOUS SITUATION".

Replace the existing final paragraph with the following:

For guidance on assessing fire HAZARDS, see IEC 60695-1-10 [17].

Subclause 11.3 – Constructional requirement for fire ENCLOSURES of ME EQUIPMENT

After the existing paragraph, insert the following:

IEC 60950-1:2005 components (i.e. power supplies, modems, network routers, etc.) are allowed to be integrated into ME EQUIPMENT when MOOP insulation has been determined to be adequate and TOUCH CURRENT limits have been satisfied.

The flame resistance requirements for wire insulation, when used inside a fire ENCLOSURE, in accordance with IEC 60950-1:2005, subclause 4.7.3.4, are:

- Minimum of FV-2 when tested in accordance with IEC 60695-11-10, or
- insulated with PVC, TFE, PTFE, FEP, polychloroprene or polyimide.

While the ME EQUIPMENT flame resistance requirement of FV-1 for wire insulation when used inside a flame enclosure is slightly more stringent than the IEC 60950-1:2005, subclause 4.7.3.4 requirement, it would seem reasonable to conclude an equivalent level of risk should IEC 60950-1 compliant wire be integrated into the ME EQUIPMENT or ME SYSTEM either:

- indirectly when IEC 60950-1 components have been integrated into the ME EQUIPMENT or ME SYSTEM, or
- directly as part of a wire selection decision for the ME EQUIPMENT or ME SYSTEM.

IEC 61010-1:2010 [22], subclause 9.3.2, requires wire insulation when used inside a fire enclosure to have a UL 2556 [72] rating of VW-1, or comply with the applicable requirements of IEC 60332-1-2 [55] or IEC 60332-2-2 [56]. These wire flame resistance requirements are considered acceptable for ME EQUIPMENT and ME SYSTEMS.

Subclause 11.6.3 – Spillage on ME EQUIPMENT and ME SYSTEMS

Replace the existing first paragraph with the following:

In addition to ME EQUIPMENT that requires the use of fluids, some ME EQUIPMENT is exposed to fluid spills as part of reasonably foreseeable misuse. In such cases spillage is considered NORMAL USE and the associated RISKS are addressed by the requirements of this subclause. Furthermore, as described in 9.4.2, tilting of TRANSPORTABLE ME EQUIPMENT to an angle of 10° is considered NORMAL USE. In such cases (as well as for ME EQUIPMENT requiring fluids) the amount and location where spills can occur vary greatly. Only a proper evaluation of the ME EQUIPMENT being tested can determine an appropriate application of the requirement. Doing such an evaluation is the responsibility of the MANUFACTURER and the results are to be provided to those performing the test (typically in the RISK MANAGEMENT FILE). This requirement would be an appropriate area for evaluation by writers of particular standards.

In accord with the principles outlined in IEC 62366, where explicit warnings or safety notices (validated as described in IEC 62366) are given, failure to heed such warnings or safety notices is considered abuse of the ME EQUIPMENT that is beyond the responsibility of the MANUFACTURER. Therefore where such warnings or safety notices are given, the liquid reservoir is filled only to the intended fill level before being tilted to 10°. Where they are not given, the liquid reservoir is filled 15% above the intended level.

Subclause 11.8 – Interruption of the power supply / SUPPLY MAINS to ME EQUIPMENT

In the existing first paragraph, replace "HAZARD" with "HAZARDOUS SITUATION" in two places, and in the final sentence, replace "HAZARDS" with "HAZARDOUS SITUATIONS".

Clause 13 – HAZARDOUS SITUATIONS and fault conditions

In the existing first paragraph, replace "HAZARDS" with "HAZARDOUS SITUATIONS".

Subclause 13.1.2 – Emissions, deformation of ENCLOSURE or exceeding maximum temperature

Replace the existing fourth paragraph with the following:

The power level (15 Watts) cited in the original publication of this standard as the threshold below which ignition will not occur in normal ambient concentrations of oxygen was found to be significantly below the level used in IEC 61010-1 [22]. The original 15 W value was not

based on any scientific test or research but came from “common practice” at some certification bodies (an anecdotal value only).

However, IEC 60950-1 provides a variety of energy levels based on voltage ranging from 100 VA to 150 VA. For purposes of simplicity, the lowest of these values (100 VA) with minimal restrictions on construction were used as an alternative to the 15 W limit (without construction restrictions) originally used in this standard for ME EQUIPMENT in general.

Direct application of the requirements for limited energy circuits in IEC 60950-1 can also be considered to assure that the RISK of fire has been reduced to acceptable levels at higher energies to demonstrate equivalent safety as well.

Since facilities where ME EQUIPMENT compliant with this standard is used will also contain information technology equipment (compliant with IEC 60950-1) and such equipment can be part of ME EQUIPMENT or ME SYSTEMS, it would be unreasonable to require the ME EQUIPMENT to offer a higher level of protection against ignition than does the other equipment used in the same environment.

After the existing fifth paragraph, insert the following:

The restrictions of this subclause are regarded to be sufficient to avoid an enduring fire. A fire is not only dangerous by itself but has side effects such as destroying isolation barriers or destroying mechanical stability. These cannot be taken into account by the above limitations. Therefore, the MANUFACTURER needs to consider the following when coming to a conclusion:

- Obviously a limitation of 100 VA cannot guarantee 'no ignition'. Smaller components such as resistors, semiconductors etc. can be exposed to levels of power that could cause ignition. However, standards such as IEC 60950-1 and IEC 61010-1 and experience demonstrate that when such situations arise if the components are mounted on FV1 rated material an enduring fire is very unlikely to occur.
- Ignition could destroy isolation barriers exposing hazardous voltages. A device which is designed according to the requirements of this subclause should still be checked to determine whether such barriers would be affected and whether additional appropriate precautions independent of the requirements of this subclause.

Since no enduring fire will occur, the mechanical stability is regarded as maintained.

Subclause 13.2.10 – Additional test criteria for motor operated ME EQUIPMENT and Table 26, last line

Replace the existing text of the rationale for this subclause with the following:

Temperature limits of motor windings in ME EQUIPMENT are determined after the first hour as an arithmetic average because experience of test houses has shown that ME EQUIPMENT for non-CONTINUOUS OPERATION reaches variable values that could temporarily differ from the maximum values. Therefore, lower temperature limits are required. The values in Table 26 are based on the requirements of IEC 60950-1:2005.

Clause 14 – PROGRAMMABLE ELECTRICAL MEDICAL SYSTEMS (PEMS)

In the last line of the existing first paragraph, replace "PROGRAMMABLE ME EQUIPMENT" with "PROGRAMMABLE MEDICAL ELECTRICAL SYSTEMS".

Replace the existing fourth paragraph with the following:

Because users of this standard are required to establish, maintain and apply a RISK MANAGEMENT PROCESS as part of compliance, this clause establishes those characteristics unique to programmable systems that should be considered as part of that PROCESS. Requirements for development and modification of medical device software have been

specified in IEC 62304. The requirements from IEC 62304 that apply to development prior to placing a product in service are included by reference for the PEMS software.

There are some requirements in IEC 62304 that apply to software maintenance after a product has been placed into service. These requirements have not been included in the reference.

Subclause 14.1 – General

After the third sentence in the existing first paragraph, insert the following:

For a software PESS, the software safety requirements from subclause 4.3 of IEC 62304:2006 and the software RISK MANAGEMENT requirements from Clause 7 of IEC 62304:2006 are included.

In the final sentence of the existing first paragraph, replace "If application of ISO 14971 shows" with " If the application of RISK MANAGEMENT shows"..

In the existing second paragraph, replace "also those of ISO 14971" with "also those of IEC 62304 that have been included by reference".

In the existing third paragraph, delete the second and third sentences.

Subclause 14.2 – Documentation

In the existing first paragraph, replace "ISO 14971" with "4.2", replace "by that standard" with "by 4.2", and delete the final sentence of the first paragraph.

After the existing first paragraph, insert the following:

Because Clause 14 addresses those RISKS associated with PEMS, this documentation is part of the RISK MANAGEMENT FILE.

Subclause 14.3 – RISK MANAGEMENT plan

In the existing first paragraph, replace "ISO 14971" with "Subclause 4.2.2".

In the existing second paragraph, replace "by ISO 14971" with "by 4.2.2".

Subclause 14.4 – PEMS DEVELOPMENT LIFE-CYCLE

*In the final sentence of the first paragraph immediately following the heading **Framework**, replace "IEC 62304 [26] describes" with "Clauses 5, 7, 8 and 9 of IEC 62304:2006 describe".*

Subclause 14.7 – Requirement specification

In the final sentence of the existing first paragraph, replace "ISO 14971" with "Subclause 4.2".

Subclause 14.8 – Architecture

In the existing first paragraph, replace "ISO 14971" with "4.2".

Subclause 14.10 – VERIFICATION

In the existing first paragraph, replace "ISO 14971" with "Subclause 4.2".

Subclause 14.13 – Connection of PEMS by NETWORK/DATA COUPLING to other equipment

Replace the existing title and text of the rationale with the following:

Subclause 14.13 – PEMS intended to be incorporated into an IT-NETWORK

In the existing first paragraph, replace "networks" with "IT-NETWORKS" in two places.

In the existing second paragraph, replace "networks" with "IT-NETWORKS", and replace "via network" with "via an IT-NETWORK".

Replace the existing third paragraph with the following:

Additional guidance on incorporation of a PEMS into an IT-NETWORK is found in Annex H.

Subclause 15.2 – Serviceability

In the existing paragraph, replace "not create a HAZARD" with "not result in a HAZARDOUS SITUATION".

Subclause 15.3.5 – Rough handling test

Replace the existing second paragraph with:

The 0,8 m/s speed together with the severity of the obstacles chosen, are considered to represent worst case reasonably foreseeable misuse levels of stress/shock energies. The intent of this test is to ascertain the risks from damage caused by rough handling shock/stress. Instability is evaluated in 9.4.

Subclause 15.3.7 – Environmental influences

Add a new item c) as follows:

- c) Corrosion of metal to metal interfaces should be taken into account.

After the rationale for subclause 15.4.3, add the following new rationale:

Subclause 15.4.3.5 – Excessive current and voltage protection

Subclause 4.7 requires the ME EQUIPMENT to be SINGLE FAULT SAFE. One possible fault could be the bridging of a CREEPAGE DISTANCE between the positive and negative terminals of an INTERNAL ELECTRICAL POWER SOURCE in the area between the INTERNAL ELECTRICAL POWER SOURCE and any protective device (see Figure A.23).

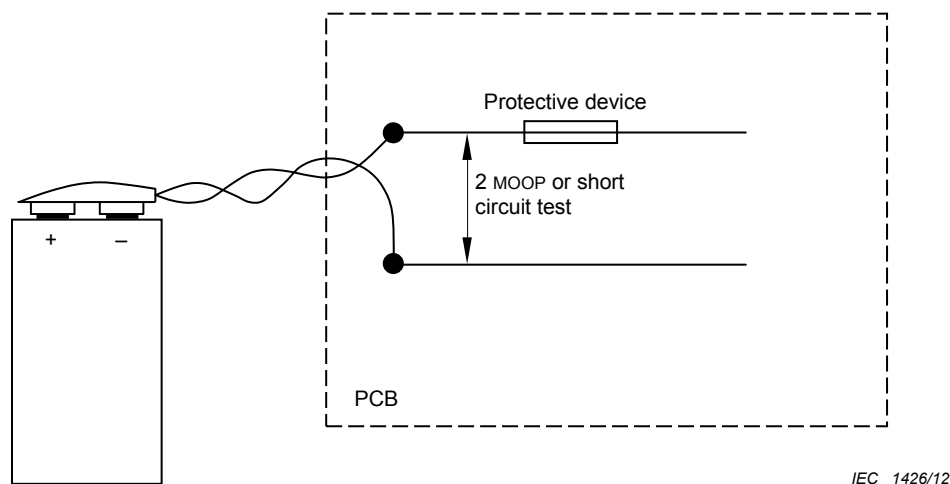


Figure A.23 – Example of the needed MEANS OF OPERATOR PROTECTION between the terminals of an INTERNAL ELECTRICAL POWER SOURCE and a subsequent protective device

A short circuit test would be required to demonstrate that this area is SINGLE FAULT SAFE unless two MEANS OF OPERATOR PROTECTION are present. In that case, the test can be omitted.

Subclause 15.5.3 – Construction of transformers used to provide separation as required by 8.5

Replace the existing first paragraph with the following:

The original citation of subclause 5.12 of IEC 61558-1 was made incorrect by modifications to the 61558 series. In reviewing the revised IEC 61558 standards, it was determined that citation of the standard would not reduce the cost of transformers or provide any other benefits since the spacing and dielectric requirements of IEC 60601-1 would be required to be applied to provide MEANS OF PATIENT PROTECTION. For this reason, the citation was deleted, and the principles of the second edition of IEC 60601-1 were employed with reference to spacing and dielectric requirements of the third edition. Where only MEANS OF OPERATOR PROTECTION is required, transformers constructed for use in IEC 60950-1 equipment should be considered acceptable.

In the existing second paragraph, replace "IEC 60950-1:2001" with "IEC 60950-1:2005", and replace "as would be traditionally be provided" with "as would traditionally be provided".

Subclause 16.2 – ACCOMPANYING DOCUMENTS of an ME SYSTEM

In the existing final paragraph, replace "is to be specified so that no HAZARD" with "are to be specified so that no HAZARDOUS SITUATION".

Subclause 16.5 – SEPARATION DEVICES

In the existing third paragraph, replace "HAZARD" with "HAZARDOUS SITUATION" and replace "HAZARDS" with "HAZARDOUS SITUATIONS".

Subclause 16.9.2.1 – MULTIPLE SOCKET-OUTLET

In the fifth sentence of the existing first paragraph, place EMERGENCY TROLLEYS in regular type.

In the first line of the existing second paragraph, place MAINS CONNECTION in regular type.

Subclause 16.9.2.1 c), 3rd dash

Delete the heading and the entire rationale.

Subclause 16.9.2.1 d)

At the end of the existing first paragraph, insert the following:

Transformers providing a greater degree of safety e.g. ones constructed to IEC 60601-1 requirements or to the isolating transformer requirements in other parts of the IEC 61558 series, e.g. IEC 61558-2-4 [63] or IEC 61558-2-23 [64], are also acceptable.

Subclause 16.9.2.2 – PROTECTIVE EARTH CONNECTIONS in ME SYSTEMS

After the existing first paragraph, insert the following:

When a MULTIPLE SOCKET-OUTLET is used to construct an ME SYSTEM, each individual item of equipment plugged in to the MULTIPLE SOCKET-OUTLET needs to meet the relevant ME EQUIPMENT or non-ME EQUIPMENT requirements. The total protective earth impedance when tested as a complete system according to 8.6.4 needs to meet the highest impedance requirement from 8.6.4. a) (i.e. shall not exceed 200 mΩ). This might require measures such as using short power supply cords within the ME SYSTEM and using a system POWER SUPPLY CORD of greater cross sectional area than is necessary for current carrying purposes to reduce its protective earth impedance. It is unacceptable practice to connect MULTIPLE SOCKET-OUTLETS in series.

Annex B – Sequence of testing**B.1 General**

In the existing third paragraph, place "alarm systems" in small capitals.

Replace the titles and text of B20 and B22 with the following:

B.20 LEAKAGE CURRENTS and PATIENT AUXILIARY CURRENTS and dielectric strength at steady-state operating temperature

See 8.4.2 and 8.7 and 8.8.3.

B.22 LEAKAGE CURRENTS and PATIENT AUXILIARY CURRENTS and dielectric strength after humidity preconditioning at ambient temperature in the laboratory

See 8.4.2 and 8.7 and 8.8.3.

Annex C – Guide to marking and labelling requirements for ME EQUIPMENT and ME SYSTEMS**Table C.1– Marking on the outside of ME EQUIPMENT, ME SYSTEMS or their parts**

In existing row 13 of Table C.1, replace "9.4.2.3" with "9.4.2.3 a) and insert a new row following existing row 13:

Prohibition against sitting or stepping: warning of	9.4.2.3 b)
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Table C.2 – Marking on the inside of ME EQUIPMENT, ME SYSTEMS or their parts

Insert a new row following existing row 3:

Marking of pressure vessels and pipes: national certification mark	9.7.5
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Table C.3 – Marking of controls and instruments

In existing row 2 of Table C.3, replace "15.4.2.2 a)" with "15.4.2.2".

Table C.4 – ACCOMPANYING DOCUMENTS, general

Replace the existing table with the following:

Description of requirement	Clause
CATEGORY AP and CATEGORY APG ME EQUIPMENT and parts	G.3.4
CLASS II ME EQUIPMENT with isolated internal screens: explanation of	8.6.9
Defibrillation voltage, any necessary recovery time	8.5.5.1 b)
Fixing of structures to floor, wall, ceiling, etc.	9.8.1
Instability excluding transport: placement and loading of doors, drawers and shelves	9.4.2.2 e)
IT-NETWORK: instructions for connecting ME EQUIPMENT	14.13
Isolation device: requirements for	8.11.4.4
Lifting points: indication of	9.4.4 a)
Mass of PATIENT, if support systems designed for less than 135 kg	9.8.3.1
Mass of PATIENT, if support systems designed for more than 135 kg	9.8.3.1
MECHANICAL PROTECTIVE DEVICE intended to function only once: instruction to call SERVICE PERSONNEL	9.8.4.3
ME EQUIPMENT: placement of SAFE WORKING LOAD	9.4.2.4.3
ME SYSTEMS: addition requirements	16.2
Noise: protective means	9.6.2.1 b)
Passing over an obstruction: instructions for	9.4.2.4.3
Working conditions	5.4 a)

Table C.5 – ACCOMPANYING DOCUMENTS, instructions for use

Insert a new row following existing row 5:

Effect of multiple cleanings/disinfections	11.6.6
--	--------

In existing row 7, replace "9.8.3.2" with "9.8.3.1" and in existing row 9 replace "9.4.2.4 a)" with "9.4.2.4.2".

Table C.6 – ACCOMPANYING DOCUMENTS, technical description

Replace the existing table with the following:

Description of requirement	Clause
External means of isolation: description of	8.11.1 b)
ME SYSTEM large transient currents: disclosure of	16.3
Non-automatic discharging device for internal capacitors: specification of	8.4.4

Annex D – Symbols on marking

In the existing fifth paragraph, replace "IEC 60878" with "IEC 60878 [60], "sign" with "signs", and "complied" with "compiled".

Table D.1 – General symbols

At the end of the table, insert the following new symbol:

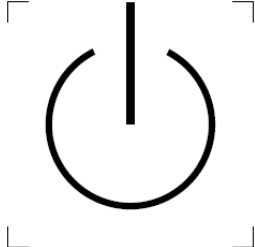
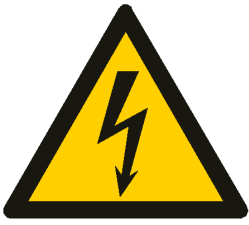
29		IEC 60417-5009	Stand-by
----	--	----------------	----------

Table D.2 – Safety signs

Replace existing Safety sign 3 with the following:

3		ISO 7010-W012	Warning, electricity
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Delete the table footnote ^a at the end of the table.

Annex G – Protection against HAZARDS of ignition of flammable anaesthetic mixtures

G.2.5 ME EQUIPMENT specified for use with FLAMMABLE ANAESTHETIC MIXTURE WITH OXYGEN OR NITROUS OXIDE

Replace "Compliance with the requirements of G.2.3 through G.2.4 (inclusive)" with "Compliance with the requirements of G.2.4 and G.2.5".

G.3 Marking, ACCOMPANYING DOCUMENTS

G.3.1 CATEGORY APG marking

Replace "IEC 60417-5332 (DB:2002-10)" with "IEC 60417-5332 (2002-10)".

G.3.2 CATEGORY AP MARKING

Replace "IEC 60417-5331 (DB:2002-10)" with "IEC 60417-5331 (2002-10)".

G.3.3 Placement of markings

Replace "The markings according to G.3.2 and G.3.3" with "The markings according to G.3.1 and G.3.2".

G.4.2 c)

Replace "inadmissible temperature or in a HAZARD in such a part (see G.6.3 a))" with "inadmissible temperature in such a part (see G.6.3 a)) or in any HAZARDOUS SITUATION".

G.5.3 * Low-energy circuits

In the first dash, replace "apparatus according to G.6" with "apparatus according to Figure G.7".

G.6.3 b)

Replace " $U_{max} \leq L_{zR}$ with a given I_{zR} , see Figure G.4, and" with " $U_{max} \leq U_{zR}$ with a given I_{zR} , see Figure G.4, and".

In the first dash, replace "apparatus according to F.8" with "apparatus according to Figure G.7", and replace "(ether volume percentage $12,2 \pm 0,4 \%$)" with "(ether volume percentage $4,3 \pm 0,2 \%$)".

Annex H – PEMS structure, PEMS DEVELOPMENT LIFE-CYCLE and documentation

H.3 Software PROCESSES

Replace the existing text of the clause, including all its subclauses, with the following:

IEC 62304 describes the processes to be included in the software development life-cycle for the development of safe medical device software.

H.5 Documentation

*Replace the existing text of the clause with "Not used.", delete Figure H.3, and replace the title of Figure H.3 with "**Figure H.3 – Not used**".*

H.6 NETWORK/DATA COUPLING

Replace the existing title and text of the clause as follows:

H.6 PEMS intended to be incorporated into an IT-NETWORK

H.6.1 General

In the context of this standard, the information transmitted over an IT-NETWORK is that intended by the MANUFACTURER to be transmittable (i.e. not through illegal or illicit actions of unauthorized persons).

H.6.2 System integration responsibilities

ME EQUIPMENT and ME SYSTEMS will sometimes be used together on an IT-NETWORK. This is likely to become more frequent with the increasing use of computers to analyze clinical data and control treatment.

Sometimes ME EQUIPMENT will have been designed by the MANUFACTURER to work on an IT-NETWORK with other ME EQUIPMENT. It will often be the case, however, that ME EQUIPMENT will not have been designed to work with all other ME EQUIPMENT on an IT-NETWORK. Someone has to be responsible for ensuring that all the separate ME EQUIPMENT on an IT-network work together satisfactorily; in other words, someone has to be responsible for designing the IT-NETWORK.

It is recognized that the IT-NETWORK integrator often has to comply with particular regulatory requirements.

In order to perform its function, the IT-NETWORK integrator needs to know:

- how the integrated IT-NETWORK is intended to be used;
- the required performance of the integrated IT-NETWORK;
- the intended configuration of the IT-NETWORK;
- the constraints on the extendibility of the IT-NETWORK;
- the specifications of all ME EQUIPMENT and other equipment to be integrated;
- the performance of each ME EQUIPMENT and other equipment; and
- the information flow in and around the IT-NETWORK.

This information will not be available to the individual MANUFACTURERS, and for this reason each individual MANUFACTURER cannot carry out the role of IT-NETWORK integrator. In any case, the IT-NETWORK integrator has to be a single person or organisation that has overall responsibility. This overall responsibility cannot be shared between several different MANUFACTURERS. The responsibility of a MANUFACTURER is limited to providing the required information on their equipment (see 14.13).

Obviously a RESPONSIBLE ORGANIZATION can employ a MANUFACTURER to integrate their IT-NETWORK. In this case the whole IT-NETWORK can become an ME SYSTEM and it will be the MANUFACTURER'S responsibility to provide a correctly integrated system. In this case the system could be separately regulated.

The IT-NETWORK integrator should be competent to assess and address the HAZARDS that are likely to arise from integrating an IT-NETWORK and to ensure that the RESIDUAL RISKS of the individual PEMS are maintained.

Typically an IT-NETWORK integrator would:

- plan the integration of any ME EQUIPMENT or ME SYSTEM and non-medical equipment in accordance with the instructions provided by the various MANUFACTURERS;
- perform RISK MANAGEMENT on the integrated IT-NETWORK; and
- pass on any MANUFACTURER'S instructions to the RESPONSIBLE ORGANIZATION where these are required for the safe operation of the integrated IT-NETWORK. These instructions should include warnings about the HAZARDS of any change of configuration.

H.7 Design considerations for NETWORK/DATA COUPLING

Replace the existing title and text of subclauses 7.1 and 7.2 with the following:

H.7 Design considerations for IT-NETWORKS

H.7.1 Overview

From the viewpoint of a PEMS MANUFACTURER, any type of an IT-NETWORK is a source of additional HAZARDOUS SITUATIONS. In principle any IT-NETWORK that is outside the control of the PEMS MANUFACTURER should never be presumed to be 100 % reliable.

H.7.2 Causes of HAZARDOUS SITUATIONS associated with IT-NETWORKS

In IT-NETWORKS, likely causes for HAZARDOUS SITUATIONS are:

- loss of data;
- inappropriate data interchange;
- corrupted data;
- inappropriate timing of data;
- unexpected receipt of data;
- unauthorized access to data.

Supplementing Annex E of ISO 14971:2007, at least the following initiating events or circumstances that can lead to HAZARDOUS SITUATIONS associated with IT-NETWORKS should be considered:

- remote servicing (external access to the network);
- operating system (compatibility of operating systems);
- modification/upgrades of software (operating systems, applications, etc.);
- interface compatibility (data collisions, data formats):
 - connections (modification of hardware, network connectors);
 - network interface boards (compatibility);
 - network protocols (DICOM, HL7, etc.);
- packet address structure/timing;
- normal network loads/bandwidth;
- peak network load;
- data media (longevity and retrievability);
- security (viruses, worms, unauthorized software updates or upgrades);
- maximum acceptable response time;
- acceptable failure rate of the network;
- availability of the network (planned and unplanned maintenance);
- inconsistency in interfaces/formats resulting in loss of fidelity during information transfer;
- heterogeneous network topologies.

Supplementing Annex C of ISO 14971:2007, the following questions should be taken into account when identifying the characteristics that could impact on safety:

a) Reasonably foreseeable misuses

Is connection to the network inconsistent with the INTENDED USE of each constituent PEMS?

b) Incorrect data flow to or from each constituent PEMS

What are the data transferred by the network used for, and to which tasks are they related? What are the consequences of a breakdown of the IT-NETWORK?

c) Deviation from the specified operational characteristics of any constituent PEMS

What are the operational characteristics of the PEMS and to what degree are they affected by the IT-NETWORK?

d) Incomplete characterization of IT-NETWORK parameters

Is the network topology, configuration, parameters (e.g. open or closed, bandwidth, transmission protocol) completely characterized? Are there any breakdown characteristics/concepts and what are these?

e) Excessive use/load of the IT-NETWORK by the network nodes

What is the planned number of network nodes and their assumed degree of use? Are the resources sufficient to meet the needs of both the IT-NETWORK itself and the devices connected to it?

f) Use errors

What skills are required by the OPERATOR for the effective operation of the system?

g) Inadequate configuration management

Do periodic service tasks alter the IT-NETWORK'S characteristics (e.g. after remote access, updates or upgrades)? Does the RESPONSIBLE ORGANIZATION ensure that modifications to each constituent PEMS are reviewed and approved?

h) Information in wrong place

Does data arrive at a convenient and predictable location? Is it accompanied by irrelevant data that could confuse the OPERATOR or obscure the wanted data? When it arrives, is its source adequately indicated?

H.7.3 Network classification based on the consequences to the PATIENT

Replace the existing title and text of subclause H.7.3 with the following:

H.7.3 Not used

Delete the existing Table H.1 and replace the title with "Table H.1 – Not used".

H.7.4 NETWORK/DATA COUPLING parameters

Replace the existing title and text of subclauses H.7.4 with the following:

H.7.4 IT-NETWORK parameters

The use of an IT-NETWORK for exchange of data either between PEMS or between PEMS and other information technology equipment requires the knowledge about both the structure of the IT-NETWORK and the PROCESSES/functions running inside them. This is important because MANUFACTURERS of PEMS or IT-NETWORKS should select the configuration of their products such that:

- they comply with internationally recognized network standards (Ethernet, Fast Ethernet, GigaBitEthernet, FDDI, etc.) and use the available bandwidth appropriately according to the INTENDED USE;
- they achieve the optimal performance for their application

A mixture of different IT-NETWORK configurations/parameter settings can emerge that are not always compatible for the different IT-NETWORK nodes in spite of the fact that they comply to valid international standards.

To avoid or at least to minimize the resulting potential of disruption, a match of a minimum set of IT-NETWORK parameters derived from the relevant standards is required.

Figure H.4 contains a list of parameters potentially required to be specified. Due to the rapid evolution of IT-NETWORK technology, this table should be seen as a starting point. It should be clear if the table should be maintained and who should be responsible for maintaining it.

In the seventh row of Figure H.4, replace "Free fields (which are used)" with "Free fields (that are used)", and in the title of Figure H.4, replace "for NETWORK/DATA COUPLING" with "for an IT-NETWORK".

Annex I – ME SYSTEMS aspects

I.1.2 Localities in a medical environment

In the existing note, replace "causing a HAZARD" with "resulting in a HAZARDOUS SITUATION" and replace "ME EQUIPMENT is of TYPE B" with "APPLIED PART is of TYPE B".

Annex J – Survey of insulation paths

Replace the existing Figure J.6 with the following:

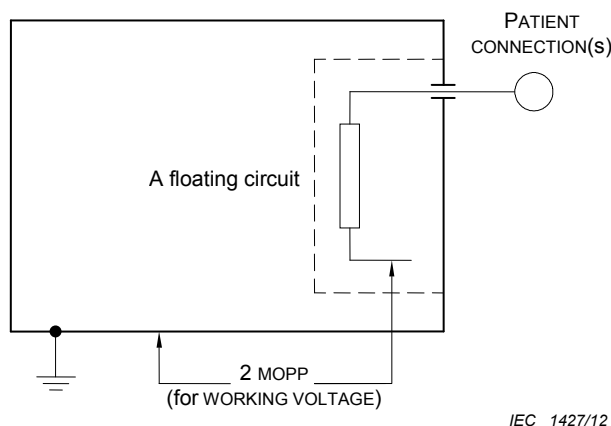


Figure J.6 – Insulation example 6

Annex L – Insulated winding wires for use without interleaved insulation

L.3 TYPE TEST

L.3.1 Dielectric strength

In the existing paragraph, replace "IEC 60851-5:1996" with "IEC 60851-5:2008".

L.3.2 Flexibility and adherence

In the existing first paragraph, replace "IEC 60851-3:1996" with "IEC 60851-3:2009" and "IEC 60851-3:1997" with "IEC 60851-3:2009".

L.3.4 Retention of electric strength after bending

In the existing note, replace "third edition" with "fourth edition".

L.4 Tests during manufacture

L.4.3 Sampling tests

In the existing first paragraph, replace "IEC 60851-5:1996" with "IEC 60851-1:2008".

Add the following new annexe:

Annex M (normative)

Reduction of pollution degrees (see 8.9.1.8)

Table M.1 shows additional protective measures that may be used to reduce the pollution degree.

**Table M.1 – Reduction of the pollution degree of internal environment
through the use of additional protection**

Additional protection	From pollution degree 2 of external environment to:	From pollution degree 3 of external environment to:
ENCLOSURE IPX4 of IEC 60529	2	2
ENCLOSURE IPX5 or IPX6 of IEC 60529	2	2
ENCLOSURE IPX7 or IPX8 of IEC 60529	2 (see note)	2 (see note)
Hermetically sealed ENCLOSURE	1	1
Constantly heated	1	1
Encapsulated	1	1
Coated	1	2
NOTE Reduction can be to pollution degree 1, if the equipment is manufactured with a low internal humidity and the manufacturer's instructions specify that, after opening the ENCLOSURE, the ENCLOSURE is reassembled in a controlled humidity environment or a desiccant is used.		

Bibliography

Replace entry [17] with:

- [17] IEC 60695-1-10, *Fire hazard testing – Part 1-10: Guidance for assessing the fire hazard of electrotechnical products – General guidelines*

Replace entry [22] with:

- [22] IEC 61010-1:2010, *Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 1: General requirements*

Replace entry [26] with:

- [26] IEC 80001-1:2010, *Application of risk management for IT-networks incorporating medical devices – Part 1: Roles, responsibilities and activities*

Replace entry [45] with:

- [45] DALZIEL, CF., LEE, WR., Re-evaluation of lethal electric currents. *IEEE Transactions on Industry and General Applications*, September/October 1968, Vol. 1 GA-4, No. 5

Add the following new entries:

- [55] IEC 60332-1-2, *Tests on electric and optical fibre cables under fire conditions Part 1-2: Test for vertical flame propagation for a single insulated wire or cable – Procedure for 1 kW pre-mixed flame*
- [56] IEC 60332-2-2, *Tests on electric and optical fibre cables under fire conditions Part 2-2: Test for vertical flame propagation for a single small insulated wire or cable – Procedure for diffusion flame*
- [57] IEC 60601-1-11, *Medical electrical equipment – Part 1-11: General requirements for basic safety and essential performance – Collateral Standard: Requirements for medical electrical equipment and medical electrical systems used in the home healthcare environment*
- [58] IEC 60601-1-12, *Medical electrical equipment – Part 1-12: General requirements for basic safety and essential performance – Collateral Standard: Requirements for medical electrical equipment and medical electrical systems used in the emergency medical services environment*
- [59] IEC 60601-2-22, *Medical electrical equipment – Part 2-22: Particular requirements for basic safety and essential performance of surgical, cosmetic, therapeutic and diagnostic laser equipment*
- [60] IEC 60878, *Graphical symbols for electrical equipment in medical practice*
- [61] IEC 61010 (all parts), *Safety requirements for electrical equipment for measurement, control, and laboratory use*
- [62] IEC 61558-1, *Safety of power transformers, power supplies, reactors and similar products – Part 1: General requirements and tests*

- [63] IEC 61558-2-4, *Safety of transformers, reactors, power supply units and similar products for supply voltages up to 1 100 V – Part 2-4: Particular requirements and tests for isolating transformers and power supply units incorporating isolating transformers*
- [64] IEC 61558-2-23, *Safety of transformers, reactors, power supply units and combinations thereof – Part 2-23: Particular requirements and tests for transformers and power supply units for construction sites*
- [65] IEC 62353, *Medical electrical equipment – Recurrent test and test after repair of medical electrical equipment*
- [66] IEC/TR 62354, *General testing procedures for medical electrical equipment*
- [67] IEC 62471:2006, *Photobiological safety of lamps and lamp systems*
- [68] ISO 7396-1, *Medical gas pipeline systems – Part 1: Pipeline systems for compressed medical gases and vacuum*
- [69] ISO 14708 (all parts), *Implants for surgery – Active implantable medical devices*
- [70] ISO 15001, *Anaesthetic and respiratory equipment – Compatibility with oxygen*
- [71] ANSI/AAMI HE-75:2009, *Human Factors Engineering – Design of Medical Devices*
- [72] UL 2556:2007, *UL Standard for Safety Wire and Cable Test Methods*

INDEX OF ABBREVIATIONS AND ACRONYMS

Add the following new abbreviations and acronyms:

Abbreviation	Term
AEL	Accessible emission limit
FV	Flammability classification according to IEC 60695-11-10
IBP	Invasive blood pressure catheter
IPS	Isolated power supply
PTFE	Polytetrafluoroethylene
PVC	Polyvinyl chloride
RFID	Radio-frequency identification
SAR	Specific absorption rate
TFE	Tetrafluoroethylene

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Aru 10
10317 Tallinn
Eesti

standard@evs.ee
www.evs.ee